CIP-B106 — Mobile and IoT Forensics

Case Studies: Case Study 2

Android 10 Pixel 3 Evidence Investigation

Submitted by

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Executive Summary

I examined the Pixel 3 Android 10 training logical copy from a mounted working copy (/media/sf_Pixel_3/Pixel 3) on Kali Linux. The handset is a Google Pixel 3 (blueline) running Android 10, SDK 29, build QQ1A.200105.003, patch level 2020-01-01. It has two Android users (0 and 11). The main Google account is thisisdfir@gmail.com, and thisisdfirtwo@gmail.com is also present. There are two Google Fi SIM records, one saved Wi-Fi network and a Bluetooth adapter named "Pixel 3".

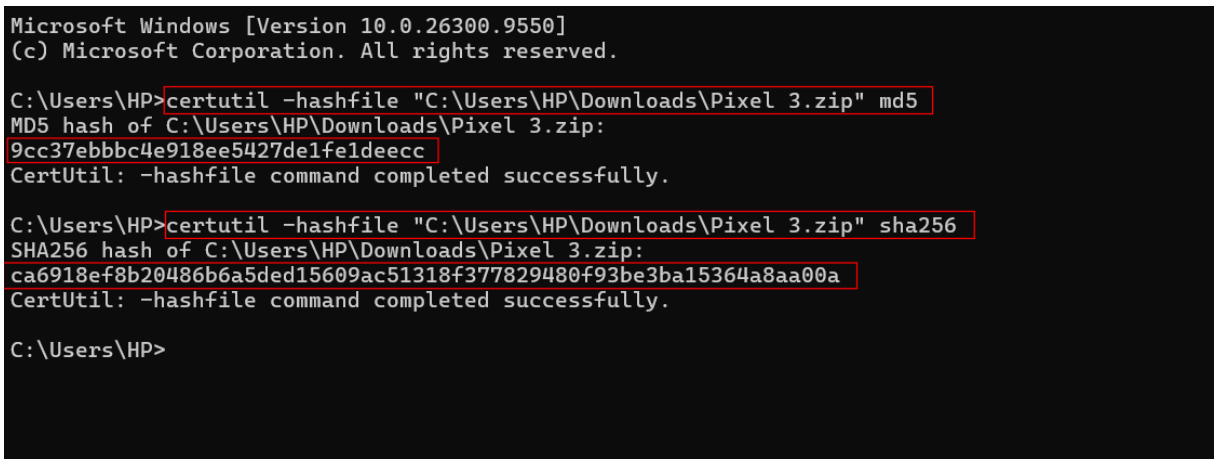
Messaging accounts were added between 29 January and 14 February 2020. Kik, TextNow, WhatsApp and Snapchat hold account data tied to the thisisdfir identity. Kik, TextNow and WhatsApp also hold messages with a contact saved as Josh Hickman. The timeline covers 13 to 15 February 2020 (UTC).

The archive SHA-256 matched the course value. The copy is logical and was taken after TWRP/Magisk rooting, so it is not a full physical image.

Evidence and Method

Evidence Register

Field	Record
Exhibit	Pixel 3 Android 10 course logical copy (extracted file system)
Working copy	/media/sf_Pixel_3/Pixel 3 (VirtualBox shared folder, read-only analysis)
Top-level content	apex, data, metadata, mnt, product, res, sbin, system, vendor, init*.rc
Expected SHA-256	ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a
Archive verified	C:\Users\HP\Downloads\Pixel 3.zip (certutil -hashfile, Windows 10.0.26300)
MD5	9cc37ebbbc4e918ee5427de1fe1deecc
SHA-256 (computed)	ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a
Integrity result	MATCH. The computed SHA-256 equals the course reference value.
Acquisition scope	Logical copy taken after TWRP root access. data/magisk_backup_* and com.topjohnwu.magisk fit the lab rooting workflow and are not treated as user activity.
Time zone	Converted times in this report are UTC.



Tools

Tool	Use
certutil (Windows)	Hashing of the evidence archive (MD5, SHA-256)
Kali Linux shell (ls, tree, cat, grep, find, xxd, date)	Navigation, keyword searches, XML review and epoch conversion
DB Browser for SQLite	Browsing tables in telephony.db, accounts, Kik, TextNow, WhatsApp, GSF and other databases
sqlite3 3.46.1	Manual queries (accounts_de.db, msgstore.db joins)
Python 3 + curl lookup script	Matching package names to Google Play app titles
EpochConverter (web)	Cross-check of millisecond epoch values

Timestamp Method

Most values are Unix epoch milliseconds. I divided by 1000 and converted with date -d '@<seconds>' -u, or datetime(x/1000,'unixepoch') in SQLite. Raw values sit beside the converted times in each table. The play_metalog value was also checked on EpochConverter and gave the same result.

Device, Account and System Profile

Image Layout and Package Inventory

The root of the copy holds the usual Android partitions. data/data lists the application sandboxes. I fed each package name through a small curl/grep loop (and a Python version of it) against Google Play to get readable names. Third-party packages include Discord, TextNow, Messenger, imo, Instagram, Threads, MeWe, Wickr, Dust, Silent Phone, Skout, Skype, Snapchat, Spotify, Gallery Vault, Magisk, X (Twitter), Venmo, Viber, WhatsApp, Wire, TikTok, LINE, Kik, Telegram, Signal and Tor Browser. A package being present only shows that the app is installed. It does not show who used it.

```
┌──(kali㉿kali) [/media/sf_Pixel_3/Pixel 3]
└─$ ls -l
total 92
drwxrwx--- 1 root vboxsf  4096 Feb 16 22:49 apex
drwxrwx--- 1 root vboxsf  4096 Feb 16 22:58 data
-rwxrwx--- 1 root vboxsf  1653 Feb 16 22:49 init.environ.rc
-rwxrwx--- 1 root vboxsf 33402 Feb 16 22:49 init.rc
-rwxrwx--- 1 root vboxsf  7690 Feb 16 22:49 init.usb.configfs.rc
-rwxrwx--- 1 root vboxsf  5649 Feb 16 22:49 init.usb.rc
-rwxrwx--- 1 root vboxsf   563 Feb 16 22:49 init.zygote32.rc
-rwxrwx--- 1 root vboxsf   981 Feb 16 22:49 init.zygote64_32.rc
drwxrwx--- 1 root vboxsf     0 Feb 16 22:58 metadata
drwxrwx--- 1 root vboxsf     0 Feb 16 22:58 mnt
drwxrwx--- 1 root vboxsf  4096 Feb 16 22:58 product
drwxrwx--- 1 root vboxsf     0 Feb 16 22:58 res
drwxrwx--- 1 root vboxsf  4096 Feb 16 22:58 sbin
drwxrwx--- 1 root vboxsf  4096 Feb 16 23:09 system
-rwxrwx--- 1 root vboxsf  2608 Feb 16 22:49 ueventd.rc
drwxrwx--- 1 root vboxsf  4096 Feb 16 23:10 vendor

┌──(kali㉿kali) [/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls -l system
total 236
drwxrwx--- 1 root vboxsf   4096 Feb 16 23:09 apex
drwxrwx--- 1 root vboxsf      0 Feb 16 23:09 app
drwxrwx--- 1 root vboxsf  65536 Feb 16 23:09 bin
-rwxrwx--- 1 root vboxsf   1970 Feb 16 23:09 build.prop
drwxrwx--- 1 root vboxsf   8192 Feb 16 23:09 etc
drwxrwx--- 1 root vboxsf 131072 Feb 16 23:09 fonts
drwxrwx--- 1 root vboxsf  12288 Feb 16 23:09 framework
drwxrwx--- 1 root vboxsf   4096 Feb 16 23:09 lib
drwxrwx--- 1 root vboxsf   4096 Feb 16 23:09 lib64
drwxrwx--- 1 root vboxsf   4096 Feb 16 23:09 priv-app
drwxrwx--- 1 root vboxsf   4096 Feb 16 23:09 usr

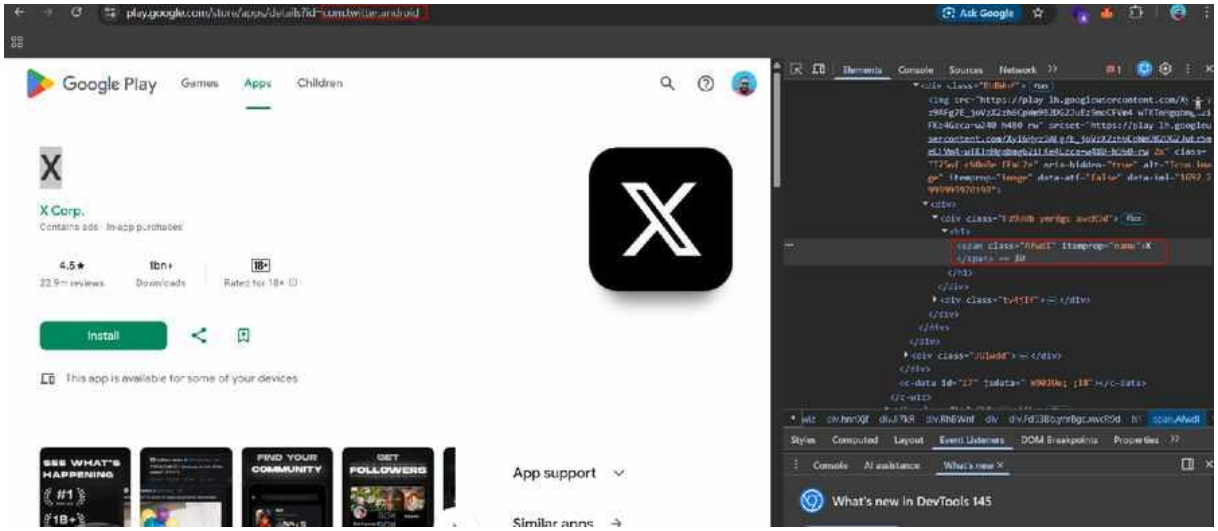
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

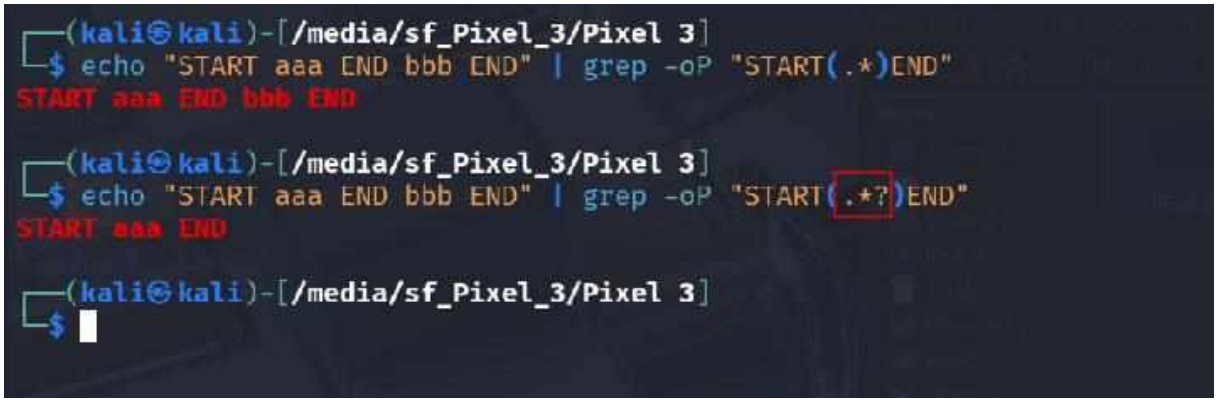
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls -l "data"
total 160
drwxrwx--- 1 root vboxsf     0 Feb 16 22:49 adb
drwxrwx--- 1 root vboxsf     0 Feb 16 22:49 apex
drwxrwx--- 1 root vboxsf 49152 Feb 16 22:50 app
drwxrwx--- 1 root vboxsf     0 Feb 16 22:50 backup
drwxrwx--- 1 root vboxsf     0 Feb 16 22:50 cache
drwxrwx--- 1 root vboxsf     0 Feb 16 22:50 dalvik-cache
drwxrwx--- 1 root vboxsf 49152 Feb 16 22:56 data
drwxrwx--- 1 root vboxsf     0 Feb 16 22:56 drm
drwxrwx--- 1 root vboxsf     0 Feb 16 22:56 magisk_backup_e132ca5af8732354d2f63dcfcd4331cddc309ae8
drwxrwx--- 1 root vboxsf     0 Feb 16 22:56 media
drwxrwx--- 1 root vboxsf  4096 Feb 16 22:57 misc
drwxrwx--- 1 root vboxsf     0 Feb 16 22:57 misc_ce
drwxrwx--- 1 root vboxsf     0 Feb 16 22:57 nfc
drwxrwx--- 1 root vboxsf     0 Feb 16 22:57 property
drwxrwx--- 1 root vboxsf 40960 Feb 16 22:57 resource-cache
drwxrwx--- 1 root vboxsf 16384 Feb 16 22:58 system
drwxrwx--- 1 root vboxsf     0 Feb 16 22:58 system_ce
drwxrwx--- 1 root vboxsf     0 Feb 16 22:58 system_de
drwxrwx--- 1 root vboxsf     0 Feb 16 22:58 tombstones
drwxrwx--- 1 root vboxsf     0 Feb 16 22:58 unencrypted
drwxrwx--- 1 root vboxsf     0 Feb 16 22:58 user_de
drwxrwx--- 1 root vboxsf  4096 Feb 16 22:58 vendor
drwxrwx--- 1 root vboxsf     0 Feb 16 22:58 vendor_ce
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls 'data/system'
appops.xml*                 diskstats_cache.json*       last-header.txt*            overlays.xml*               packages.xml*               shortcut_service.xml*  users/
battery-history/            display-manager-state.xml*  locksettings.db*            package_cache/              package-usage.list*         slice/                 watchlist_report.db*
batterystats.bin*           dropbox/                    log-files.xml*              package-cstats.list*        package-watchdog.xml*       sound_model.db*        watchlist_settings.xml*
batterystats-checkin.bin*   entropy.dat*                netpolicy.xml*              package-dcl.list*           procstats/                  st_sound_model.db*
batterystats-daily.xml*     graphicsstats/              netstats/                   package-dex-usage.list*     recoverablekeystore.db*     sync/
cachequota.xml*             install_sessions.xml*       notification_log.db*        packages.list*              screen_on_time*             uiderrors.txt*
device_policies.xml*        job/                        notification_policy.xml*    packages-warnings.xml*      sensor_service/             usagestats/
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls -l 'data/user_de'
total 52
drwxrwx--- 1 root vboxsf 40960 Feb 16 22:58 0
drwxrwx--- 1 root vboxsf 12288 Feb 16 22:58 11
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls 'data/data'
com.android.chrome                          com.google.android.apps.naps                com.google.android.googlequicksearchbox     com.instagram.threadsapp
com.android.hotwordenrollment.okgoogle      com.google.android.apps.messaging           com.google.android.gsf                      com.mewe
com.android.hotwordenrollment.xgoogle       com.google.android.apps.nexuslauncher       com.google.android.ims                      com.mywickr.wickr2
com.android.keychain                        com.google.android.apps.photos              com.google.android.inputmethod.latin        com.qti.ltebc
com.android.mtp                             com.google.android.apps.pixelmigrate        com.google.android.keep                     com.radicalapps.cyberdust
com.android.nfc                             com.google.android.apps.safetyhub           com.google.android.markup                   com.silentcircle.silentphone
com.android.providers.calendar              com.google.android.apps.tachyon             com.google.android.music                    com.skout.android
com.android.providers.contacts              com.google.android.apps.tips                com.google.android.onetimeinitializer       com.skype.raider
com.android.providers.downloads             com.google.android.apps.turbo               com.google.android.partnersetup             com.snapchat.android
com.android.providers.media                 com.google.android.apps.tycho               com.google.android.pixel.setupwizard        com.spotify.music
com.android.providers.telephony             com.google.android.apps.walletnfcrel        com.google.android.projection.gearhead      com.thinkyeah.galleryvault
com.android.providers.userdictionary        com.google.android.apps.wallpaper           com.google.android.settings.intelligence    com.topjohnwu.magisk
com.android.service.ims.presence            com.google.android.apps.wellbeing           com.google.android.setupwizard              com.twitter.android
com.android.traceur                         com.google.android.apps.work.oobconfig      com.google.android.syncadapters.contacts    com.venmo
com.android.vending                         com.google.android.apps.youtube.music       com.google.android.tts                      com.viber.voip
com.discord                                 com.google.android.as                       com.google.android.videos                   com.whatsapp
com.enflick.android.TextNow                 com.google.android.calendar                 com.google.android.webview                  com.wire
com.facebook.orca                           com.google.android.configupdater            com.google.android.wfcactivation            com.zhiliaoapp.musically
com.google.android.apps.chromecast.app      com.google.android.contacts                 com.google.android.youtube                  jp.naver.line.android
com.google.android.apps.docs                com.google.android.deskclock                com.google.ar.core                          kik.android
com.google.android.apps.docs.editors.docs   com.google.android.dialer                   com.google.intelligence.sense               org.telegram.messenger
com.google.android.apps.dreamliner          com.google.android.documentsui              com.google.vr.apps.ornament                 org.thoughtcrime.securesms
com.google.android.apps.enterprise.dmagent  com.google.android.ext.services             com.google.vr.vrcore                        org.torproject.torbrowser
com.google.android.apps.gcs                 com.google.android.gm                       com.imgur.mobile
com.google.android.apps.helprtc             com.google.android.gms                      com.imo.android.imous
com.google.android.apps.magazines           com.google.android.GoogleCamera             com.instagram.android
```


Google Play
Games
Apps
Children
X
X Corp.
Install
This app is available for some of your devices
SEE WHAT'S HAPPENING
FIND YOUR COMMUNITY
GET FOLLOWERS
App support
Similar apps
Elements
Console
Sources
Network
Styles
Computed
Layout
Event Listeners
DOM Breakpoints
Properties
What's new in DevTools 145

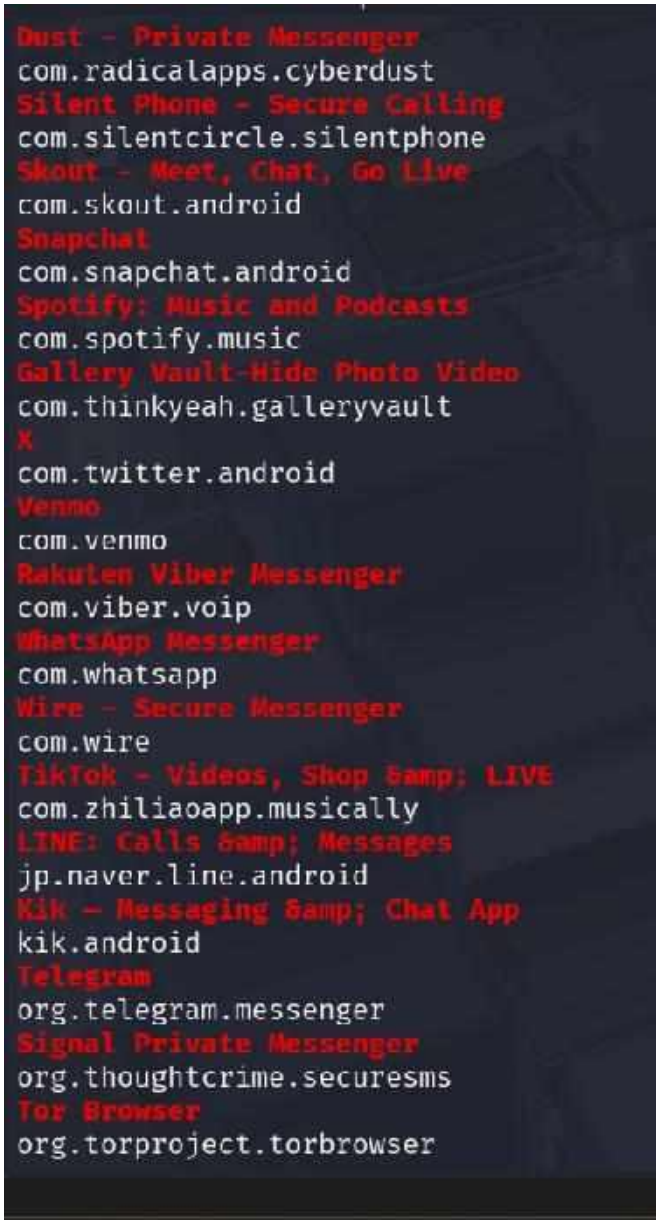

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ echo "START aaa END bbb END" | grep -oP "START(.*)END"
START aaa END bbb END
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ echo "START aaa END bbb END" | grep -oP "START(.*?)END"
START aaa END
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$


┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ curl --no-progress-meter https://play.google.com/store/apps/details?id=com.twitter.android | grep -Pio 'X'
X


┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ curl --no-progress-meter https://play.google.com/store/apps/details?id=com.twitter.android | grep -Pio '\K.+?(?=</sp
n>)'
X
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$

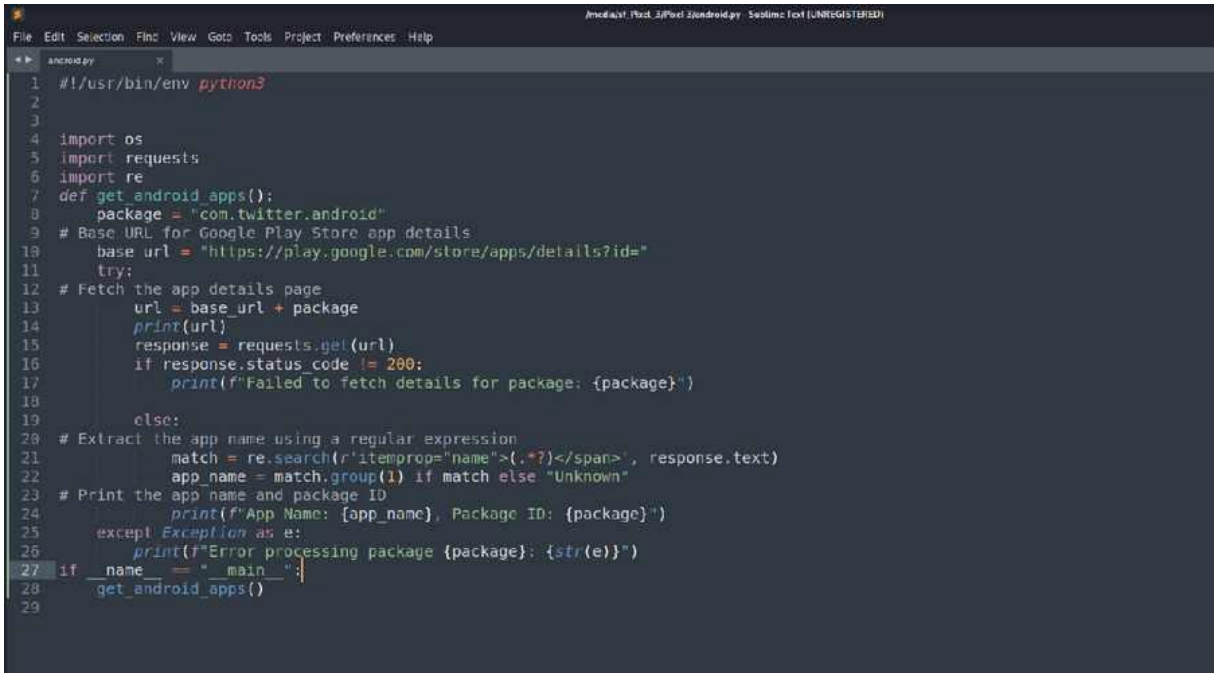
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls 'data/data' \
| while read package; do \
curl --no-progress-meter https://play.google.com/store/apps/details?id=$package \
| grep -Pio '<span class="AfwdI" itemprop="name">\K.*?(?=</span>)' \
&& echo $package; \
done

Google Chrome
com.android.chrome
Discord - Talk, Play, Hang Out
com.discord
TextNow: Call + Text Unlimited
com.enflick.android.TextNow
Messenger
com.facebook.orca
Google Home
com.google.android.apps.chromecast.app
Google Drive
com.google.android.apps.docs
Google Docs
com.google.android.apps.docs.editors.docs
Google Support Services
com.google.android.apps.helprtc
Google News - Daily Headlines
com.google.android.apps.magazines
Google Maps
com.google.android.apps.maps
Google Messages
com.google.android.apps.messaging
Google Photos
```




```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls 'data/data' | while read package; do
  curl --no-progress-meter "https://play.google.com/store/apps/details?id=$package" \
    | grep -Pio '<div jsname="sngebd">\K.*?(?=</div>)' \
    | grep -io money && echo "$package"
done


┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```





Build, Hardware, SIM, Users and Accounts

Item	Value	Source
Manufacturer / model	Google / Pixel 3	product/build.prop (ro.product.product.*)
Codename / device	blueline	product/build.prop
Android / SDK	10 / 29	ro.product.build.version.release / .sdk
Build ID / incremental	QQ1A.200105.003 / 6079943	product/build.prop
Build date	1576611252 → 2019-12-17 19:34:12 UTC	ro.product.build.date.utc
Security patch	2020-01-01	system/build.prop

Hardware serial	8CEX1N716	mnt/vendor/persist/sensors/tmd2725_als_factory.txt
SIM records	ICCID 8901260971148676693 (MCC 310 / MNC 260); ICCID 89011203004056803842 (310 / 120); carrier Google Fi	user_de/0/com.android.providers.telephony/databases/telephony.db › siminfo
Blocked number	+19195121037	blockednumbers.db › blocked
Users	0 ("This Is") and 11 ("User 2"); deviceOwnerUserId -10000	data/system/users/userlist.xml, 0.xml, 11.xml
User 0 lastLoggedIn	1581695517124 → 2020-02-14 15:51:57 UTC	users/0.xml
User 11 created / last in	1581644358241 → 2020-02-14 01:39:18; 1581684601922 → 12:50:01 UTC	users/11.xml
Google accounts	thisisdfir@gmail.com (608 hits); thisisdfirtwo@gmail.com (120 hits)	grep across data; Gmail.xml active-account
Bluetooth	Name "Pixel 3", address 7C:D9:5C:AC:A2:CE	users/0/settings_secure.xml
Wi-Fi	SSID "CookiesDcastle85" (WPA-PSK), mapped MAC ae:f6:c7:a8:a5:c2	data/misc/wifi/WifiConfigStore.xml

accounts_de.db (last_password_entry_time_millis_epoch) shows when each account was added. The Google account came first on 2020-01-29 15:41:07 UTC. Telegram, Signal, imo, TikTok, Messenger, Twitter and WhatsApp followed on the same day. Silent Circle and Skype came on 1 February, TextNow and Viber (+19195794674) on 8 February, and Duo on 2020-02-14 01:19:15 UTC. These are account-creation events on the handset and do not prove the user was present each time. I did not attempt to recover any password.

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'product/build.prop'

# begin common build properties
# autogenerated by build/make/tools/buildinfo_common.sh
ro.product.build.date=Tue Dec 17 19:34:12 UTC 2019
ro.product.build.date.utc=1576611252
ro.product.build.fingerprint=google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
ro.product.build.id=QQ1A.200105.003
ro.product.build.tags=release-keys
ro.product.build.type=user
ro.product.build.version.incremental=6079943
ro.product.build.version.release=10
ro.product.build.version.sdk=29
ro.product.product.brand=google
ro.product.product.device=blueline
ro.product.product.manufacturer=Google
ro.product.product.model=Pixel 3
ro.product.product.name=blueline
# end common build properties

#
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep patch  'system/build.prop'

ro.build.version.security_patch=2020-01-01

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ product/build.prop
product/build.prop: 4: ro.product.build.date=Tue: not found
product/build.prop: 5: ro.product.build.date.utc=1576611252: not found
product/build.prop: 6: ro.product.build.fingerprint=google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-ke
product/build.prop: 7: ro.product.build.id=QQ1A.200105.003: not found
product/build.prop: 8: ro.product.build.tags=release-keys: not found
product/build.prop: 9: ro.product.build.type=user: not found
product/build.prop: 10: ro.product.build.version.incremental=6079943: not found
product/build.prop: 11: ro.product.build.version.release=10: not found
product/build.prop: 12: ro.product.build.version.sdk=29: not found
product/build.prop: 13: ro.product.product.brand=google: not found
product/build.prop: 14: ro.product.product.device=blueline: not found
product/build.prop: 15: ro.product.product.manufacturer=Google: not found
product/build.prop: 16: ro.product.product.model=Pixel: not found
product/build.prop: 17: ro.product.product.name=blueline: not found
product/build.prop: 24: persist.rild.nitz_plmn=: not found
product/build.prop: 25: persist.rild.nitz_long_ons_0=: not found
product/build.prop: 26: persist.rild.nitz_long_ons_1=: not found
product/build.prop: 27: persist.rild.nitz_long_ons_2=: not found
product/build.prop: 28: persist.rild.nitz_long_ons_3=: not found
product/build.prop: 29: persist.rild.nitz_short_ons_0=: not found
product/build.prop: 30: persist.rild.nitz_short_ons_1=: not found
product/build.prop: 31: persist.rild.nitz_short_ons_2=: not found
product/build.prop: 32: persist.rild.nitz_short_ons_3=: not found
product/build.prop: 36: ro.telephony.default_network=10: not found
product/build.prop: 38: debug.sf.hw=1: not found
product/build.prop: 39: debug.gralloc.enable_fb_ubwc=1: not found
product/build.prop: 42: persist.vendor.cne.feature=1: not found
product/build.prop: 45: media.stagefright.enable-player=true: not found
product/build.prop: 46: media.stagefright.enable-http=true: not found
```

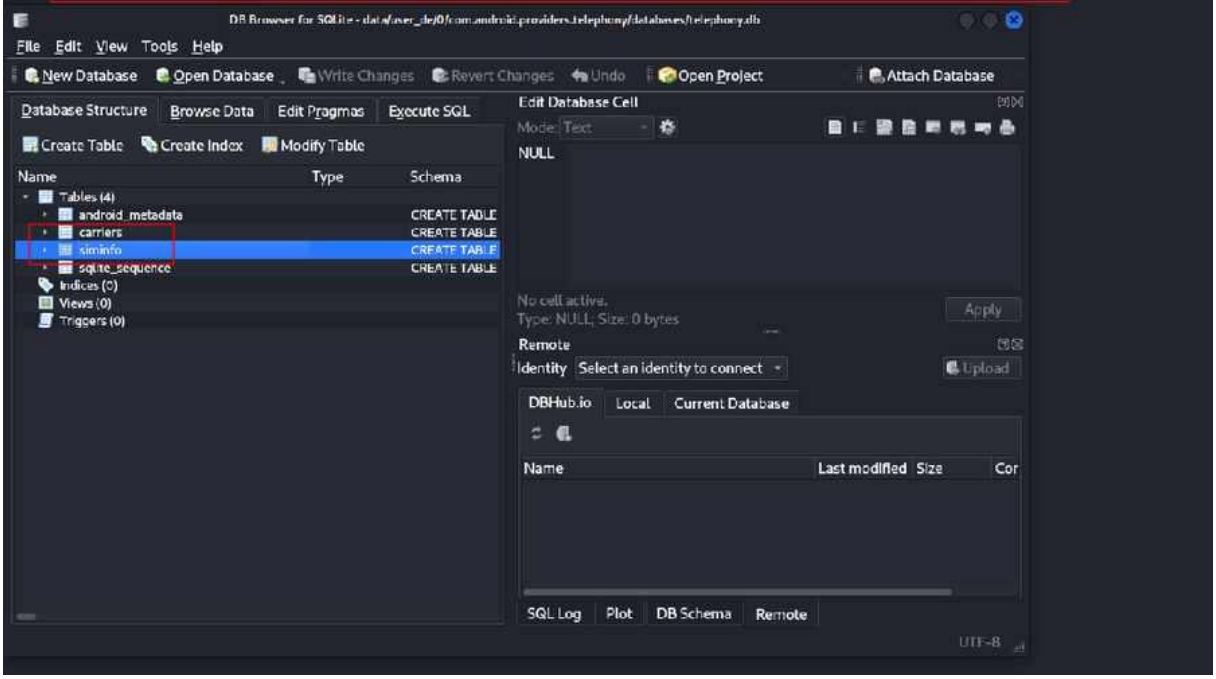
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/user_de/0/com.android.providers.telephony/'

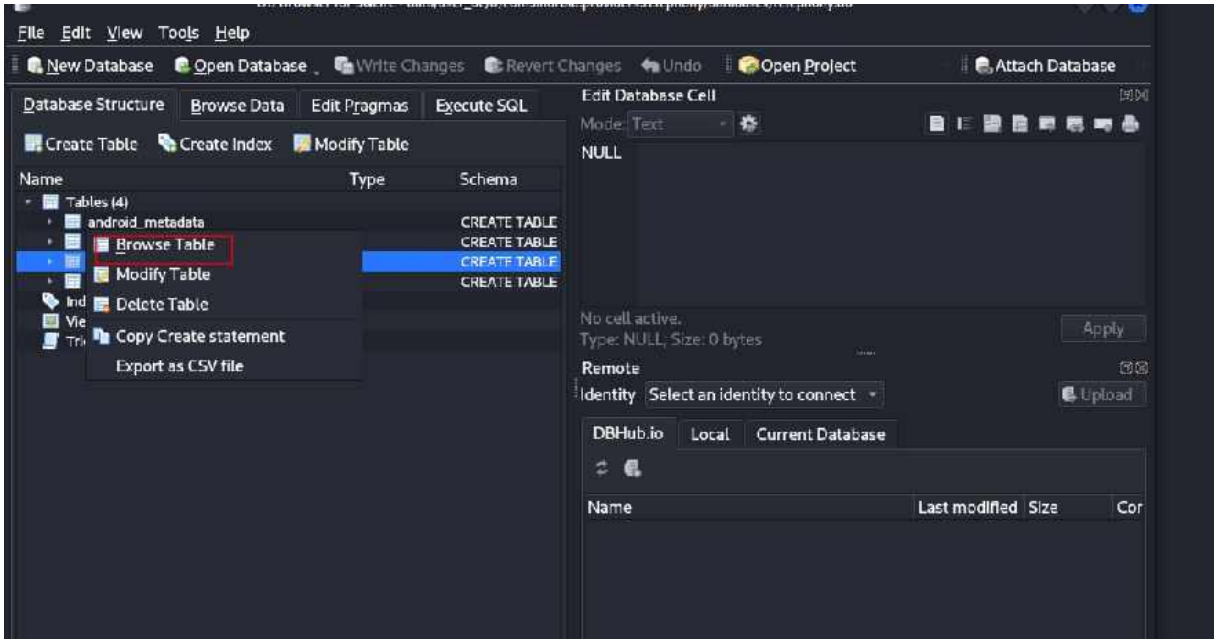
data/user_de/0/com.android.providers.telephony/
├── databases
│   ├── carrierIdentification.db
│   ├── CarrierInformation.db
│   ├── cellbroadcasts.db
│   ├── HbpcdLookup.db
│   ├── mmssms.db
│   └── telephony.db
└── shared_prefs
    ├── build-id.xml
    ├── CarrierIdProvider.xml
    ├── preferred-apn.xml
    └── preferred-full-apn.xml

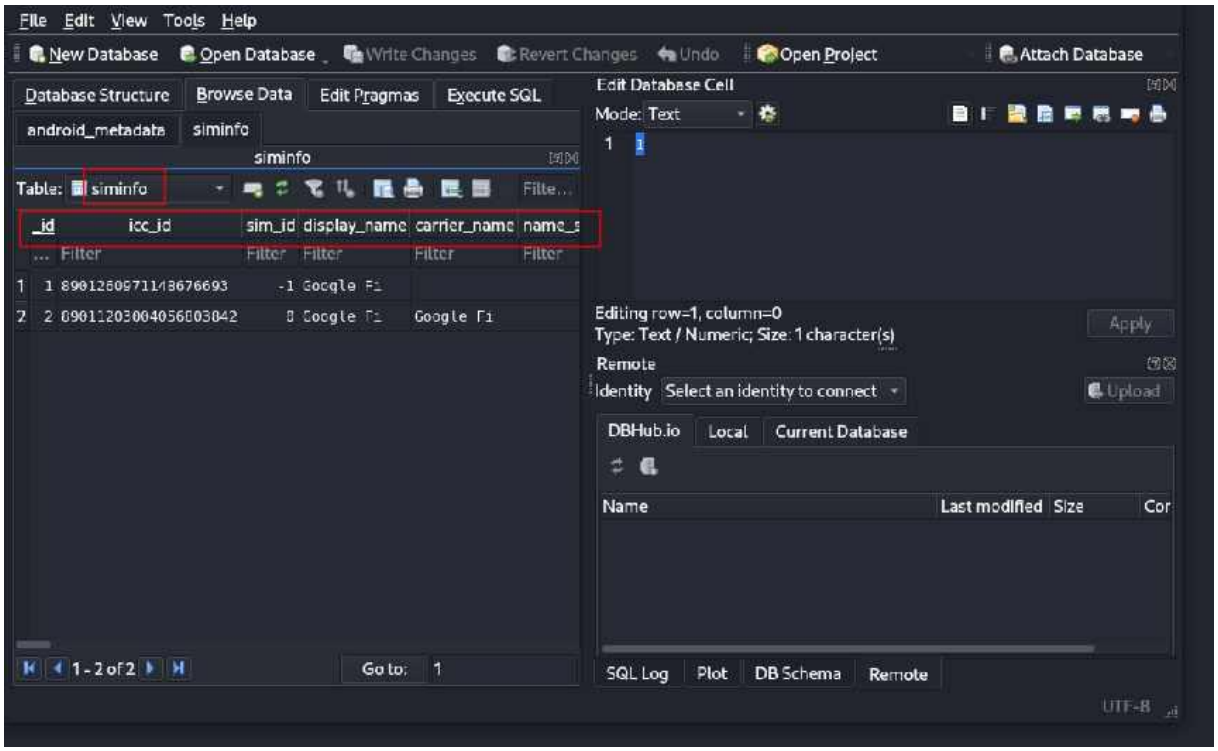
3 directories, 10 files

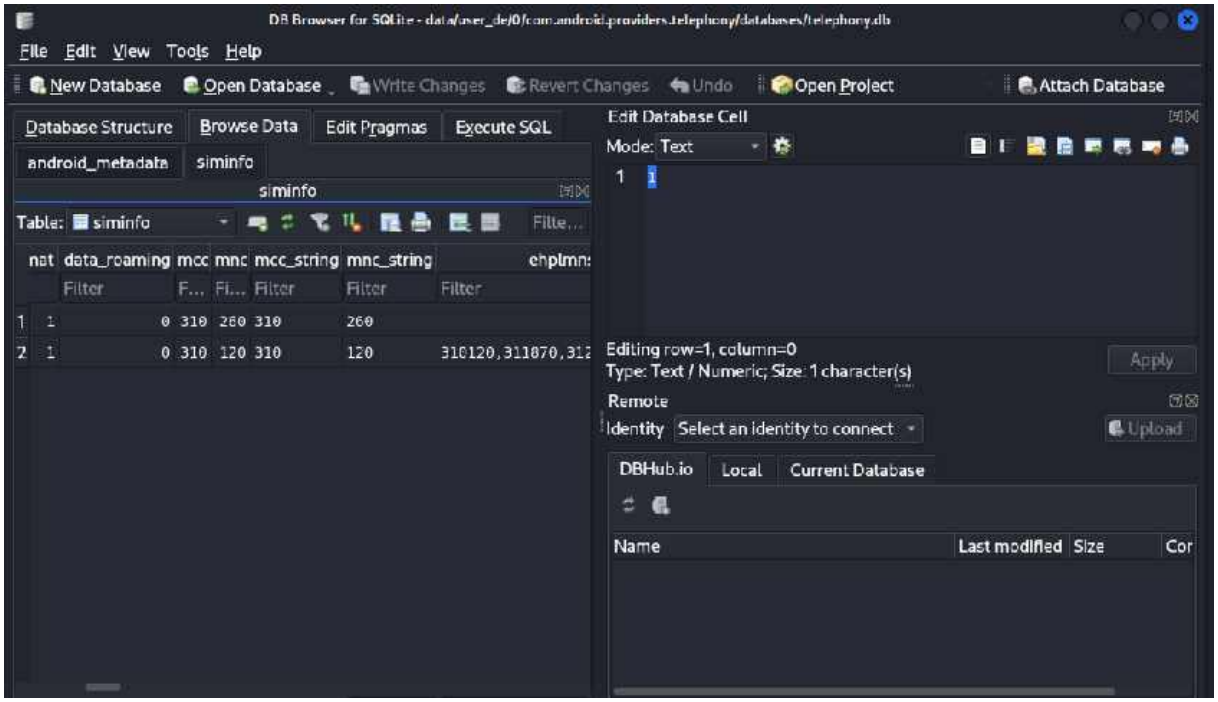
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

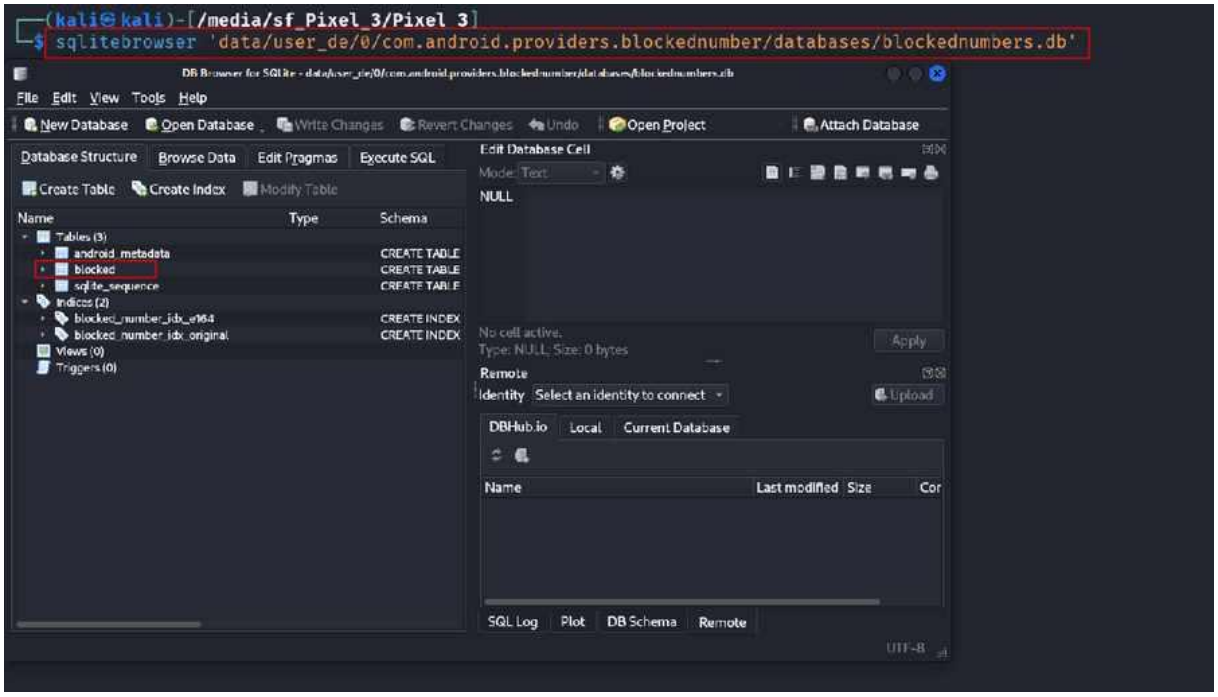
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/user_de/0/com.android.providers.telephony/databases/telephony.db'
```

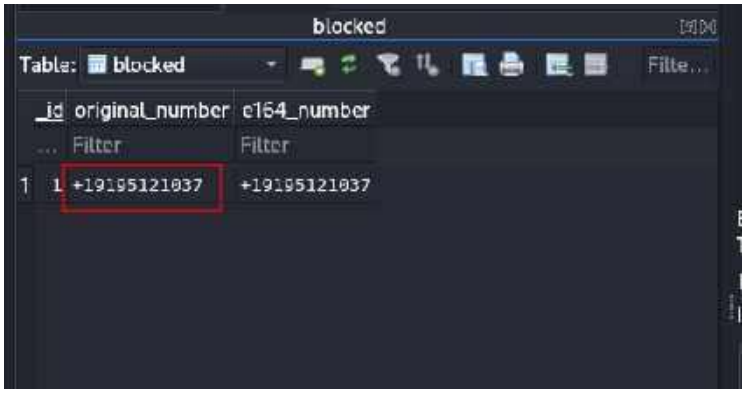



File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Attach Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
Create Table
Create Index
Modify Table
Name
Type
Schema
Tables (4)
android_metadata
CREATE TABLE
Browse Table
Modify Table
Delete Table
Copy Create statement
Export as CSV file
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size


File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Attach Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
android_metadata
siminfo
Table: siminfo
_id
icc_id
sim_id
display_name
carrier_name
Edit Database Cell
Mode: Text
Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
1 - 2 of 2
Go to:
SQL Log
Plot
DB Schema
Remote
UTF-8


DB Browser for SQLite - data/user_de/0/com.android.providers.telephony/databases/telephony.db
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Attach Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
android_metadata
siminfo
siminfo
Table: siminfo
nat data_roaming mcc mnc mcc_string mnc_string ehplmn
1 1 0 310 260 310 260
2 1 0 310 120 310 120 310120,311870,312
Edit Database Cell
Mode: Text
1 1
Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
Cor


(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$ sqlitebrowser 'data/user_de/0/com.android.providers.blockednumber/databases/blockednumbers.db'
DB Browser for SQLite - data/user_de/0/com.android.providers.blockednumber/databases/blockednumbers.db
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Attach Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
Create Table
Create Index
Modify Table
Name
Type
Schema
Tables (3)
android_metadata
blocked
sqlite_sequence
Indices (2)
blocked_number_idx_e164
blocked_number_idx_original
Views (0)
Triggers (0)
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
Cor
SQL Log
Plot
DB Schema
Remote
UTF-8




```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/user_de/0/com.android.phone/'

data/user_de/0/com.android.phone/
├── files
│   ├── carrierconfig-com.google.android.apps.tycho-890112030040568038 42-10015.xml
│   ├── carrierconfig-com.google.android.carrier-890112030040568038 42-10015.xml
│   └── carrierconfig-com.google.android.carrier-890126097114867669 3-10014.xml
└── shared_prefs
    ├── com.android.phone_preferences.xml
    └── _has_set_default_values.xml

3 directories, 5 files

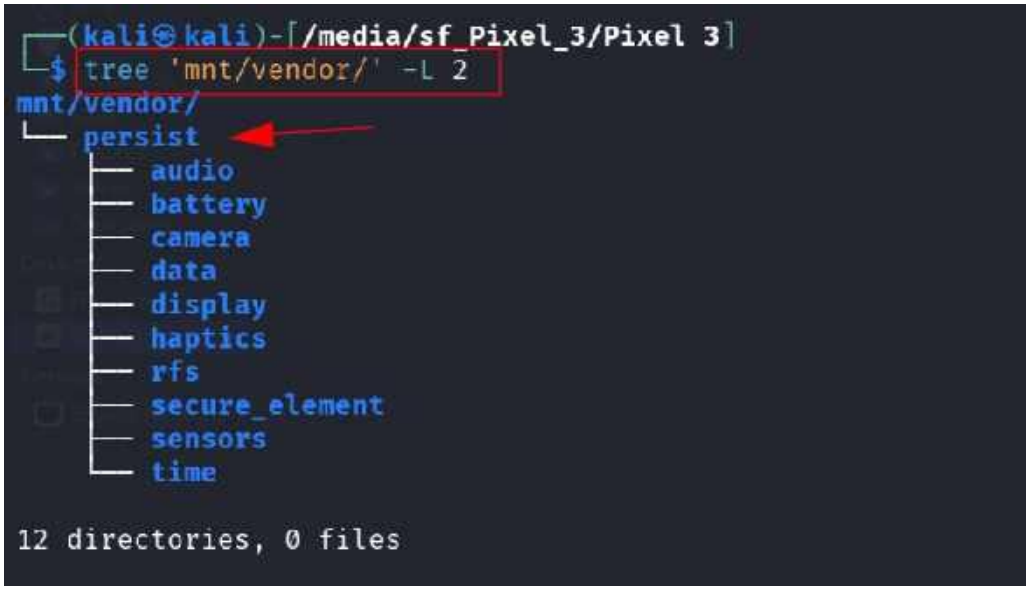
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/user_de/0/com.android.phone/files/carrierconfig-com.google.android.carrier-890112030040568038 42-10015.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<carrier_config>
<package_version>8</package_version>
<bundle_data>
<string name="gps.lpp_profile">2</string>
<string-array name="enable_apps_string_array" num="0" />
<boolean name="carrier_app_required_during_setup_bool" value="true" />
<boolean name="config_telephony_use_own_number_for_voicemail_bool" value="true" />
<string name="carrier_config_version_string">fi_us-12000000333.16</string>
<string name="carrier_name_string">Google Fi</string>
<boolean name="carrier_name_override_bool" value="true" />
</bundle_data>
</carrier_config>

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/user_de/0/com.android.phone/files/carrierconfig-com.google.android.carrier-890126097114867669 3-10014.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<carrier_config>
<package_version>8</package_version>
<bundle_data>
<string name="gps.lpp_profile">2</string>
<string-array name="enable_apps_string_array" num="0" />
<boolean name="carrier_app_required_during_setup_bool" value="true" />
<boolean name="config_telephony_use_own_number_for_voicemail_bool" value="true" />
<string name="carrier_config_version_string">fi_us-12000000333.16</string>
<string name="carrier_name_string">Google Fi</string>
<boolean name="carrier_name_override_bool" value="true" />
</bundle_data>
</carrier_config>
```


(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$ tree 'mnt/vendor/' -L 2
mnt/vendor/
└── persist
    ├── audio
    ├── battery
    ├── camera
    ├── data
    ├── display
    ├── haptics
    ├── rfs
    ├── secure_element
    ├── sensors
    └── time
12 directories, 0 files


(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$ grep -ri 'serial number' 'mnt/vendor/'
mnt/vendor/persist/sensors/tmd2725_als_factory.txt:Serial Number: 8CEX1N716
(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$ grep -ri 'EX1N715' 'mnt/vendor/'
mnt/vendor/persist/camera/front_stereo.cal: "id": "8CEX1N716",
mnt/vendor/persist/camera/front_stereo.cal: "debug_path": "/home/innorev/camera_clifford/destination/output/debug/debug-2018-12-14/8CEX1N716-15447
79510.32"
mnt/vendor/persist/camera/front_stereo.cal: "/home/innorev/camera_clifford/destination/8CEX1N716",
mnt/vendor/persist/camera/front_stereo.cal: "8CEX1N716",
mnt/vendor/persist/sensors/edgesense_roller_cal/4_results/measurements.csv:SYS_SN,8CEX1N716
mnt/vendor/persist/sensors/tmd2725_als_factory.txt:Serial Number: 8CEX1N716
(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/system/users'
data/system/users
├── 0
│   ├── app_idle_stats.xml
│   ├── appwidgets.xml
│   ├── package-restrictions.xml
│   ├── photo.png
│   ├── registered_services
│   │   ├── android.accounts.AccountAuthenticator.xml
│   │   └── android.content.SyncAdapter.xml
│   ├── roles.xml
│   ├── runtime-permissions.xml
│   ├── settings_config.xml
│   ├── settings_global.xml
│   ├── settings_secure.xml
│   ├── settings_ssaid.xml
│   ├── settings_system.xml
│   ├── wallpaper
│   ├── wallpaper_info.xml
│   └── wallpaper_orig
├── 0.xml
├── 11
│   ├── app_idle_stats.xml
│   ├── appwidgets.xml
│   ├── device_policies.xml
│   ├── package-restrictions.xml
│   ├── photo.png
│   ├── registered_services
│   │   ├── android.accounts.AccountAuthenticator.xml
│   │   └── android.content.SyncAdapter.xml
│   ├── roles.xml
│   ├── runtime-permissions.xml
│   ├── settings_secure.xml
│   ├── settings_ssaid.xml
│   ├── settings_system.xml
│   ├── wallpaper
│   ├── wallpaper_info.xml
│   └── wallpaper_orig
├── 11.xml
└── userlist.xml
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/system/users/userlist.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users nextSerialNumber="12" version="7">
    <guestRestrictions>
        <restrictions no_sms="true" no_install_unknown_sources="true" no_config_wifi="true" no_outgoing_calls="true" />
    </guestRestrictions>
    <deviceOwnerUserId id="-10000" />
    <user id="0" />
    <user id="11" />
</users>

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/system/users/0.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="0" serialNumber="0" flags="19" created="0" lastLoggedIn="1581695517124" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys" icon="/data/system/users/0/photo.png" profileBadge="0">
    <name>This Is</name>
    <restrictions />
</user>

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/system/users/11.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="11" serialNumber="11" flags="16" created="1581644358241" lastLoggedIn="1581684601922" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys" icon="/data/system/users/11/photo.png" profileBadge="0">
    <name>User 2</name>
    <restrictions />
</user>

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d @1581695517 -u
Fri Feb 14 03:51:57 PM UTC 2020

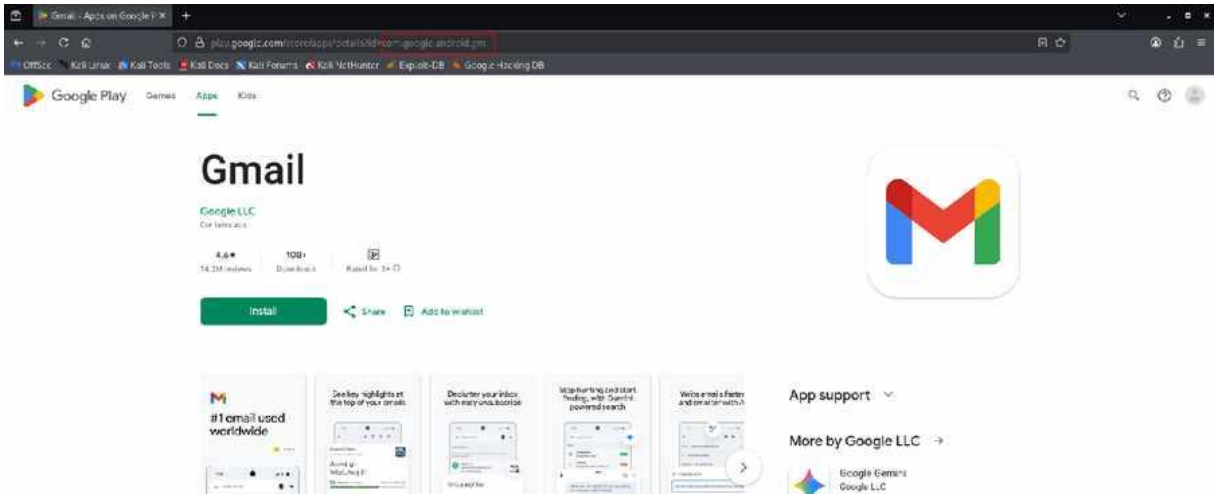
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

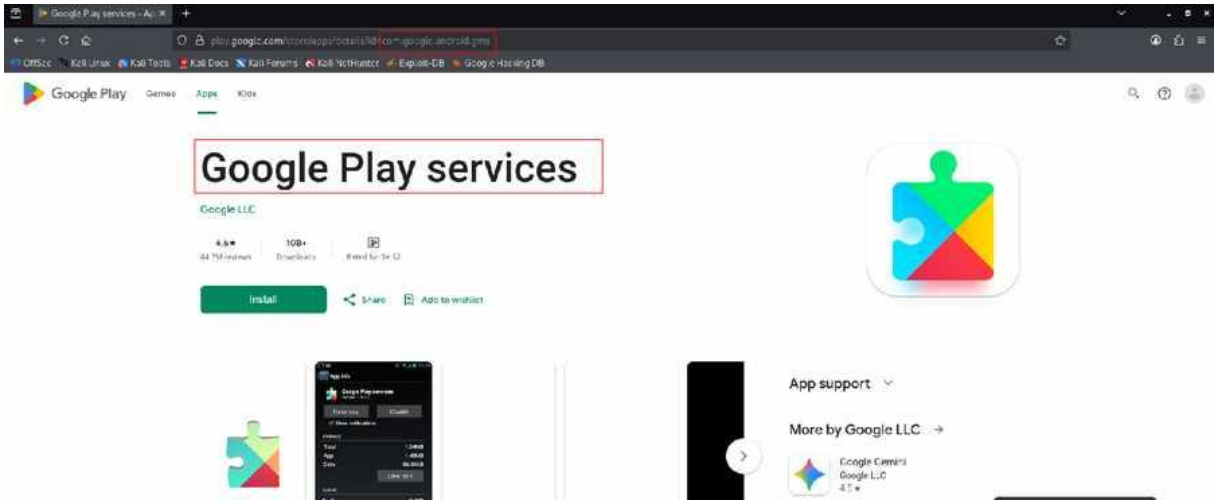
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep -EiorhI '([[:alnum:]_.-]+@[[:alnum:]_.-]+?\.[[:alpha:].]{2,6})' | sort|uniq -c | sort -nr | head -n 20

    608 thisisdfir@gmail.com
    176 extension-devs@eff.org
    120 thisisdfirtwo@gmail.com
     96 19842352054_1oHEUWGxDxo499MnzUJStpXDQ1YoMyeXX4O6Faqv4Acw@prod.tncp.textnow.com
     40 android.hardware.graphics.mapper@2.1.so
     40 android.hardware.graphics.mapper@2.0.so
     40 android.hardware.graphics.common@1.1.so
     40 android.hardware.graphics.common@1.0.so
     38 support@enflick.com
     38 android.hardware.renderscript@1.0.so
     38 19842032223@prod.tncp.textnow.com
     32 ff7a7f86a2e8481abacfb5e046d5046a@patrol-sg.snssdk.com
     24 android.hidl.safe_union@1.0.so
     24 android.hidl.memory.token@1.0.so
     24 android.hidl.memory@1.0.so
     24 android.hardware.graphics.mapper@3.0.so
     24 android.hardware.graphics.common@1.2.so
     20 libstagefright_bufferpool@2.0.1.so
     18 android.system.suspend@1.0.so
     18 android.hidl.token@1.0-utils.so
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep -EiorhI '([[:alnum:]_.-]+@[[:alnum:]_.-]+\.[[:alpha:].]{2,6})' . | sort | uniq -c | sort -nr | wc -l
640

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```






```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep -EiorhI '([[:alnum:]_.-]+@gmail\.com)' | sort|uniq -c | sort -nr | head

    608 thisisdfir@gmail.com
    120 thisisdfirtwo@gmail.com
     10 nc_1_mail_thisisdfir@gmail.com
      6 feature_list_thisisdfir@gmail.com
      4 ncg_1_account_thisisdfirtwo@gmail.com
      4 ncg_1_account_thisisdfir@gmail.com
      4 flagsLastLogged-thisisdfir@gmail.com
      2 waa-enabledthisisdfir@gmail.com
      2 user-settings-cachethisisdfir@gmail.com
      2 toc-cookiethisisdfir@gmail.com

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.google.android.gm' -L 1

data/data/com.google.android.gm
├── app_webview
├── cache
├── databases
├── files
├── no_backup
└── shared_prefs

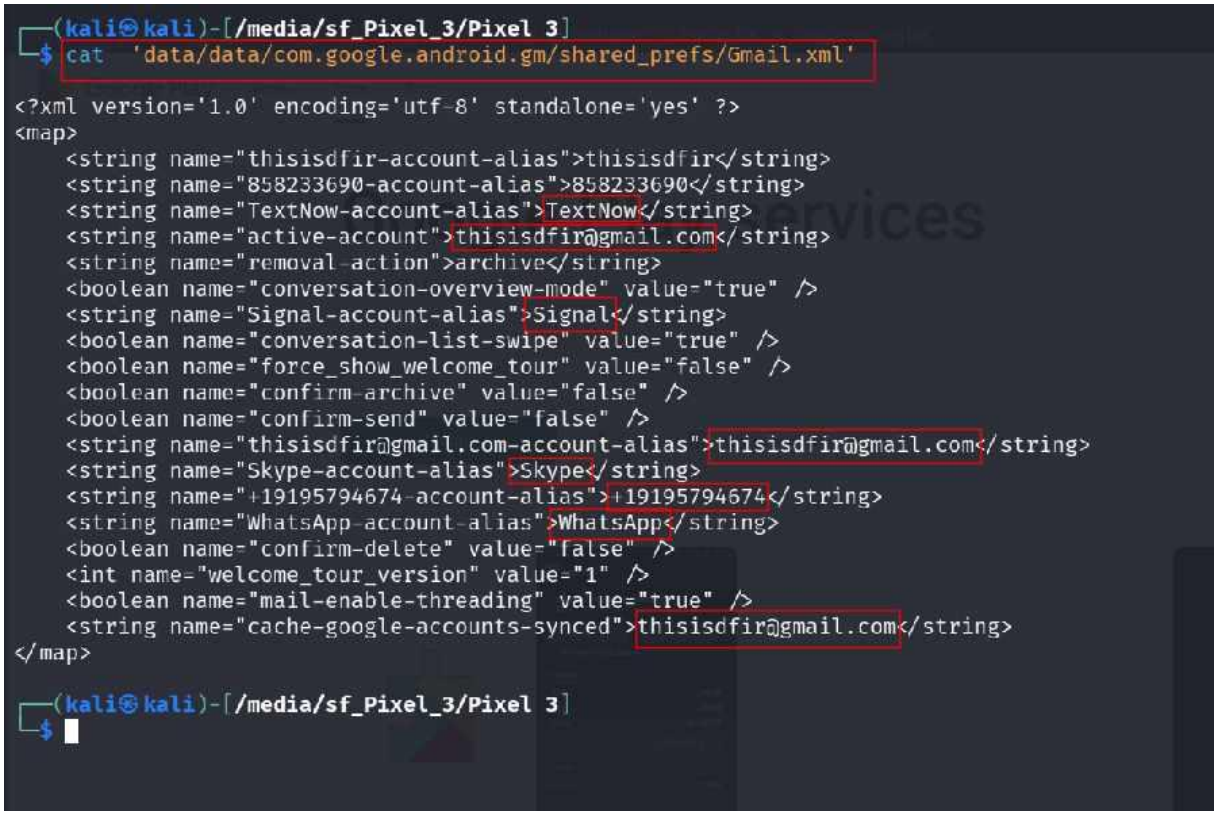
7 directories, 0 files

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.google.android.gm/shared_prefs' -L 1

data/data/com.google.android.gm/shared_prefs
├── ACCOUNT_COUNT_MAP.xml
├── Account-thisisdfir@gmail.com.xml
├── AndroidMail.Main.xml
├── ANIMATABLE_LOGO_VIEW_FILE_KEY.xml
├── attachmentManager_-1294433372.xml
├── c9edm_-1294433372.xml
├── com.google.android.apps.gmail.accountDumpState&#58;thisisdfir@gmail.com&#58;com.google.xml
├── com.google.android.apps.gmail.dumpState.xml
├── com.google.android.apps.gmail.notifications&#58;thisisdfir@gmail.com.xml
├── com.google.android.gm_preferences.xml
├── com.google.android.gms.appid.xml
├── COMPOSE_UPLOADERS.xml
├── FirebaseAppHeartBeat.xml
├── FlagPrefs.xml
├── G6yPrefs.xml
├── GigSyncEngine_thisisdfir@gmail.com.xml
├── GMAIL_NETWORK_LOGGING_DEPOT.xml
├── Gmail.xml
├── _has_set_default_values.xml
├── MailAppProvider.xml
├── phenotype__com.google.android.libraries.social.populous.xml
├── timeZoneDbVersion.xml
├── triaged_notification_receive_times_thisisdfir@gmail.com.xml
├── UnifiedEmail.xml
├── warm_accounts_preferences.xml
├── watermark_notification_tokens_thisisdfir@gmail.com.xml
└── WebViewChromiumPrefs.xml

1 directory, 27 files
```


(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$ cat 'data/data/com.google.android.gm/shared_prefs/Gmail.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
<string name="thisisdfir-account-alias">thisisdfir</string>
<string name="858233690-account-alias">858233690</string>
<string name="TextNow-account-alias">TextNow</string>
<string name="active-account">thisisdfir@gmail.com</string>
<string name="removal-action">archive</string>
<boolean name="conversation-overview-mode" value="true" />
<string name="Signal-account-alias">Signal</string>
<boolean name="conversation-list-swipe" value="true" />
<boolean name="force_show_welcome_tour" value="false" />
<boolean name="confirm-archive" value="false" />
<boolean name="confirm-send" value="false" />
<string name="thisisdfir@gmail.com-account-alias">thisisdfir@gmail.com</string>
<string name="Skype-account-alias">Skype</string>
<string name="+19195794674-account-alias">+19195794674</string>
<string name="WhatsApp-account-alias">WhatsApp</string>
<boolean name="confirm-delete" value="false" />
<int name="welcome_tour_version" value="1" />
<boolean name="mail-enable-threading" value="true" />
<string name="cache-google-accounts-synced">thisisdfir@gmail.com</string>
</map>
(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/data/com.google.android.gm/shared_prefs/UnifiedEmail.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <int name="offline_search_index_corpus_version" value="2" />
    <boolean name="email-accounts-import-tried" value="true" />
    <long name="analytics-send-nb_accounts-epoch" value="1581644062878" />
    <int name="auto-advance-mode" value="0" />
    <int name="offline-search-index-schema-version" value="2" />
    <set name="display_images" />
    <boolean name="conversation-photo-teaser-shown-three" value="true" />
    <boolean name="sc-onboarding-shown-v2" value="true" />
    <boolean name="confirm-archive" value="false" />
    <string name="gm-density">DEFAULT</string>
    <long name="dols_animation_timestamp" value="1581643980962" />
    <long name="owners_name_cache_updated_timestamp" value="1581643980626" />
    <string name="email_address_prefix:thisisdfir@gmail.com">This Is DFIR</string>
    <string name="swipe-left-action">archive</string>
    <boolean name="dark-theme-active" value="true" />
    <int name="snap_header_mode" value="0" />
    <string name="default-reply-action">unset</string>
    <string name="conversation_source">show-conversation-from-previous-conversation</string>
    <boolean name="non-release-build" value="true" />
    <string name="custom-swipe-actions-onboarding-card-state">pending</string>
    <boolean name="conversation_overview_mode" value="true" />
    <set name="display_sender_images_patterns_set" />
    <boolean name="confirm-send" value="false" />
    <boolean name="conversation-list-sender-image" value="true" />
    <string name="swipe-right-action">archive</string>
    <boolean name="confirm_delete" value="false" />
    <int name="migrated-version" value="4" />
    <int name="prefs-version-number" value="4" />
</map>

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.android.vending/cache/logs/'

data/data/com.android.vending/cache/logs/
├── com.google.thisisdfir%40gmail.com.metalog
│   └── play_metalog.log
└── null_account.metalog
    ├── logs_upload_attempt.log
    └── play_metalog.log

3 directories, 3 files

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ xxd 'data/data/com.android.vending/cache/logs/com.google.thisisdfir%40gmail.com.metalog/play_metalog.log'
00000000: 08f4 cede b08a 2e60 da46 6831           ......`.Fh

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

Convert epoch to human-readable date and vice versa

1581771242040 Timestamp to Human date [batch convert]

Supports Unix timestamps in seconds, milliseconds, microseconds and nanoseconds.

Assuming that this timestamp is in **milliseconds**:

GMT : Saturday, February 15, 2020 12:54:02.040 PM

Your time zone : Saturday, February 15, 2020 7:54:02.040 AM GMT-05:00

Relative : 6 years ago

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep -Eio '([[:alnum:]_.-]+@[[:alnum:]_.-]+\.[[:alpha:]]{2,6})'   'data/system/job/jobs.xml' | sort | uniq -c
     43 thisisdfir@gmail.com
     36 thisisdfirtwo@gmail.com

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree -L 1  'data/system_ce/0/'
data/system_ce/0/
├── accounts_ce.db
├── launch_params
├── recent_images
├── recent_tasks
├── shortcut_service
└── snapshots

5 directories, 1 file

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlite3 data/system_ce/0/accounts_ce.db
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> SELECT * FROM accounts;
1|thisisdfir@gmail.com|com.google|aas_et/AKppINYHk3843SOlirhnRUtCveD5dMEkDbJE-UM6ATDHh15WhYU1lcZNba5nhbrKUmsSReDEDBsCeGL7JPU4q-Yg4xnvw0l3EIJFTafNRpTQp9VWTc60f96ZOmBpTtd9rcFCO31RD2qnXx2XDpWJ7u0jnSIu78gwiCGPaUpaegazIAQxH1Pa-1h0VwGXElenMabM5yUWpAIhzpatx3xMFoA=
2|858233690|org.telegram.messenger|
3|Signal|org.thoughtcrime.securesms|
4|imo HD|com.imo.android.imous|
5|TikTok|com.zhiliaoapp.musically|
6|Messenger|com.facebook.messenger|
7|TDfir|com.twitter.android.auth.login|
8|WhatsApp|com.whatsapp|
10|thisisdfir|com.silentcircle.account|dummyPassword
11|Skype|com.skype.raider|
12|TextNow|com.enflick.android.TextNow.account|
13|+19195794674|com.viber.voip|961ea18e43b4a2b35216e208d9414212e4b699c3
14|Duo|com.google.android.apps.tachyon|
sqlite>
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree -L 1  'data/system_de/0/'

data/system_de/0/
├── accounts_de.db
├── persisted_taskIds.txt
├── ringtones
└── spblob

3 directories, 2 files

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlite3 'data/system_de/0/accounts_de.db'

SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> select name, type, datetime(last_password_entry_time_millis_epoch/1000, 'UNIXEPOCH') from accounts;
thisisdfir@gmail.com|com.google|2020-01-29 15:41:07
858233690|org.telegram.messenger|2020-01-29 18:31:04
Signal|org.thoughtcrime.securesms|2020-01-29 18:31:40
imo HD|com.imo.android.imous|2020-01-29 18:32:00
TikTok|com.zhiliaoapp.musically|2020-01-29 18:43:16
Messenger|com.facebook.messenger|2020-01-29 18:51:02
TDfir|com.twitter.android.auth.login|2020-01-29 19:42:51
WhatsApp|com.whatsapp|2020-01-29 21:10:02
thisisdfir|com.silentcircle.account|2020-02-01 20:41:31
Skype|com.skype.raider|2020-02-01 21:08:20
TextNow|com.enflick.android.TextNow.account|2020-02-08 19:54:52
+19195794674|com.viber.voip|2020-02-08 20:15:38
Duo|com.google.android.apps.tachyon|2020-02-14 01:19:15
sqlite>
```

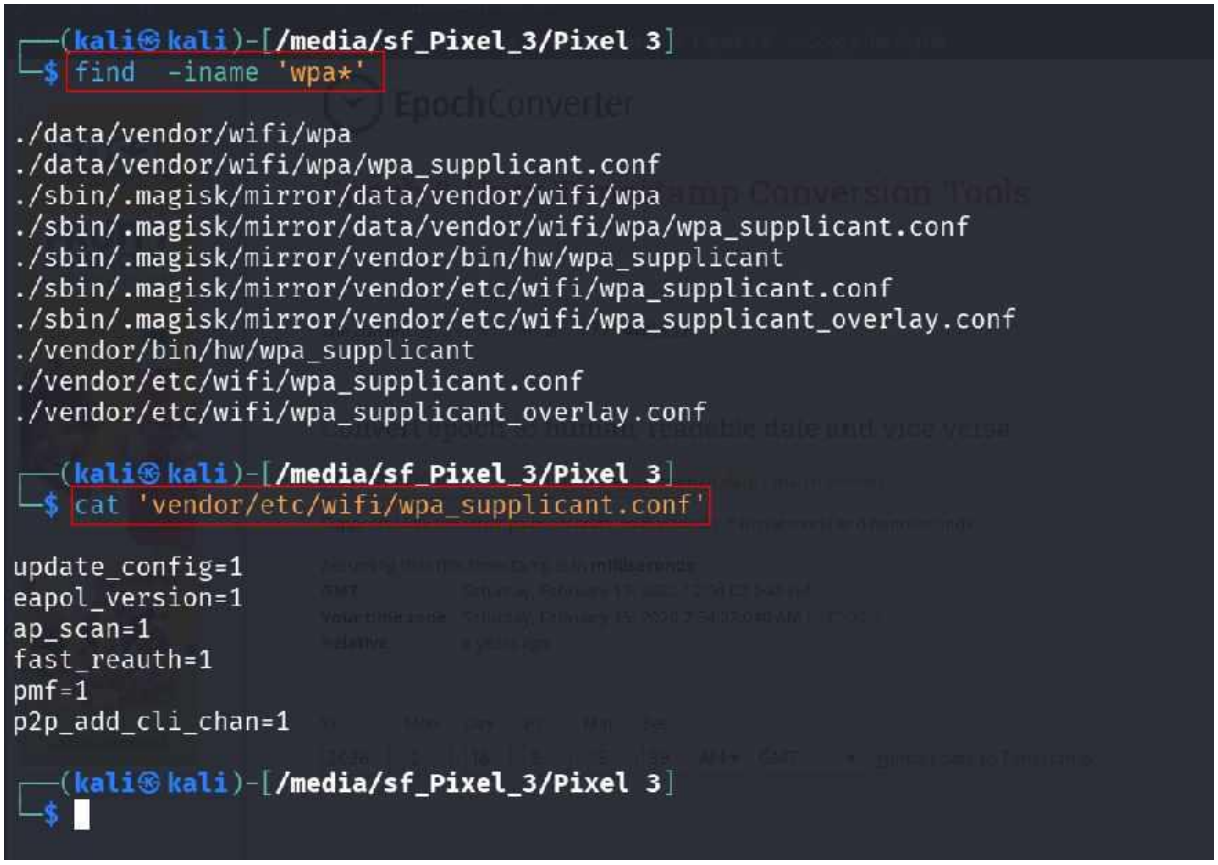
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep -i 'bluetooth' 'data/system/users/0/settings_secure.xml'

  <setting id="65" name="bluetooth_name" value="Pixel 3" package="android" defaultValue="Pixel 3" defaultSysSet="true" />
  <setting id="71" name="bluetooth_address" value="7C:D9:5C:AC:A2:CE" package="android" defaultValue="7C:D9:5C:AC:A2:CE" defaultSysSet=
"true" />

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat data/misc/wifi/WifiConfigStore.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<WifiConfigStoreData>
<int name="Version" value="2" />
<Integrity>
<byte-array name="EncryptedData" num="48">e71394aa95f2d767a1c46e787ca99c0478e1246724153a8551005969fd38294e8a4082529d69a15d5bc49f28ca21
a54</byte-array>
<byte-array name="IV" num="12">18234dad790da5cb3ba8eceb</byte-array>
</Integrity>
<NetworkList />
<MacAddressMap>
<map name="MacMapEntry">
<string name="&quot;CcookiesDcastleR5&quot;WPA_PSK">ae:f6:c7:a8:a5:c2</string>
</map>
</MacAddressMap>
<PasspointConfigData>
<long name="ProviderIndex" value="0" />
</PasspointConfigData>
</WifiConfigStoreData>

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```



System Settings

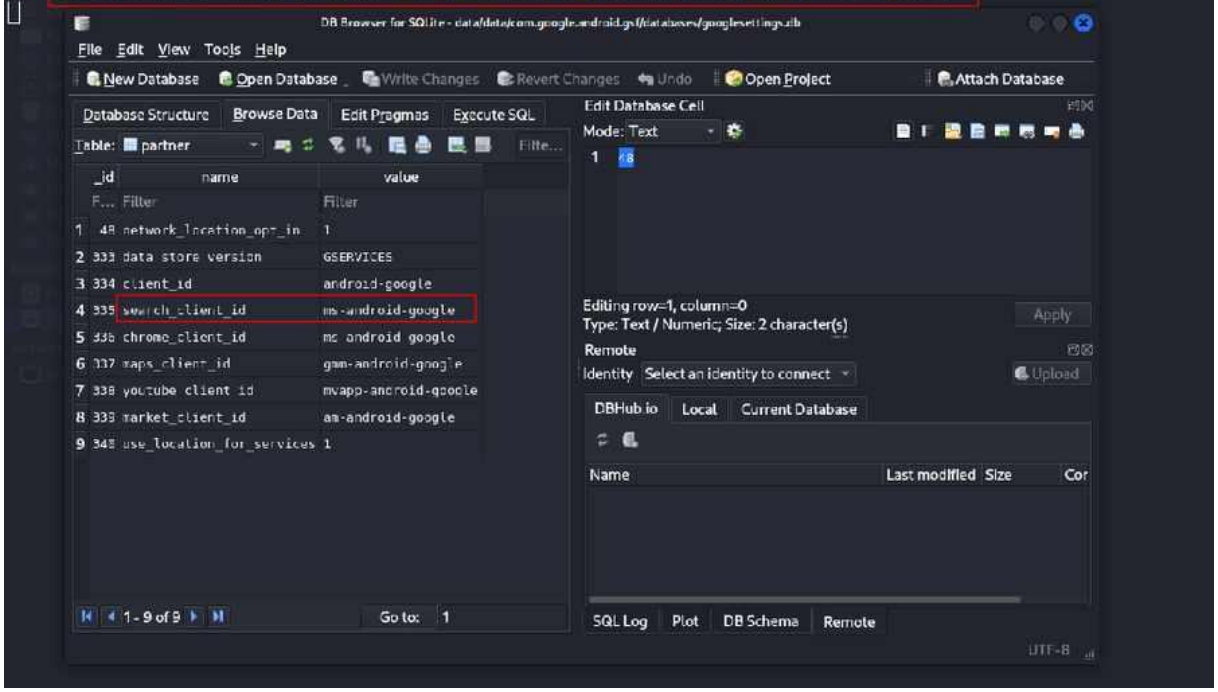
I reviewed googlesettings.db (partner table), gservices.db (main table, 1,914 rows) and the GMS shared_prefs. The Maps API preference file holds LegalCountry = US and Cohort = 887. These are configuration values and say nothing about user activity on their own.

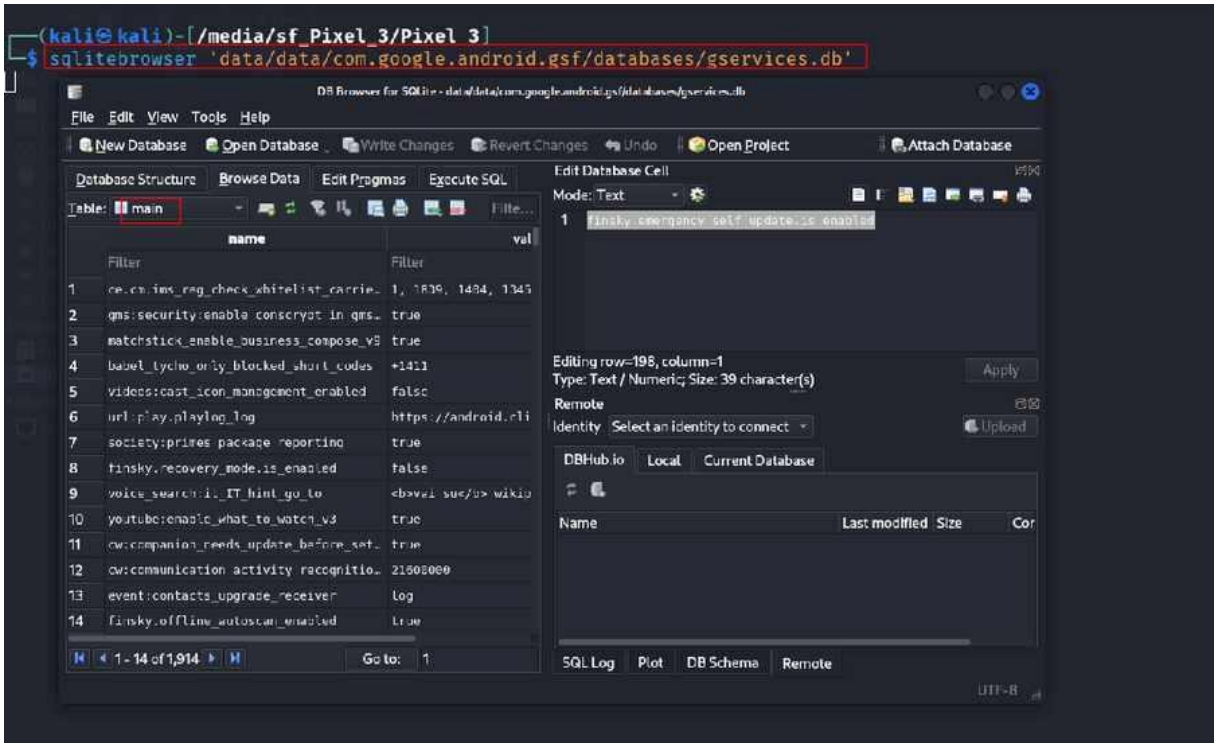
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.google.android.gsf/'
data/data/com.google.android.gsf/
├── databases
│   ├── googlesettings.db
│   ├── gservices.db
│   └── subscribedfeeds.db
├── files
│   ├── security_token
│   └── signatures
│       ├── sig1142668760
│       ├── sig2101071397
│       ├── sig-319209697
│       ├── sig-473270056
│       └── sig828861265
└── shared_prefs
    └── CheckinService.xml

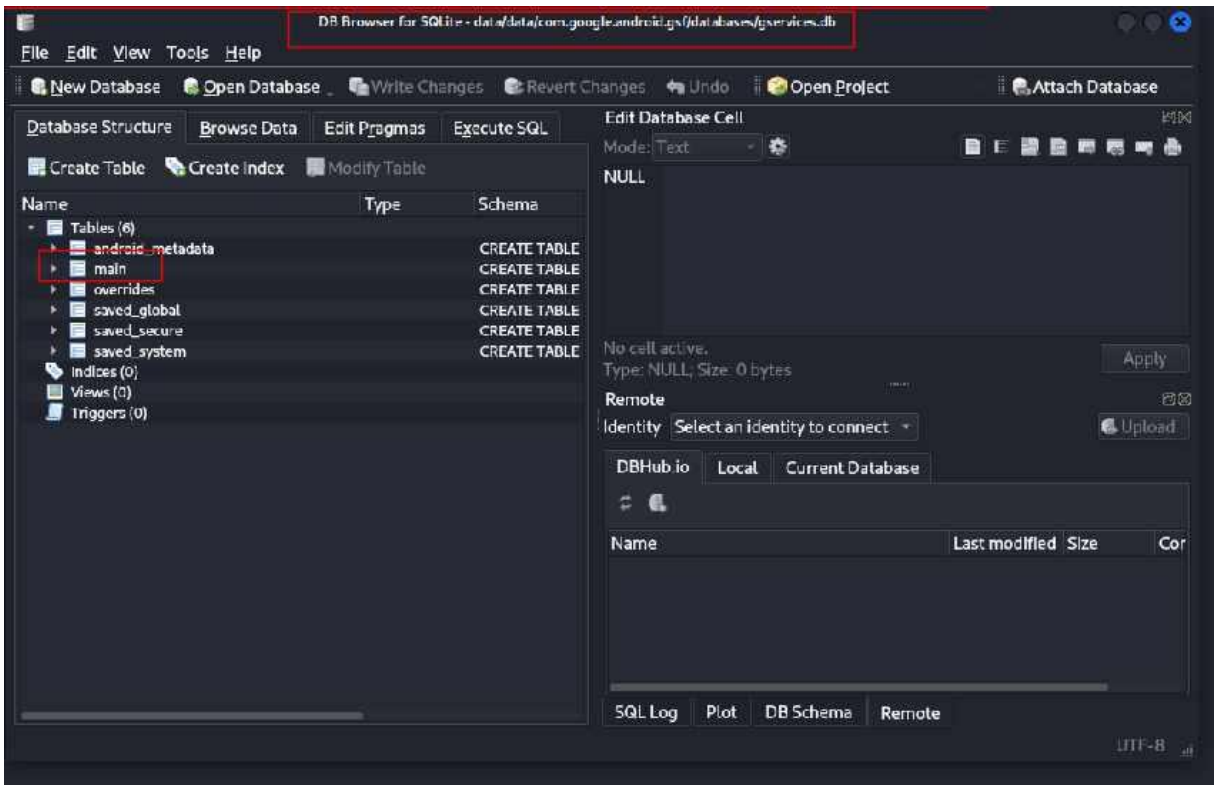
5 directories, 10 files

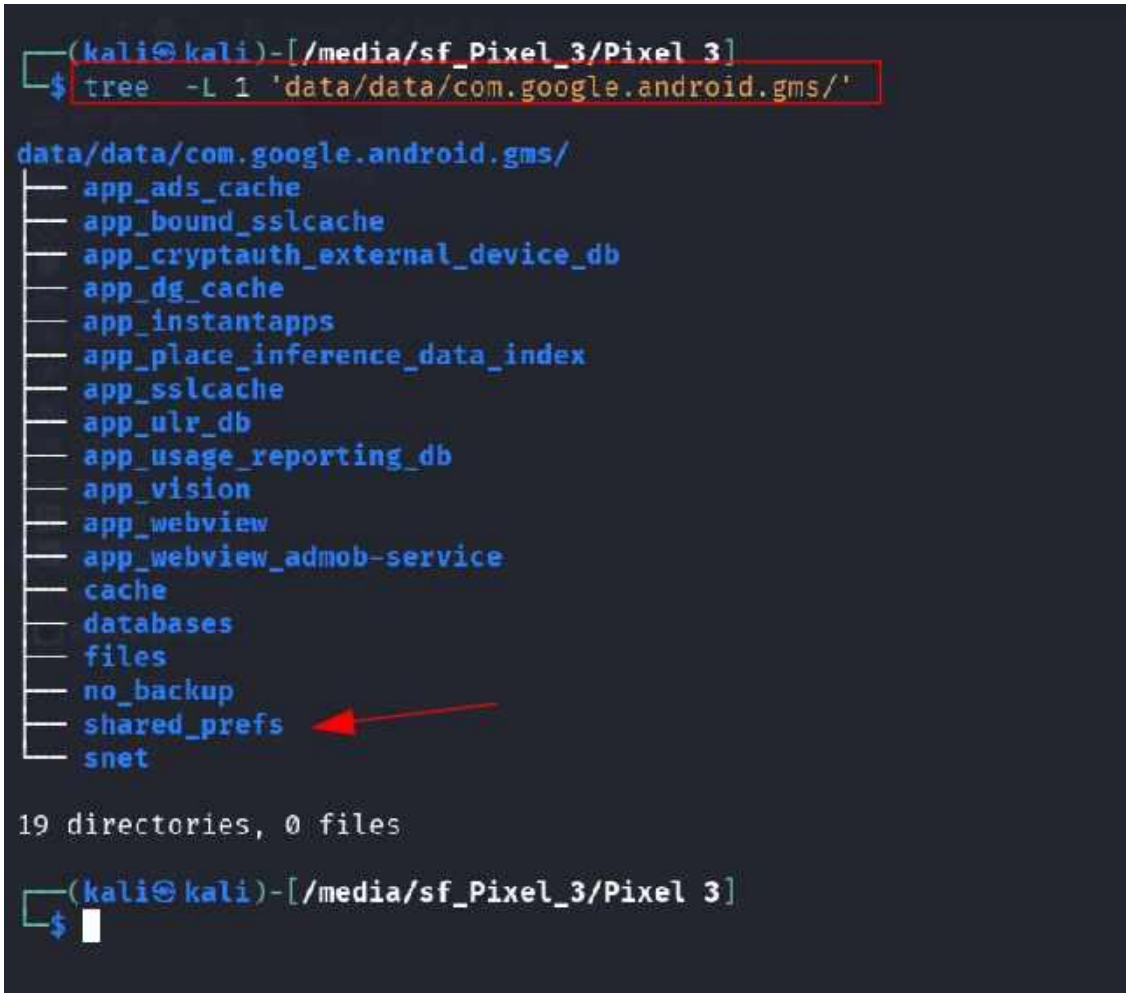
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.google.android.gsf/databases/googlesettings.db'
```




(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$ sqlitebrowser 'data/data/com.google.android.gsf/databases/gservices.db'
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Attach Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
Table: main
name
val
Edit Database Cell
Mode: Text
Editing row=198, column=1
Type: Text / Numeric; Size: 39 character(s)
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
SQL Log
Plot
DB Schema
Remote
UTF-8


File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Attach Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
Create Table
Create Index
Modify Table
Name
Type
Schema
Tables (6)
android_metadata
main
overrides
saved_global
saved_secure
saved_system
CREATE TABLE
Indices (0)
Views (0)
Triggers (0)
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
SQL Log
Plot
DB Schema
Remote
UTF-8


┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree -L 1 'data/data/com.google.android.gms/'
data/data/com.google.android.gms/
├── app_ads_cache
├── app_bound_sslcache
├── app_cryptauth_external_device_db
├── app_dg_cache
├── app_instantapps
├── app_place_inference_data_index
├── app_sslcache
├── app_ulr_db
├── app_usage_reporting_db
├── app_vision
├── app_webview
├── app_webview_admob-service
├── cache
├── databases
├── files
├── no_backup
├── shared_prefs
└── snet
19 directories, 0 files
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls 'data/data/com.google.android.gms/shared_prefs'

ActionsUploadSharedPrefs.xml
activityRecognitionAccuracyModeName.xml
activity.transitionStateSharedPreferences.xml
adid_settings.xml
admob_user_agent.xml
AppDataSearch-main-config.xml
AppDataSearch-main-icing-settings.xml
AppsIndexerSharedPrefs.xml
AppUsageEventWatcher.xml
auth.account.visibility.whitelist.prefs.xml
auth_cron_chimera_service_storage.xml
BackupDeviceState.xml
Batterystats.xml
BEACON_STATE.xml
bluetooth_addresses_prefs.xml
CarservicePH.xml
carservice.xml
CAST_ACTIVE_NETWORK_MAP.xml
CheckinLogging.xml
Checkin.xml
coffee_preferences.xml
com.google.android.apps.walletnfcrel.AppInviteReferral.xml
com.google.android.gcm.xml
com.google.android.gms.accountsettings_preferences.xml
com.google.android.gms.analytics.prefs.xml
com.google.android.gms.appid.xml
com.google.android.gms.auth.accounts.addaccount.setupwizard.SetupWizardPreferences.xml
com.google.android.gms.auth.api.credentials.save_confirmation_dialog_manager.xml
com.google.android.gms.auth.authzen.cryptauth.DeviceStateSyncManager.xml
com.google.android.gms.auth.authzen.cryptauth.SyncManager.proximity_features.xml
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/data/com.google.android.gms/shared_prefs/com.google.maps.api.android.lib6.drd.PREFERENCES_FILE.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <string name="Cohort">887</string>
    <string name="LegalCountry">US</string>
</map>

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/data/com.google.android.gms/shared_prefs/Fingerprintstats.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <boolean name=":requiresCharging" value="true" />
    <long name=":lastRunMsec" value="1581560776867" />
    <long name=":recordIntervalSecs" value="-1" />
</map>

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d @1581560776 -u
Thu Feb 13 02:26:16 AM UTC 2020

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/data/com.google.android.gms/shared_prefs/Launcherstats.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <boolean name=":requiresCharging" value="true" />
    <long name=":lastRunMsec" value="1581560777469" />
    <long name=":recordIntervalSecs" value="-1" />
</map>

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d @1581560777 -u
Thu Feb 13 02:26:17 AM UTC 2020

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls 'data/system'

appops.xml                  display-manager-state.xml  log-files.xml           package-dcl.list        recoverablekeystore.db  uiderrors.txt
battery-history             dropbox                    netpolicy.xml           package-dex-usage.list  screen_on_time          usagestats
batterystats.bin            entropy.dat                netstats                packages.list           sensor_service          users
batterystats-checkin.bin    graphicsstats              notification_log.db     packages-warnings.xml   shortcut_service.xml    watchlist_report.db
batterystats-daily.xml      install_sessions.xml       notification_policy.xml packages.xml           slice                   watchlist_settings.xml
cachequota.xml              job                        overlays.xml            package-usage.list      sound_model.db
device_policies.xml         last-header.txt            package_cache           package-watchdog.xml    st_sound_model.db
diskstats_cache.json        locksettings.db            package-cstats.list     procstats               sync

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/system/locksettings.db'
```

DB Browser for SQLite - data/system/locksettings.db

File Edit View Tools Help

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Attach Database

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: locksettings

	_id	name	user
	Filter	Filter	Filter
1	1	lockscreen.disabled	0
2	2	migrated	0
3	3	migrated_user_specific	0
4	4	lockscreen.password_type_alternate	0
5	5	migrated_biometric_weak	0
6	6	migrated_lockscreen_disabled	0
7	7	migrated_frp	0
8	8	lockscreen.enabledtrustagents	0
9	20	lockscreen.password_type	0
10	22	sp_handle	0
11	23	lockscreen.profilechallenge	0
12	24	lockscreen.passwordhistory	0
13	125	lockscreen.istrustusuallymanaged	0
14	148	lockscreen.enabledtrustagents	11

1 - 14 of 19 | Go to: 1

Edit Database Cell

Mode: Text

1

Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)

Apply

Remote

Identity: Select an identity to connect

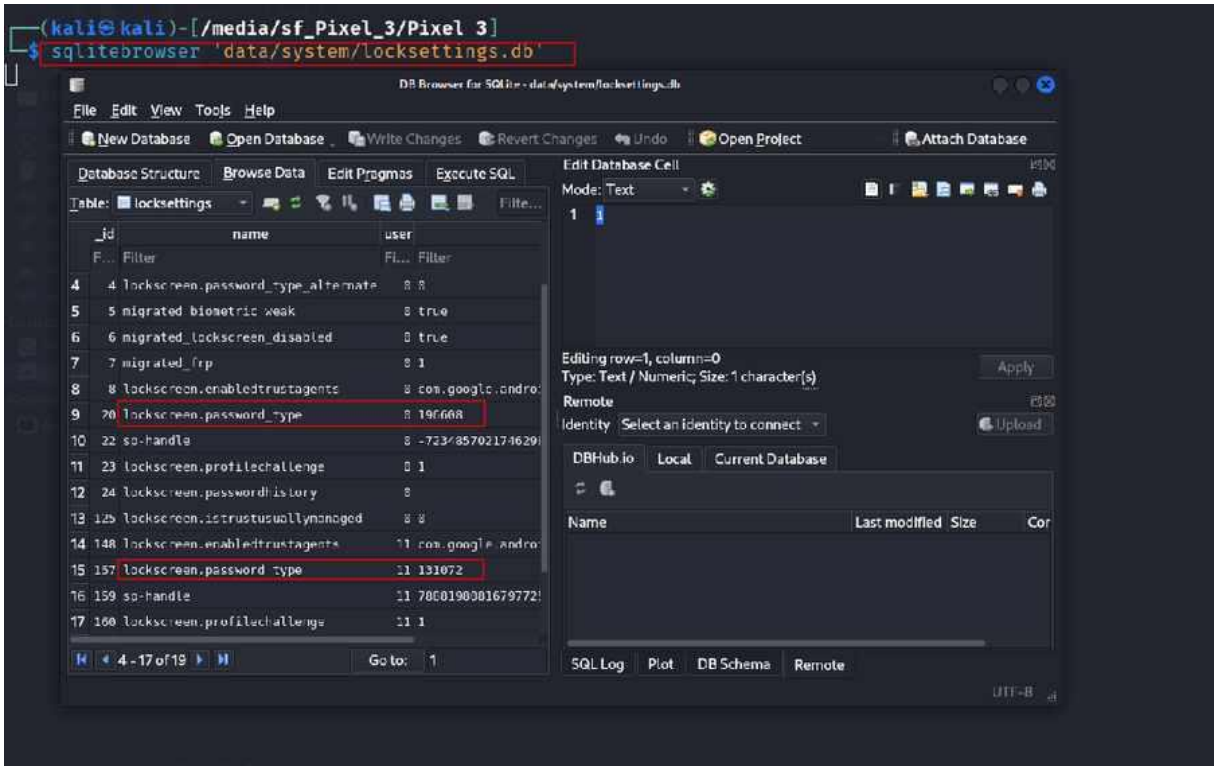
Upload

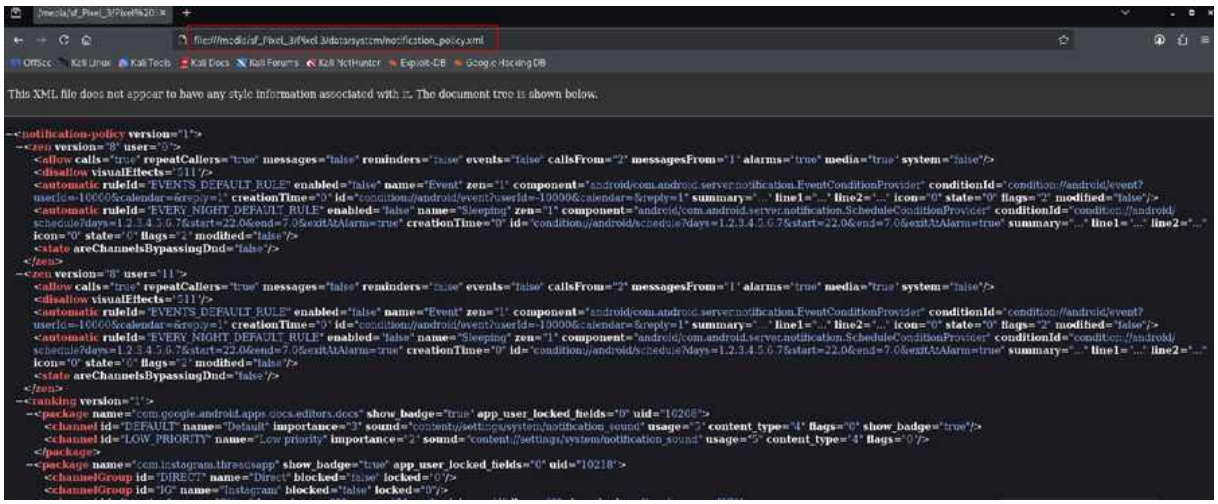
DBHub.io | Local | Current Database

Name | Last modified | Size | Cor

SQL Log | Plot | DB Schema | Remote

UTF-8


(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$ sqlitebrowser 'data/system/locksettings.db'
DB Browser for SQLite - data/system/locksettings.db
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Attach Database
Database Structure Browse Data Edit Pragmas Execute SQL
Table: locksettings
_id name user
4 4 lockscreen.password_type_alternate 0 0
5 5 migrated_biometric_weak 0 true
6 6 migrated_lockscreen_disabled 0 true
7 7 migrated_frp 0 1
8 8 lockscreen.enabledtrustagents 0 com.google.andro
9 20 lockscreen.password_type 0 196608
10 22 sp-handle 0 -723485702174629
11 23 lockscreen.profilechallenge 0 1
12 24 lockscreen.passwordhistory 0
13 125 lockscreen.istrustusuallymanaged 0 0
14 148 lockscreen.enabledtrustagents 11 com.google.andro
15 157 lockscreen.password_type 11 131072
16 159 sp-handle 11 70E0190301679772
17 160 lockscreen.profilechallenge 11 1
4 - 17 of 19
Go to: 1
Edit Database Cell
Mode: Text
1 1
Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)
Apply
Remote
Identity Select an identity to connect
Upload
DBHub.io Local Current Database
Name Last modified Size Cor
SQL Log Plot DB Schema Remote
UTF-8


This XML file does not appear to have any style information associated with it. The document tree is shown below.
<notification-policy version="1">
<zen version="8" user="0">
<allow calls="true" repeatCallers="true" messages="false" reminders="false" events="false" callsFrom="2" messagesFrom="1" alarms="true" media="true" system="false"/>
<disallow visualEffects="511"/>
<automatic ruleId="EVENTS_DEFAULT_RULE" enabled="false" name="Event" zen="1" component="android/com.android.server.notification.EventConditionProvider" conditionId="condition://android/event?userId=-10000&calendar=&reply=1" creationTime="0" id="condition://android/event?userId=-10000&calendar=&reply=1" summary="..." line1="..." line2="..." icon="0" state="0" flags="2" modified="false"/>
<automatic ruleId="EVERY_NIGHT_DEFAULT_RULE" enabled="false" name="Sleeping" zen="1" component="android/com.android.server.notification.ScheduleConditionProvider" conditionId="condition://android/schedule?days=1.2.3.4.5.6.7&start=22.0&end=7.0&exitAtAlarm=true" creationTime="0" id="condition://android/schedule?days=1.2.3.4.5.6.7&start=22.0&end=7.0&exitAtAlarm=true" summary="..." line1="..." line2="..." icon="0" state="0" flags="2" modified="false"/>
<state areChannelsBypassingDnd="false"/>
</zen>
<zen version="8" user="11">
<allow calls="true" repeatCallers="true" messages="false" reminders="false" events="false" callsFrom="2" messagesFrom="1" alarms="true" media="true" system="false"/>
<disallow visualEffects="511"/>
<automatic ruleId="EVENTS_DEFAULT_RULE" enabled="false" name="Event" zen="1" component="android/com.android.server.notification.EventConditionProvider" conditionId="condition://android/event?userId=-10000&calendar=&reply=1" creationTime="0" id="condition://android/event?userId=-10000&calendar=&reply=1" summary="..." line1="..." line2="..." icon="0" state="0" flags="2" modified="false"/>
<automatic ruleId="EVERY_NIGHT_DEFAULT_RULE" enabled="false" name="Sleeping" zen="1" component="android/com.android.server.notification.ScheduleConditionProvider" conditionId="condition://android/schedule?days=1.2.3.4.5.6.7&start=22.0&end=7.0&exitAtAlarm=true" creationTime="0" id="condition://android/schedule?days=1.2.3.4.5.6.7&start=22.0&end=7.0&exitAtAlarm=true" summary="..." line1="..." line2="..." icon="0" state="0" flags="2" modified="false"/>
<state areChannelsBypassingDnd="false"/>
</zen>
<ranking version="1">
<package name="com.google.android.apps.docs.editors.docs" show_badge="true" app_user_locked_fields="0" uid="10268">
<channel id="DEFAULT" name="Default" importance="3" sound="content://settings/system/notification_sound" usage="5" content_type="4" flags="0" show_badge="true"/>
<channel id="LOW_PRIORITY" name="Low priority" importance="2" sound="content://settings/system/notification_sound" usage="5" content_type="4" flags="0"/>
</package>
<package name="com.instagram.threadsapp" show_badge="true" app_user_locked_fields="0" uid="10218">
<channelGroup id="DIRECT" name="Direct" blocked="false" locked="0"/>
<channelGroup id="IG" name="Instagram" blocked="false" locked="0"/>

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep -Pio 'package name=".+?"'  'data/system/notification_policy.xml'

package name="com.google.android.apps.docs.editors.docs"
package name="com.instagram.threadsapp"
package name="com.google.android.gm"
package name="com.zhiliaoapp.musically"
package name="com.google.android.music"
package name="com.google.android.apps.docs"
package name="com.discord"
package name="com.google.android.music"
package name="com.google.android.dialer"
package name="com.google.android.apps.tips"
package name="com.instagram.android"
package name="com.google.android.apps.messaging"
package name="com.google.android.googlequicksearchbox"
package name="com.google.android.apps.youtube.music"
package name="com.silentcircle.silentphone"
package name="com.android.bluetooth"
package name="com.topjohnwu.magisk"
package name="com.radicalapps.cyberdust"
package name="jp.naver.line.android"
package name="com.android.printspooler"
package name="com.android.cellbroadcastreceiver"
package name="com.enflick.android.TextNow"
package name="android"
package name="com.google.android.inputmethod.latin"
package name="com.whatsapp"
package name="com.mywickr.wickr2"
package name="org.torproject.torbrowser"
package name="com.snapchat.android"
package name="com.android.vending"
package name="com.android.server.telecom"
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep -Pio 'package name=".+?"'  'data/system/notification_policy.xml' | wc -l

97

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep gms 'data/system/packages.list'
com.google.android.gms 10141 0 /data/user/0/com.google.android.gms google:privapp:targetSdkVersion=29 2001,1005,3002,3003,3001,3007,1007,3006 0 200414037
com.google.android.gms.location.history 10191 0 /data/user/0/com.google.android.gms.location.history default:targetSdkVersion=28 none 0 1

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat "data/system/netpolicy.xml"
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<policy-list version="11" restrictBackground="false">
<network-policy networkTemplate="1" subscriberId="310260974867669" cycleStart="2019-01-16T00:00-05:00[America/New_York]" cyclePeriod="P1M" warningBytes="-1"
limitBytes="-1" lastWarningSnooze="-1" lastLimitSnooze="-1" metered="true" inferred="true" />
<uid-policy uid="10067" policy="4" />
<uid-policy uid="10111" policy="4" />
<uid-policy uid="10134" policy="4" />
<uid-policy uid="10141" policy="4" />
<uid-policy uid="10151" policy="4" />
<uid-policy uid="1110067" policy="4" />
<uid-policy uid="1110111" policy="4" />
<uid-policy uid="1110134" policy="4" />
<uid-policy uid="1110141" policy="4" />
<uid-policy uid="1110151" policy="4" />
</policy-list>
<whitelist />
```

Index of file:///media/sf_Pixel_3/Pixel 3/data/system/users/0

Up to higher level directory

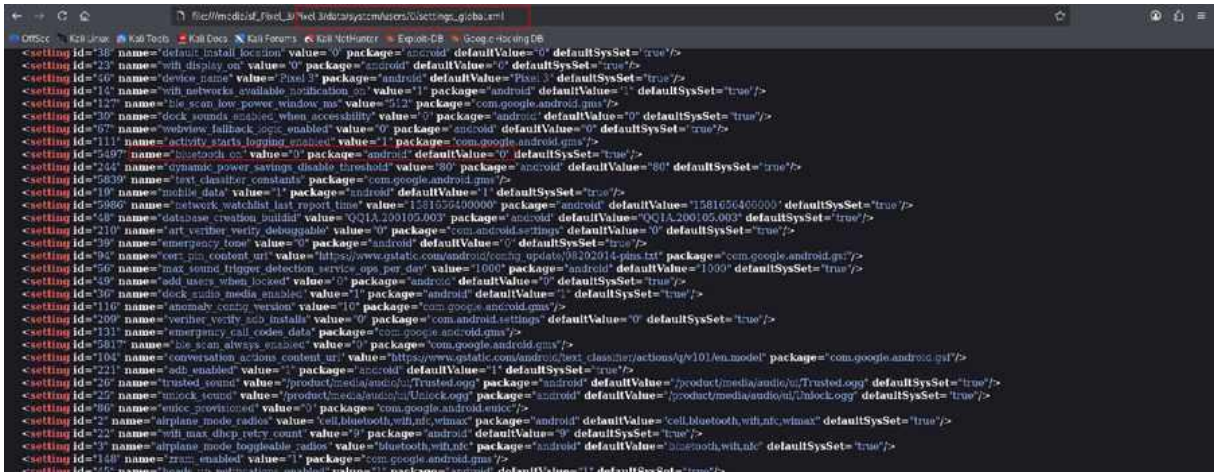
Name	Size	Last Modified
app_idle_stats.xml	51 KB	2/16/26 10:58:00 PM EST
appwidgets.xml	1 KB	2/16/26 10:58:00 PM EST
package_restrictions.xml	116 KB	2/16/26 10:58:00 PM EST
photo.png	5 KB	2/16/26 10:58:00 PM EST
registered_services		2/16/26 10:58:00 PM EST
roles.xml	1 KB	2/16/26 10:58:00 PM EST
runtime-permissions.xml	74 KB	2/16/26 10:58:00 PM EST
settings_config.xml	33 KB	2/16/26 10:58:00 PM EST
settings_global.xml	25 KB	2/16/26 10:58:00 PM EST
settings_secure.xml	22 KB	2/16/26 10:58:00 PM EST
settings_ssaid.xml	6 KB	2/16/26 10:58:00 PM EST
settings_system.xml	6 KB	2/16/26 10:58:00 PM EST
wallpaper	3503 KB	2/16/26 10:58:00 PM EST
wallpaper_info.xml	1 KB	2/16/26 10:58:00 PM EST
wallpaper_orig	3503 KB	2/16/26 10:58:00 PM EST

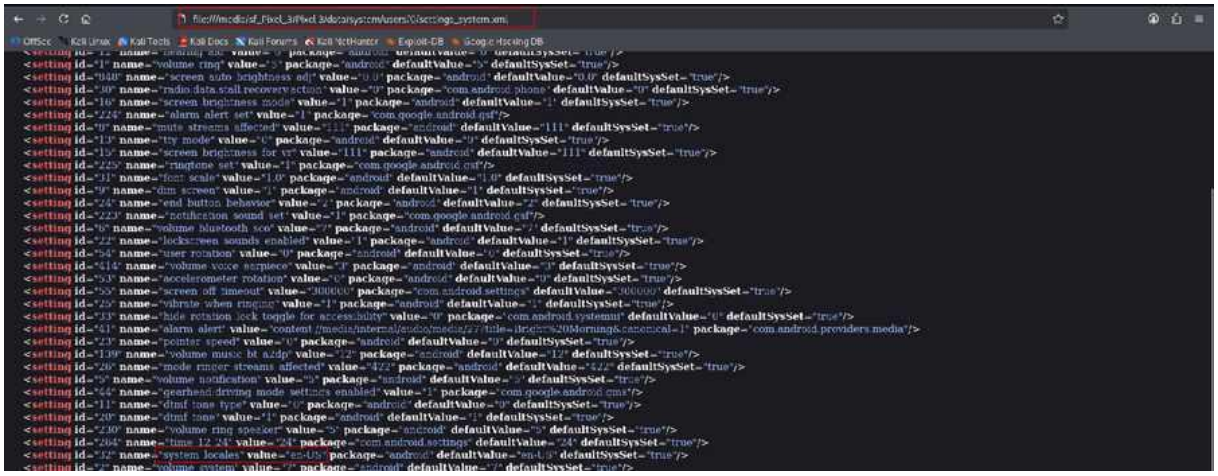
file:///media/sf_Pixel_3/Pixel 3/data/system/users/0/settings_secure.xml

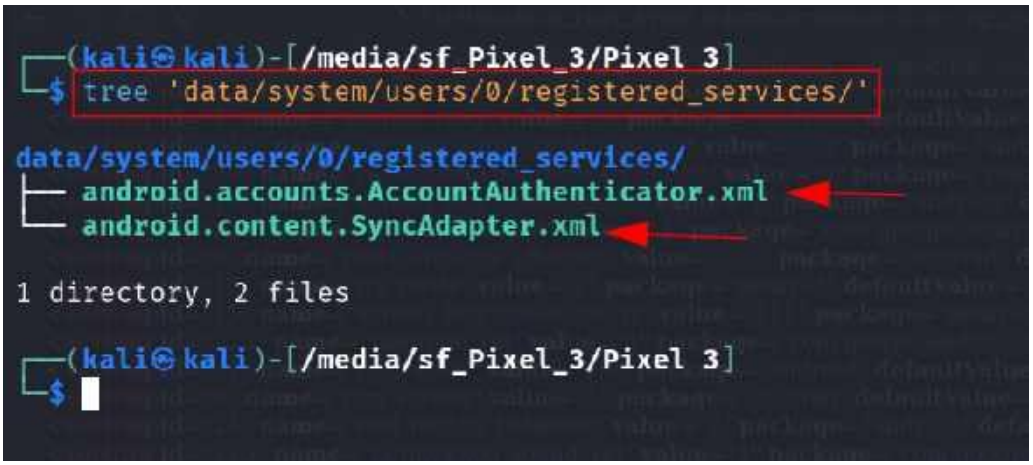
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This XML file does not appear to have any style information associated with it. The document tree is shown below.


```
-<settings version="183">
  <setting id="22" name="lock_screen_show_notifications" value="1" package="android" defaultValue="1" defaultSysSet="true" />
  <setting id="48" name="aware_lock_enabled" value="0" package="android" defaultValue="0" defaultSysSet="true" />
  <setting id="32" name="unknown_sources_default_reversed" value="1" package="android" defaultValue="1" defaultSysSet="true" />
  <setting id="83" name="sysui_volume_up_silent" package="com.android.systemui" />
  <setting id="138" name="usb_migration_state" value="0" package="com.google.android.apps.pixelmigrate" />
  <setting id="312" name="media_button_receiver" value="" package="android" defaultValue="" defaultSysSet="true" />
  <setting id="150" name="allowed_geolocation_origins" value="http://www.google.com http://www.google.co.uk" package="com.google.android.setupwizard" defaultValue="http://www.google.com http://www.google.co.uk"
  defaultSysSet="true" />
  <setting id="193" name="immersive_mode_confirmations" value="confirmed" package="android" defaultValue="confirmed" defaultSysSet="true" />
  <setting id="157" name="dark_mode_dialog_seen" value="1" package="com.android.settings" defaultValue="1" defaultSysSet="true" />
  <setting id="112" name="autofill_user_data_max_category_count" value="20" package="com.google.android.gms" />
  <setting id="146" name="location_mode" value="3" package="com.google.android.gms" defaultValue="3" defaultSysSet="true" />
  <setting id="143" name="user_full_data_backup_aware" value="1" package="com.google.android.gms" />
  <setting id="79" name="default_input_method" value="com.google.android.inputmethod.latin/com.android.inputmethod.latin.LatinIME" package="android" defaultValue="com.google.android.inputmethod.latin/
  com.android.inputmethod.latin.LatinIME" defaultSysSet="true" />
  <setting id="30" name="show_zen_settings_suggestion" value="1" package="android" defaultValue="1" defaultSysSet="true" />
  <setting id="114" name="config_update_certificate"
  value="MIID4TCCAsmgAwIBAgIJAMsyDT4FwMuvMA0GCSqGSIb3DQEBBQUAMIGGMQswCQYDVQQGEwJVUzETMBEGA1UECAwKQ2FsaWZvcm5pYTEWMBQGA1UEBwwNTW91bnRhaW4gVmlldzETMBEGA1UECgwKR29vZ2xlIE
  ..."
  <setting id="398" name="zen_settings_updated" value="1" package="com.android.settings" defaultValue="1" defaultSysSet="true" />
  <setting id="397" name="location_providers_allowed" value="gps,network" package="android" defaultValue="gps,network" defaultSysSet="true" />
  <setting id="23" name="lock_screen_allow_private_notifications" value="1" package="android" defaultValue="1" defaultSysSet="true" />
```





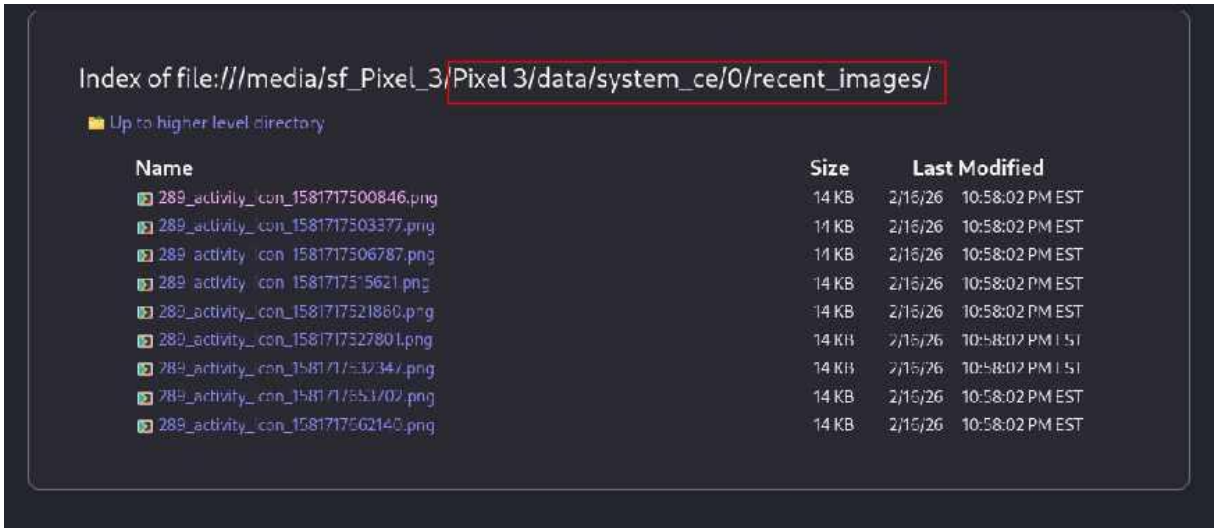

(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$ tree 'data/system/users/0/registered_services/'
data/system/users/0/registered_services/
android.accounts.AccountAuthenticator.xml
android.content.SyncAdapter.xml
1 directory, 2 files
(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$

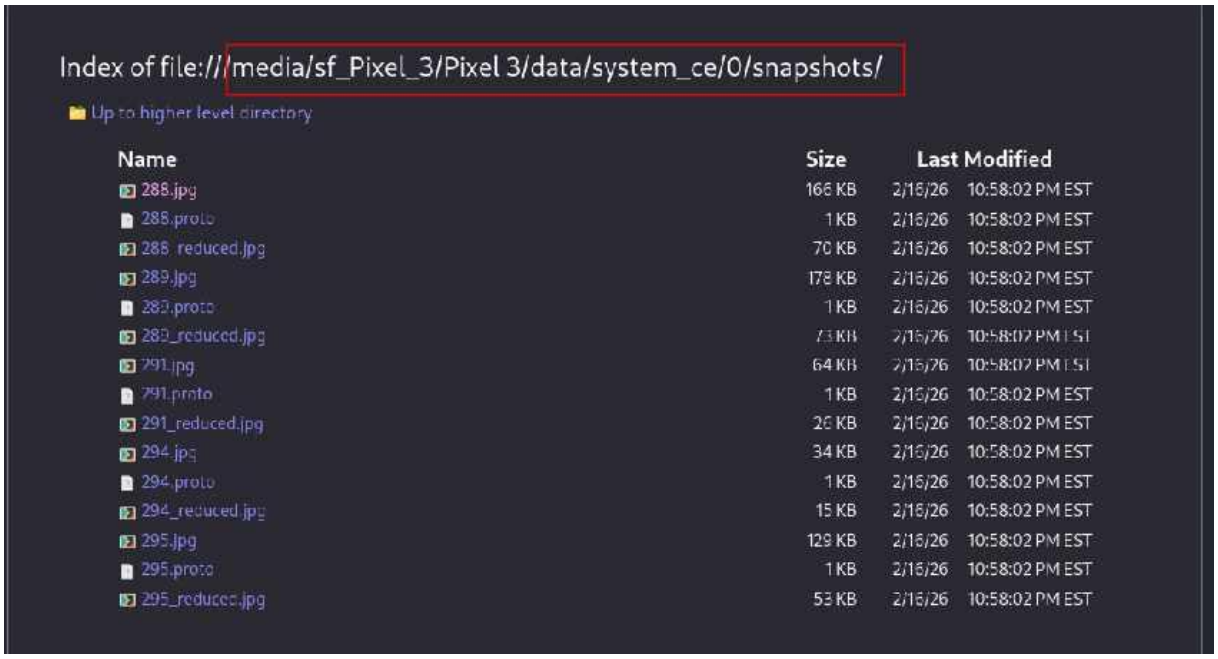
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/system/users/0/registered_services/android.accounts.AccountAuthenticator.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<services>
    <service uid="10239" type="com.enflick.android.TextNow.account" />
    <service uid="10244" type="com.whatsapp" />
    <service uid="10187" type="com.google.android.gm.pop3" />
    <service uid="10141" type="com.google" />
    <service uid="10209" type="com.google.android.apps.tachyon" />
    <service uid="10238" type="com.zhiliaoapp.account" />
    <service uid="10245" type="com.twitter.android.auth.login" />
    <service uid="10224" type="com.silentcircle.account" />
    <service uid="10236" type="com.skype.raider" />
    <service uid="10241" type="com.facebook.messenger" />
    <service uid="10187" type="com.google.android.gm.exchange" />
    <service uid="10238" type="com.zhiliaoapp.musically" />
    <service uid="10231" type="com.viber.voip" />
    <service uid="10141" type="com.google.android.gms.matchstick" />
    <service uid="10240" type="com.imo.android.imous" />
    <service uid="10233" type="org.thoughtcrime.securesms" />
    <service uid="10234" type="org.telegram.messenger" />
    <service uid="10187" type="com.google.android.gm.legacyimap" />
</services>
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/system/users/0/registered_services/android.content.SyncAdapter.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<services>
    <service uid="10141" authority="com.google.android.gms.people" accountType="com.google" />
    <service uid="10187" authority="com.google.android.gm.email.provider" accountType="com.google.android.gm.pop3" />
    <service uid="10187" authority="com.google.android.gm.tasks.provider" accountType="com.google.android.gm.exchange" />
    <service uid="10233" authority="com.android.contacts" accountType="org.thoughtcrime.securesms" />
    <service uid="10212" authority="com.google.android.apps.magazines" accountType="com.google" />
    <service uid="10208" authority="com.google.android.apps.docs.editors.kix" accountType="com.google" />
    <service uid="10178" authority="com.google.android.calendar.tasks" accountType="com.google" />
    <service uid="10231" authority="com.android.contacts" accountType="com.viber.voip" />
    <service uid="10141" authority="com.android.contacts" accountType="com.google" />
    <service uid="10245" authority="com.twitter.android.provider.TwitterProvider" accountType="com.twitter.android.auth.login" />
    <service uid="10224" authority="com.android.contacts" accountType="com.silentcircle.account" />
    <service uid="10211" authority="com.google.android.keep" accountType="com.google" />
    <service uid="10169" authority="com.google.android.videos.sync" accountType="com.google" />
    <service uid="10175" authority="com.google.android.apps.tycho.provider" accountType="com.google" />
    <service uid="10240" authority="com.imo.android.imous.syncadapter.stubprovider" accountType="com.imo.android.imous" />
    <service uid="10141" authority="com.google.android.gms.auth.accountstate" accountType="com.google" />
    <service uid="10240" authority="com.android.contacts" accountType="com.imo.android.imous" />
    <service uid="10141" authority="com.google.android.gms.games.background" accountType="com.google" />
    <service uid="10141" authority="subscribedfeeds" accountType="com.google" />
    <service uid="10171" authority="com.google.android.music.MusicContent" accountType="com.google" />
    <service uid="10234" authority="com.android.contacts" accountType="org.telegram.messenger" />
    <service uid="10238" authority="com.zhiliaoapp.musically.AccountProvider" accountType="com.zhiliaoapp.musically" />
    <service uid="10209" authority="com.android.contacts" accountType="com.google.android.apps.tachyon" />
    <service uid="10178" authority="com.google.android.calendar" accountType="com.google" />
    <service uid="10141" authority="com.google.android.location.reporting" accountType="com.google" />
    <service uid="10244" authority="com.android.contacts" accountType="com.whatsapp" />
    <service uid="10178" authority="com.android.calendar" accountType="com.google" />
    <service uid="10239" authority="com.enflick.android.TextNow.TNContentProvider" accountType="com.enflick.android.TextNow.account" />
    <service uid="10187" authority="com.android.contacts" accountType="com.google.android.gm.exchange" />
    <service uid="10187" authority="com.google.android.gm.email.provider" accountType="com.google.android.gm.legacyimap" />
    <service uid="10141" authority="com.google.android.gms.appstate" accountType="com.google" />
    <service uid="10141" authority="com.google.android.gms.auth.confirm" accountType="com.google" />
    <service uid="10187" authority="com.google.android.gm.email.provider" accountType="com.google.android.gm.exchange" />
    <service uid="10141" authority="com.google.android.gms.subscribedfeeds" accountType="com.google" />
    <service uid="10141" authority="com.google.android.gms.drive.sync" accountType="com.google" />
    <service uid="10241" authority="com.android.contacts" accountType="com.facebook.messenger" />
    <service uid="10236" authority="com.android.contacts" accountType="com.skype.raider" />
    <service uid="10208" authority="com.google.android.apps.docs" accountType="com.google" />
    <service uid="10141" authority="com.google.android.gms.plus.action" accountType="com.google" />
    <service uid="1019" authority="com.android.chrome" accountType="com.google" />
    <service uid="10141" authority="com.google.android.gms.reminders" accountType="com.google" />
    <service uid="10141" authority="com.google.android.gms.fitness" accountType="com.google" />
    <service uid="10141" authority="com.android.contacts" accountType="com.google.android.gms.matchstick" />
    <service uid="10141" authority="com.google.android.gms.chromesync" accountType="com.google" />
    <service uid="10141" authority="com.google.android.gms.games" accountType="com.google" />
    <service uid="10187" authority="com.android.calendar" accountType="com.google.android.gm.exchange" />
    <service uid="10187" authority="gmail-ls" accountType="com.google" />
</services>
```





Application Life Cycle and Usage

I compared package metadata (packages.list, packages.xml) with usage records (package-usage.list, usagestats) to follow apps from install to last use. Cross-source check: Kik's install time in its measurement prefs (2020-01-29 17:08:42 UTC) agrees with the Branch install time in branch_referral_shared_pref.xml and comes about 80 minutes before the Kik login.

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ find  -iname  'package*'

./data/media/0/Android/data/jp.naver.line.android/files/yuki/3_1/package
./data/system/package-cstats.list
./data/system/package-dcl.list
./data/system/package-dex-usage.list
./data/system/package-usage.list
./data/system/package-watchdog.xml
./data/system/packages-warnings.xml
./data/system/packages.list
./data/system/packages.xml
./data/system/package_cache
./data/system/users/0/package-restrictions.xml
./data/system/users/11/package-restrictions.xml
./sbin/.magisk/mirror/data/media/0/Android/data/jp.naver.line.android/files/yuki/3_1/package
./sbin/.magisk/mirror/data/system/package-cstats.list
./sbin/.magisk/mirror/data/system/package-dcl.list
./sbin/.magisk/mirror/data/system/package-dex-usage.list
./sbin/.magisk/mirror/data/system/package-usage.list
./sbin/.magisk/mirror/data/system/package-watchdog.xml
./sbin/.magisk/mirror/data/system/packages-warnings.xml
./sbin/.magisk/mirror/data/system/packages.list
./sbin/.magisk/mirror/data/system/packages.xml
./sbin/.magisk/mirror/data/system/package_cache
./sbin/.magisk/mirror/data/system/users/0/package-restrictions.xml
./sbin/.magisk/mirror/data/system/users/11/package-restrictions.xml
```

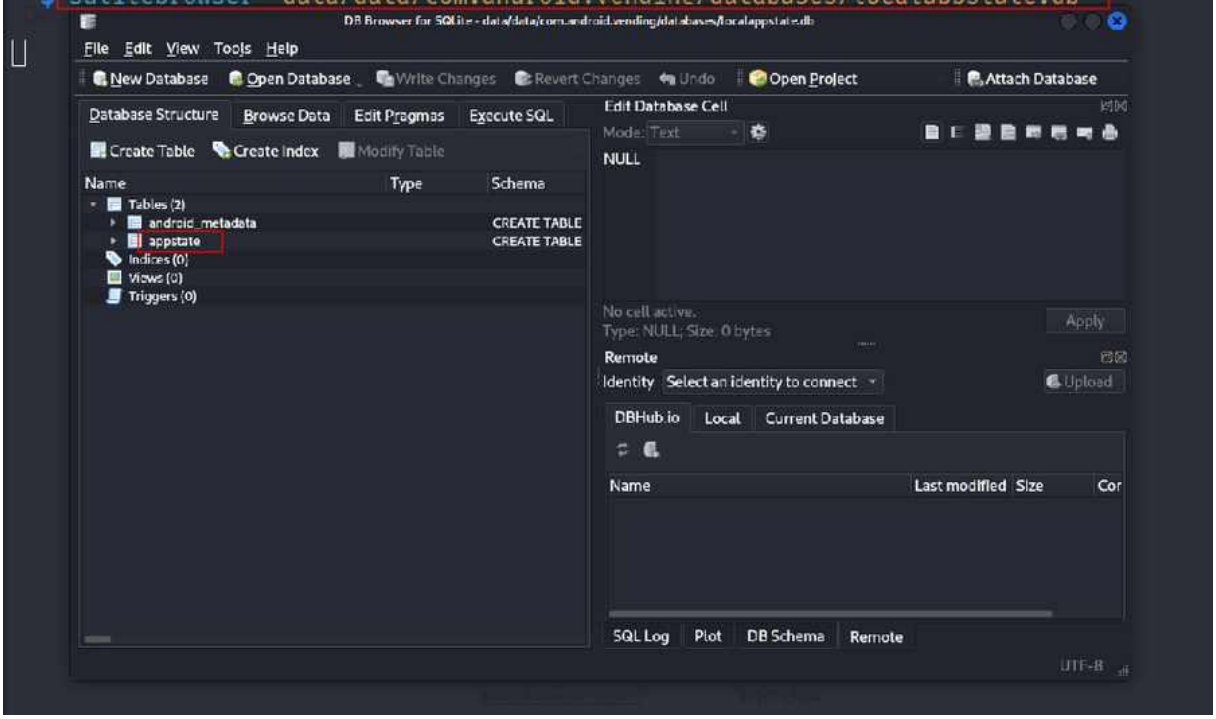
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls -l 'data/data/com.android.vending'

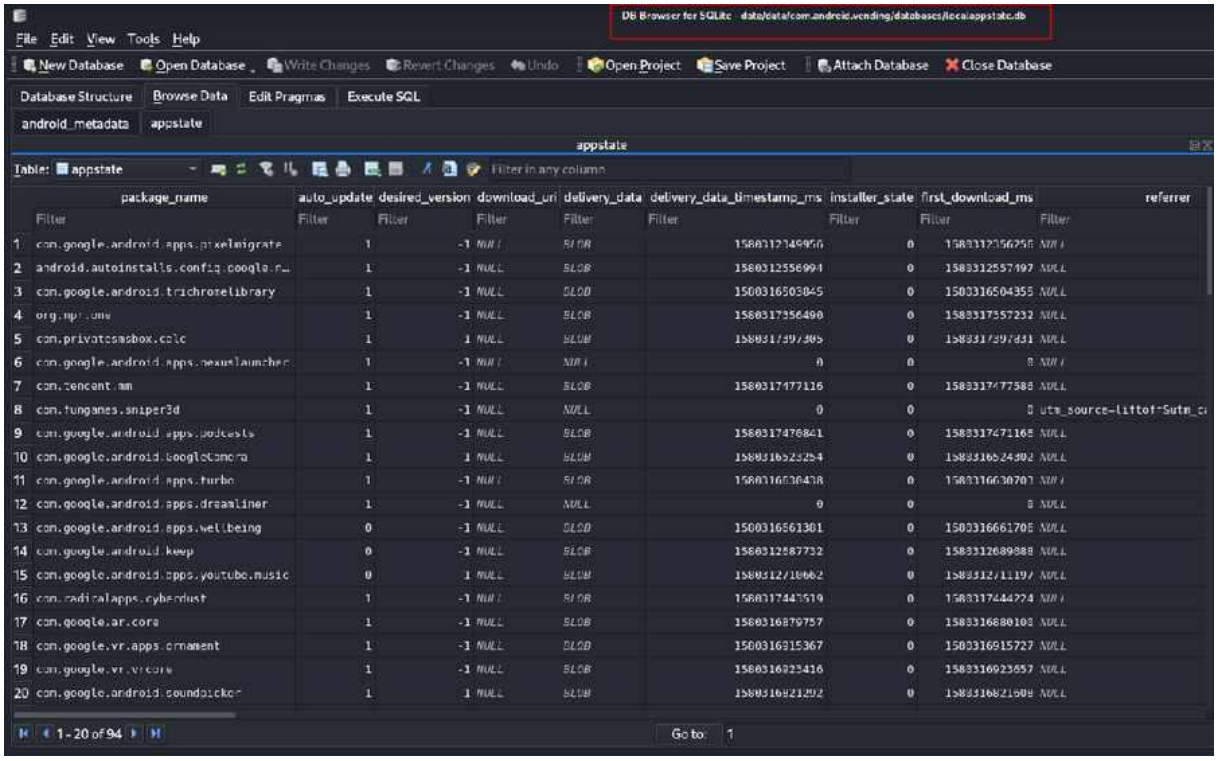
total 104
drwxrwx--- 1 root vboxsf 81920 Feb 16 22:50 app_commerce_acquire_cache
drwxrwx--- 1 root vboxsf     0 Feb 16 22:50 cache
drwxrwx--- 1 root vboxsf  8192 Feb 16 22:50 databases
drwxrwx--- 1 root vboxsf  4096 Feb 16 22:50 files
drwxrwx--- 1 root vboxsf     0 Feb 16 22:50 no_backup
drwxrwx--- 1 root vboxsf 12288 Feb 16 22:50 shared_prefs
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls -l 'data/data/com.android.vending/databases'

total 2720
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 asset_module_service
-rwxrwx--- 1 root vboxsf  32768 Feb 16 22:50 auto_update.db
-rwxrwx--- 1 root vboxsf  20480 Feb 16 22:50 counters.db
-rwxrwx--- 1 root vboxsf 155648 Feb 16 22:50 data_usage.db
-rwxrwx--- 1 root vboxsf  20480 Feb 16 22:50 download_service
-rwxrwx--- 1 root vboxsf  28672 Feb 16 22:50 fetch_suggestions.db
-rwxrwx--- 1 root vboxsf 212992 Feb 16 22:50 frosting.db
-rwxrwx--- 1 root vboxsf  20480 Feb 16 22:50 group_install.db
-rwxrwx--- 1 root vboxsf  77824 Feb 16 22:50 install_queue.db
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 install_service
-rwxrwx--- 1 root vboxsf  32768 Feb 16 22:50 install_source.db
-rwxrwx--- 1 root vboxsf  49152 Feb 16 22:50 keyed_app_states.db
-rwxrwx--- 1 root vboxsf  49152 Feb 16 22:50 library.db
-rwxrwx--- 1 root vboxsf 376832 Feb 16 22:50 localappstate.db
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 notification_cache
-rwxrwx--- 1 root vboxsf  45056 Feb 16 22:50 notification_clickability
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 offline_acquire_store.db
-rwxrwx--- 1 root vboxsf 995328 Feb 16 22:50 phenotype.db
-rwxrwx--- 1 root vboxsf  86016 Feb 16 22:50 prefetch_recommendation_store.db
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 scheduled_acquisitions.db
-rwxrwx--- 1 root vboxsf  20480 Feb 16 22:50 scheduler
-rwxrwx--- 1 root vboxsf  73728 Feb 16 22:50 split_install_service
-rwxrwx--- 1 root vboxsf  20480 Feb 16 22:50 suggestions.db
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 user_languages_store
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.android.vending/databases/localappstate.db'
```






```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlite3 'data/data/com.android.vending/databases/localappstate.db'

SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .tables
android_metadata  appstate
sqlite> PRAGMA table_info(appstate);
0|package_name|STRING|0||1
1|auto_update|INTEGER|0||0
2|desired_version|INTEGER|0||0
3|download_uri|STRING|0||0
4|delivery_data|BLOB|0||0
5|delivery_data_timestamp_ms|INTEGER|0||0
6|installer_state|INTEGER|0||0
7|first_download_ms|INTEGER|0||0
8|referrer|STRING|0||0
9|account|STRING|0||0
10|title|STRING|0||0
11|flags|INTEGER|0||0
12|continue_url|STRING|0||0
13|last_notified_version|INTEGER|0||0
14|last_update_timestamp_ms|INTEGER|0||0
15|account_for_update|STRING|0||0
16|auto_acquire_tags|STRING|0||0
17|external_referrer_timestamp_ms|INTEGER|0||0
18|persistent_flags|INTEGER|0||0
19|permissions_version|INTEGER|0||0
```

```
17|external_referrer_timestamp_ms|INTEGER|0||0
18|persistent_flags|INTEGER|0||0
19|permissions_version|INTEGER|0||0
20|delivery_token|STRING|0||0
21|completed_split_ids|STRING|0||0
22|active_split_id|STRING|0||0
23|request_id|STRING|0||0
24|managed_configuration_token|BLOB|0||0
25|total_completed_bytes_downloaded|INTEGER|0||0
26|app_details|BLOB|0||0
27|install_client_event_id|INTEGER|0||0
28|last_client_event_id|INTEGER|0||0
29|requesting_package_name|STRING|0||0
30|update_discovered_timestamp_ms|INTEGER|0||0
31|install_logging_context|BLOB|0||0
32|logging_context|BLOB|0||0
33|install_request_timestamp_ms|INTEGER|0||0
34|desired_derived_apk_id|INTEGER|0||0
35|notification_intent|STRING|0||0
36|auto_update_flow_policy|BLOB|0||0
37|install_reason|STRING|0||0
38|sandbox_version|INTEGER|0||0
39|requested_modules|TEXT|0||0
40|install_request_data|BLOB|0||0
41|active_accelerator_index|INTEGER|0||0
42|update_discovered_version_code|INTEGER|0||0
43|desired_frosting_id|INTEGER|0||0
44|active_resource_id|BLOB|0||0
sqlite>
```

```
sqlite> select  package_name, delivery_data_timestamp_ms, first_download_ms, account, title from appstate limit 20;
com.google.android.apps.pixelmigrate|1580312349956|1580312356256||Data Transfer Tool
android.autoinstalls.config.google.nexus|1580312556994|1580312557497|thisisdfir@gmail.com|android.autoinstalls.config.google.nexus
com.google.android.trichromelibrary|1580316503845|1580316504355|thisisdfir@gmail.com|Google Chrome: Fast & Secure
org.npr.one|1580317356490|1580317357232|thisisdfir@gmail.com|NPR One
com.privatesmsbox.calc|1580317397305|1580317397831|thisisdfir@gmail.com|Calculator Pro+ - Private Message & Call Screening
com.google.android.apps.nexuslauncher|0|0||
com.tencent.mm|1580317477116|1580317477586|thisisdfir@gmail.com|WeChat
com.fungames.sniper3d|0|0||
com.google.android.apps.podcasts|1580317470841|1580317471166|thisisdfir@gmail.com|Google Podcasts: Discover free & trending podcasts
com.google.android.GoogleCamera|1580316523254|1580316524302|thisisdfir@gmail.com|Google Camera
com.google.android.apps.turbo|1580316630438|1580316630703|thisisdfir@gmail.com|Device Health Services
com.google.android.apps.dreamliner|0|0||
com.google.android.apps.wellbeing|1580316661381|1580316661706|thisisdfir@gmail.com|Digital Wellbeing
com.google.android.keep|1580312687732|1580312689088|thisisdfir@gmail.com|Google Keep - Notes and Lists
com.google.android.apps.youtube.music|1580312710662|1580312711197|thisisdfir@gmail.com|YouTube Music - Stream Songs & Music Videos
com.radicalapps.cyberdust|1580317443519|1580317444224|thisisdfir@gmail.com|Dust - a safer place to text
com.google.ar.core|1580316879757|1580316880100|thisisdfir@gmail.com|Google Play Services for AR
com.google.vr.apps.ornament|1580316915367|1580316915727|thisisdfir@gmail.com|Playground (was AR Stickers)
com.google.vr.vrcore|1580316923416|1580316923657|thisisdfir@gmail.com|Google VR Services
com.google.android.soundpicker|1580316821292|1580316821608|thisisdfir@gmail.com|Sounds
sqlite>
```

```
sqlite> select  package_name, delivery_data_timestamp_ms, first_download_ms, account, title from appstate where lower(title) like '%cha
t%';
com.tencent.mm|1580317477116|1580317477586|thisisdfir@gmail.com|WeChat
com.imo.android.imous|1580317694844|1580317695279|thisisdfir@gmail.com|imo free HD video calls and chat
com.snapchat.android|1580317778348|1580317779105|thisisdfir@gmail.com|Snapchat
com.discord|1580317364216|1580317364610|thisisdfir@gmail.com|Discord - Chat for Gamers
com.viber.voip|1580317540884|1580317541312|thisisdfir@gmail.com|Viber Messenger - Messages, Group Chats & Calls
com.skout.android|1580317631281|1580317631748|thisisdfir@gmail.com|SKOUT - Meet, Chat, Go Live
sqlite>
```

```
sqlite> .header on
sqlite> .mode csv
sqlite> .output appstate.csv
sqlite> select  package_name, delivery_data_timestamp_ms, first_download_ms, account, title from appstate;
sqlite> .quit
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls -l appstate.csv
-rwxrwx--- 1 root vboxsf 9263 Feb 18 17:00 appstate.csv

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep -i wechat appstate.csv
com.tencent.mm,1580317477116,1580317477586,thisisdfir@gmail.com,WeChat

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d '@1580317477' -u
Wed Jan 29 05:04:37 PM UTC 2020

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
sqlite> select  first_download_ms, account, title from appstate where first_download_ms > 100 order by first_download_ms limit 3;
1580312356256||Data Transfer Tool
1580312557497|thisisdfir@gmail.com|android.autoinstalls.config.google.nexus
1580312649019|thisisdfir@gmail.com|Google Docs
sqlite> select  first_download_ms, account, title from appstate where first_download_ms > 100 order by first_download_ms desc limit 3;
1580317923118|thisisdfir@gmail.com|Imgur: Find funny GIFs, memes & watch viral videos
1580317779105|thisisdfir@gmail.com|Snapchat
1580317762141|thisisdfir@gmail.com|Twitter
sqlite>
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d '@1580312356' -u
Wed Jan 29 03:39:16 PM UTC 2020

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d '@1580317923' -u
Wed Jan 29 05:12:03 PM UTC 2020
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls -l 'data/data/com.android.vending/databases'
total 2720
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 asset_module_service
-rwxrwx--- 1 root vboxsf  32768 Feb 16 22:50 auto_update.db
-rwxrwx--- 1 root vboxsf  20480 Feb 16 22:50 counters.db
-rwxrwx--- 1 root vboxsf 155648 Feb 16 22:50 data_usage.db
-rwxrwx--- 1 root vboxsf  20480 Feb 16 22:50 download_service
-rwxrwx--- 1 root vboxsf  28672 Feb 16 22:50 fetch_suggestions.db
-rwxrwx--- 1 root vboxsf 212992 Feb 16 22:50 frosting.db
-rwxrwx--- 1 root vboxsf  20480 Feb 16 22:50 group_install.db
-rwxrwx--- 1 root vboxsf  77824 Feb 16 22:50 install_queue.db
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 install_service
-rwxrwx--- 1 root vboxsf  32768 Feb 16 22:50 install_source.db
-rwxrwx--- 1 root vboxsf  49152 Feb 16 22:50 keyed_app_states.db
-rwxrwx--- 1 root vboxsf  49152 Feb 16 22:50 library.db
-rwxrwx--- 1 root vboxsf 376832 Feb 16 22:50 localappstate.db
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 notification_cache
-rwxrwx--- 1 root vboxsf  45056 Feb 16 22:50 notification_clickability
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 offline_acquire_store.db
-rwxrwx--- 1 root vboxsf 995328 Feb 16 22:50 phenotype.db
-rwxrwx--- 1 root vboxsf  86016 Feb 16 22:50 prefetch_recommendation_store.db
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 scheduled_acquisitions.db
-rwxrwx--- 1 root vboxsf  20480 Feb 16 22:50 scheduler
-rwxrwx--- 1 root vboxsf  73728 Feb 16 22:50 split_install_service
-rwxrwx--- 1 root vboxsf  20480 Feb 16 22:50 suggestions.db
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:50 user_languages_store
-rwxrwx--- 1 root vboxsf 294912 Feb 16 22:50 verify_apps.db
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlite3 'data/data/com.android.vending/databases/localappstate.db'
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> select  package_name, delivery_data_timestamp_ms, first_download_ms, account, title from appstate where lower(title) like '%wha
tsapp%';
com.whatsapp|1580317743713|1580317744179|thisisdfir@gmail.com|WhatsApp Messenger
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlite3 'data/data/com.android.vending/databases/library.db'
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> select doc_id, document_hash, purchase_time from ownership where doc_id like '%whatsapp%';
com.whatsapp|-2208265135048976340|1543518467665
sqlite> .quit
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d '@1543518467' -u
Thu Nov 29 07:07:47 PM UTC 2018
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -u \
--date=@$(sqlite3 'data/data/com.android.vending/databases/library.db'  \
"SELECT purchase_time/1000 FROM ownership WHERE doc_id   like '%whatsapp%'")

Thu Nov 29 07:07:47 PM UTC 2018
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls 'data/data/'
```

```
com.android.chrome
com.android.hotwordenrollment.okgoogle
com.android.hotwordenrollment.xgoogle
com.android.keychain
com.android.mtp
com.android.nfc
com.android.providers.calendar
com.android.providers.contacts
com.android.providers.downloads
com.android.providers.media
com.android.providers.telephony
com.android.providers.userdictionary
com.android.service.ims.presence
com.android.traceur
com.android.vending
com.discord
com.enflick.android.TextNow
com.facebook.orca
com.google.android.apps.chromecast.app
com.google.android.apps.docs
com.google.android.apps.docs.editors.docs
com.google.android.apps.dreamliner
com.google.android.apps.enterprise.dmagent
com.google.android.apps.gcs
com.google.android.apps.helprtc
com.google.android.apps.magazines
com.google.android.apps.maps
com.google.android.apps.messaging
com.google.android.apps.nexuslauncher
com.google.android.apps.photos
com.google.android.apps.pixelmigrate
com.google.android.apps.safetyhub
com.google.android.apps.tachyon
com.google.android.apps.tips
com.google.android.apps.turbo
com.google.android.apps.tycho
com.google.android.apps.walletnfcrel
com.google.android.apps.wallpaper
com.google.android.apps.wellbeing
com.google.android.apps.work.oobconfig
com.google.android.apps.youtube.music
com.google.android.as
com.google.android.calendar
com.google.android.configupdater
com.google.android.contacts
com.google.android.deskclock
com.google.android.dialer
com.google.android.documentsui
com.google.android.ext.services
com.google.android.gm
com.google.android.gms
com.google.android.GoogleCamera
com.google.android.googlequicksearchbox
com.google.android.gsf
com.google.android.ims
com.google.android.inputmethod.latin
com.google.android.keep
com.google.android.markup
com.google.android.music
com.google.android.onetimeinitializer
com.google.android.partnersetup
com.google.android.pixel.setupwizard
com.google.android.projection.gearhead
com.google.android.settings.intelligence
com.google.android.setupwizard
com.google.android.syncadapters.contacts
com.google.android.tts
com.google.android.videos
com.google.android.webview
com.google.android.wfcactivation
com.google.android.youtube
com.google.ar.core
com.google.intelligence.sense
com.google.vr.apps.ornament
com.google.vr.vrcore
com.imgur.mobile
com.imo.android.imous
com.instagram.android
com.instagram.threadsapp
com.mewe
com.mywickr.wickr2
com.qti.ltebc
com.radicalapps.cyberdust
com.silentcircle.silentphone
com.skout.android
com.skype.raider
com.snapchat.android
com.spotify.music
com.thinkyeah.galleryvault
com.topjohnwu.magisk
com.twitter.android
com.venmo
com.viber.voip
com.whatsapp
com.wire
com.zhiliaoapp.musically
jp.naver.line.android
kik.android
org.telegram.messenger
org.thoughtcrime.securesms
org.torproject.torbrowser
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls 'data/data/' | grep -i com.google
```

```
com.google.android.apps.chromecast.app
com.google.android.apps.docs
com.google.android.apps.docs.editors.docs
com.google.android.apps.dreamliner
com.google.android.apps.enterprise.dmagent
com.google.android.apps.gcs
com.google.android.apps.helprtc
com.google.android.apps.magazines
com.google.android.apps.maps
com.google.android.apps.messaging
com.google.android.apps.nexuslauncher
com.google.android.apps.photos
com.google.android.apps.pixelmigrate
com.google.android.apps.safetyhub
com.google.android.apps.tachyon
com.google.android.apps.tips
com.google.android.apps.turbo
com.google.android.apps.tycho
com.google.android.apps.walletnfcrel
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls 'data/data/' | grep -i com.android

com.android.chrome
com.android.hotwordenrollment.okgoogle
com.android.hotwordenrollment.xgoogle
com.android.keychain
com.android.mtp
com.android.nfc
com.android.providers.calendar
com.android.providers.contacts
com.android.providers.downloads
com.android.providers.media
com.android.providers.telephony
com.android.providers.userdictionary
com.android.service.ims.presence
com.android.traceur
com.android.vending
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls 'data/data/' | grep -Eiv 'com.google|com.android'
com.discord
com.enflick.android.TextNow
com.facebook.orca
com.imgur.mobile
com.imo.android.imous
com.instagram.android
com.instagram.threadsapp
com.mewe
com.mywickr.wickr2
com.qti.ltebc
com.radicalapps.cyberdust
com.silentcircle.silentphone
com.skout.android
com.skype.raider
com.snapchat.android
com.spotify.music
com.thinkyeah.galleryvault
com.topjohnwu.magisk
com.twitter.android
com.venmo
com.viber.voip
com.whatsapp
com.wire
com.zhiliaoapp.musically
jp.naver.line.android
kik.android
org.telegram.messenger
org.thoughtcrime.securesms
org.torproject.torbrowser
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ find -iname  'package*'

./data/media/0/Android/data/jp.naver.line.android/files/yuki/3_1/package
./data/system/package-cstats.list
./data/system/package-dcl.list
./data/system/package-dex-usage.list
./data/system/package-usage.list
./data/system/package-watchdog.xml
./data/system/packages-warnings.xml
./data/system/packages.list
./data/system/packages.xml
./data/system/package_cache
./data/system/users/0/package-restrictions.xml
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/system/packages.list' | grep -Ei '\.gm'
com.google.android.gm 10187 0 /data/user/0/com.google.android.gm google:targetSdkVersion=29 3003 0 62049238
com.google.android.gms 10141 0 /data/user/0/com.google.android.gms google:privapp:targetSdkVersion=29 2001,1005,3002,3003,3001,3007,1007,3006 0 200414037
com.google.android.gms.location.history 10191 0 /data/user/0/com.google.android.gms.location.history default:targetSdkVersion=28 none 0 1
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep -in 'package name="com.google.android.gm' 'data/system/packages.xml'
3962:    <package name="com.google.android.gm" codePath="/data/app/com.google.android.gm-QTglY_VBwLxA5dDcCqrCw==" nativeLibraryPath="/data/app/com.google.android.gm-QTglY_VBwLxA5dDcCqrCw==/lib" publicFlags="944486133" privateFlags="524288" ft="16ff237b230" it="11e8f7d4c00" ut="16ff237b83e" version="62049238" userId="10187" installer="com.android.vending" isOrphaned="true" categoryHint="4">
6485:    <package name="com.google.android.gms" codePath="/data/app/com.google.android.gms-ZpkWIkavN2y4dNnqJLme0g==" nativeLibraryPath="/data/app/com.google.android.gms-ZpkWIkavN2y4dNnqJLme0g==/lib" primaryCpuAbi="arm64-v8a" secondaryCpuAbi="armeabi-v7a" publicFlags="-1605714235" privateFlags="524296" ft="17043c42548" it="11e8f7d4c00" ut="17043c42752" version="200414037" sharedUserId="10141" installer="com.android.vending" isOrphaned="true" categoryHint="7">
9310:    <package name="com.google.android.gms.location.history" codePath="/product/app/LocationHistoryPrebuilt" nativeLibraryPath="/product/app/LocationHistoryPrebuilt/lib" publicFlags="810040901" privateFlags="524288" ft="11e8f7d4c00" it="11e8f7d4c00" ut="11e8f7d4c00" version="1" userId="10191" isOrphaned="true">
10408:    <updated-package name="com.google.android.gm" codePath="/product/app/PrebuiltGmail" ft="11e8f/d4c00" it="11e8f/d4c00" ut="11e8f/d4c00" version="61499782" nativeLibraryPath="/product/app/PrebuiltGmail/lib" userId="10187">
10733:    <updated-package name="com.google.android.gms" codePath="/product/priv-app/PrebuiltGmsCore" ft="11e8f7d4c00" it="11e8f7d4c00" ut="11e8f7d4c00" version="17786048" nativeLibraryPath="/product/priv-app/PrebuiltGmsCore/lib" primaryCpuAbi="arm64-v8a" secondaryCpuAbi="armeabi-v7a" sharedUserId="10141" />
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep -Ei '\.gm' 'data/system/package-usage.list'
com.google.android.gm 1580519249465 1580523580618 1580497732035 1580523580614 1580523574402 0 0 0
com.google.android.gms 1580523599326 1580529895520 1580395544983 1580529394178 1580526326157 0 1580529874202 0
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/system/usagestats/' -L 2
data/system/usagestats/
├── 0
│   ├── daily
│   ├── monthly
│   ├── version
│   ├── weekly
│   └── yearly
└── 11
    ├── daily
    ├── monthly
    ├── version
    ├── weekly
    └── yearly

11 directories, 2 files
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/system/usagestats/0' -L 2

data/system/usagestats/0
├── daily
│   ├── 1580917965671
│   ├── 1581004373881
│   ├── 1581090879070
│   ├── 1581177279272
│   ├── 1581263873464
│   ├── 1581350284755
│   ├── 1581436707859
│   ├── 1581523178051
│   ├── 1581609587089
│   └── 1581696001773
├── monthly
│   └── 1580312256162
├── version
├── weekly
│   ├── 1580312256162
│   ├── 1580917965671
│   └── 1581523178051
└── yearly
    └── 1580312256162
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d '@1580917965' -u
Wed Feb  5 03:52:45 PM UTC 2020
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ git clone https://github.com/abrignoni/Android-Usagestats-XML-Protobuf.git

Cloning into 'Android-Usagestats-XML-Protobuf' ...
remote: Enumerating objects: 83, done.
remote: Total 83 (delta 0), reused 0 (delta 0), pack-reused 83 (from 1)
Receiving objects: 100% (83/83), 90.31 KiB | 167.00 KiB/s, done.
Resolving deltas: 100% (37/37), done.
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cp -avr 'data/system/usagestats/' Android-Usagestats-XML-Protobuf

'data/system/usagestats/' → 'Android-Usagestats-XML-Protobuf/usagestats'
'data/system/usagestats/0' → 'Android-Usagestats-XML-Protobuf/usagestats/0'
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/Android-Usagestats-XML-Protobuf/protobuf]
└─$ cd "/media/sf_Pixel_3/Pixel 3/Android-Usagestats-XML-Protobuf"
python3 usagestats_conv.py

Android Usagestats XML Parser
By: @AlexisBrignoni
Web: abrignoni.com

Files:
Processing: /media/sf_Pixel_3/Pixel 3/Android-Usagestats-XML-Protobuf/usagestats/0/daily/1580917965671

Processing: /media/sf_Pixel_3/Pixel 3/Android-Usagestats-XML-Protobuf/usagestats/0/daily/1581004373881

Processing: /media/sf_Pixel_3/Pixel 3/Android-Usagestats-XML-Protobuf/usagestats/0/daily/1581090879070
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/Android-Usagestats-XML-Protobuf]
└─$ sqlite3 'usagestats.db'
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .tables
data
sqlite> pragma table_info(data);
0|usage_type|TEXT|0||0
1|lastime|INTEGER|0||0
2|timeactive|INTEGER|0||0
3|last_time_service_used|INTEGER|0||0
4|last_time_visible|INTEGER|0||0
5|total_time_visible|INTEGER|0||0
6|app_launch_count|INTEGER|0||0
7|package|TEXT|0||0
8|types|TEXT|0||0
9|classs|TEXT|0||0
10|source|TEXT|0||0
11|fullatt|TEXT|0||0
sqlite>
```

Android Usagestats report (Dates are localtime!)

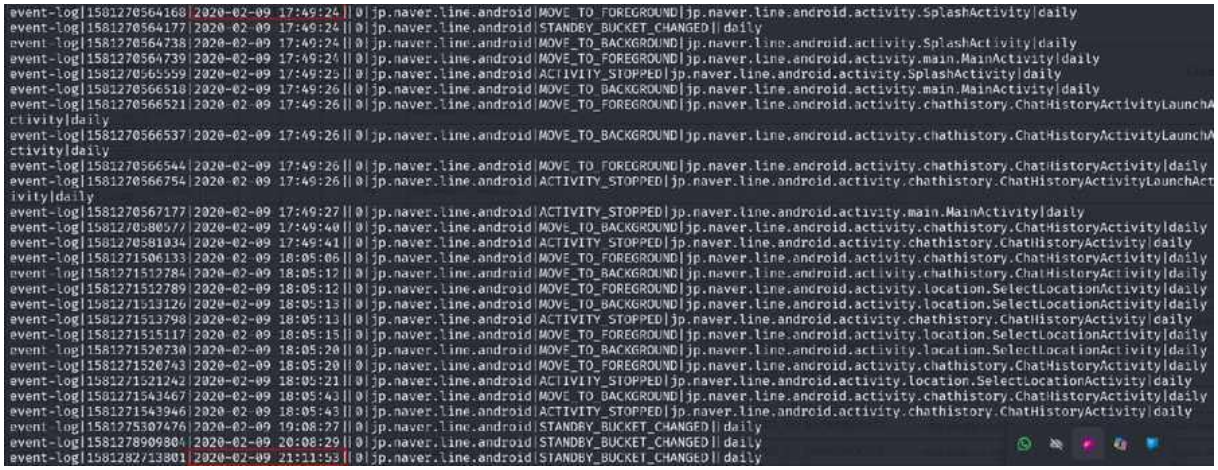
Usage Type	Last Time Active	Time Active In Msecs	Time Active In Secs	Last Time Service Used	Last Time Visible	Total Time Visible	App Launch Count	Package	Types	Class
packages	2020-02-14 18:42:20	10997	10			3		com.android.settings		
packages	2020-02-14 18:42:20	4995624	4995			65		com.android.settings		
packages	2020-02-14 18:42:20	219617	219			11		com.android.settings		
packages	2020-02-14 18:42:20	1995624	1995			65		com.android.settings		
event log	2020-02-14 18:42:19		0					com.android.settings	MOVE_TO_BACKGROUND	com.android.settings.SubSettings
event log	2020-02-14 18:42:19		0					com.android.settings	MOVE_TO_FORGROUND	com.android.settings.SubSettings

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlite3
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
Connected to a transient in-memory database.
Use ".open FILENAME" to reopen on a persistent database.
sqlite> .open 'data/data/com.android.vending/databases/localappstate.db'
sqlite> select  package_name, first_download_ms, account, title from appstate where package_name like '%line%'  ;
com.google.android.apps.creamliner|0||
jp.naver.line.android|1580317512405|thisisdfir@gmail.com|LINE: Free Calls & Messages
sqlite>
```

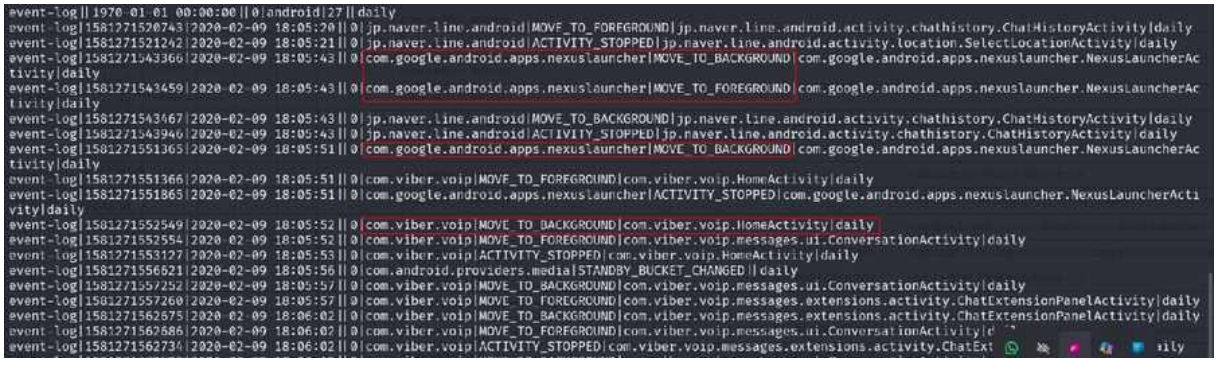
```
sqlite> .open 'usagestats.db'
sqlite> SELECT
  usage_type,
  datetime(lastime/1000, 'UNIXEPOCH') as lasttimeactive,
  timeactive as time_Active_in_msecs,
  timeactive/1000 as timeactive_in_secs,
  package,
CASE types
     WHEN '1' THEN 'MOVE_TO_FOREGROUND'
     WHEN '2' THEN 'MOVE_TO_BACKGROUND'
     WHEN '5' THEN 'CONFIGURATION_CHANGE'
     WHEN '7' THEN 'USER_INTERACTION'
     WHEN '8' THEN 'SHORTCUT_INVOCATION'
     WHEN '23' THEN 'ACTIVITY_STOPPED'
     ELSE types
END types,
  classs,
  source
FROM data
where package = 'jp.naver.line.android'   and usage_type ='event-log'
ORDER BY lasttimeactive ;
```

```
event-log|2020-02-09 17:49:24||0|jp.naver.line.android|MOVE_TO_FOREGROUND|jp.naver.line.android.activity.SplashActivity|daily
event-log|2020-02-09 17:49:24||0|jp.naver.line.android|STANDBY_BUCKET_CHANGED||daily
event-log|2020-02-09 17:49:24||0|jp.naver.line.android|MOVE_TO_BACKGROUND|jp.naver.line.android.activity.SplashActivity|daily
event-log|2020-02-09 17:49:24||0|jp.naver.line.android|MOVE_TO_FOREGROUND|jp.naver.line.android.activity.main.MainActivity|daily
event-log|2020-02-09 17:49:25||0|jp.naver.line.android|ACTIVITY_STOPPED|jp.naver.line.android.activity.SplashActivity|daily
event-log|2020-02-09 17:49:26||0|jp.naver.line.android|MOVE_TO_BACKGROUND|jp.naver.line.android.activity.main.MainActivity|daily
event-log|2020-02-09 17:49:26||0|jp.naver.line.android|MOVE_TO_FOREGROUND|jp.naver.line.android.activity.chathistory.ChatHistoryActivityLaunchActivity|daily
event-log|2020-02-09 17:49:26||0|jp.naver.line.android|MOVE_TO_BACKGROUND|jp.naver.line.android.activity.chathistory.ChatHistoryActivityLaunchActivity|daily
event-log|2020-02-09 17:49:26||0|jp.naver.line.android|MOVE_TO_FOREGROUND|jp.naver.line.android.activity.chathistory.ChatHistoryActivity|daily
event-log|2020-02-09 17:49:26||0|jp.naver.line.android|ACTIVITY_STOPPED|jp.naver.line.android.activity.chathistory.ChatHistoryActivityLaunchActivity|daily
event-log|2020-02-09 17:49:27||0|jp.naver.line.android|ACTIVITY_STOPPED|jp.naver.line.android.activity.main.MainActivity|daily
event-log|2020-02-09 17:49:40||0|jp.naver.line.android|MOVE_TO_BACKGROUND|jp.naver.line.android.activity.chathistory.ChatHistoryActivity|daily
event-log|2020-02-09 17:49:41||0|jp.naver.line.android|ACTIVITY_STOPPED|jp.naver.line.android.activity.chathistory.ChatHistoryActivity|daily
event-log|2020-02-09 18:05:06||0|jp.naver.line.android|MOVE_TO_FOREGROUND|jp.naver.line.android.activity.chathistory.ChatHistoryActivity|daily
event-log|2020-02-09 18:05:12||0|jp.naver.line.android|MOVE_TO_BACKGROUND|jp.naver.line.android.activity.chathistory.ChatHistoryActivity|daily
event-log|2020-02-09 18:05:12||0|jp.naver.line.android|MOVE_TO_FOREGROUND|jp.naver.line.android.activity.location.SelectLocationActivity|daily
event-log|2020-02-09 18:05:13||0|jp.naver.line.android|MOVE_TO_BACKGROUND|jp.naver.line.android.activity.location.SelectLocationActivity|daily
event-log|2020-02-09 18:05:13||0|jp.naver.line.android|ACTIVITY_STOPPED|jp.naver.line.android.activity.chathistory.ChatHistoryActivity|daily
event-log|2020-02-09 18:05:15||0|jp.naver.line.android|MOVE_TO_FOREGROUND|jp.naver.line.android.activity.location.SelectLocationActivity|daily
event-log|2020-02-09 18:05:20||0|jp.naver.line.android|MOVE_TO_BACKGROUND|jp.naver.line.android.activity.location.SelectLocationActivity|daily
event-log|2020-02-09 18:05:20||0|jp.naver.line.android|MOVE_TO_FOREGROUND|jp.naver.line.android.activity.chathistory.ChatHistoryActivity|daily
event-log|2020-02-09 18:05:21||0|jp.naver.line.android|ACTIVITY_STOPPED|jp.naver.line.android.activity.location.SelectLocationActivity|daily
event-log|2020-02-09 18:05:43||0|jp.naver.line.android|MOVE_TO_BACKGROUND|jp.naver.line.android.activity.chathistory.ChatHistoryActivity|daily
event-log|2020-02-09 18:05:43||0|jp.naver.line.android|ACTIVITY_STOPPED|jp.naver.line.android.activity.chathistory.ChatHistoryActivity|daily
event-log|2020-02-09 19:08:27||0|jp.naver.line.android|STANDBY_BUCKET_CHANGED||daily
event-log|2020-02-09 20:08:29||0|jp.naver.line.android|STANDBY_BUCKET_CHANGED||daily
event-log|2020-02-09 21:11:53||0|jp.naver.line.android|STANDBY_BUCKET_CHANGED||daily
```

```
sqlite> SELECT
  usage_type,
  lastime,
  datetime(lastime/1000, 'UNIXEPOCH') as lasttimeactive,
  timeactive as time_Active_in_msecs,
  timeactive/1000 as timeactive_in_secs,
  package,
CASE types
     WHEN '1' THEN 'MOVE_TO_FOREGROUND'
     WHEN '2' THEN 'MOVE_TO_BACKGROUND'
     WHEN '5' THEN 'CONFIGURATION_CHANGE'
     WHEN '7' THEN 'USER_INTERACTION'
     WHEN '8' THEN 'SHORTCUT_INVOCATION'
     WHEN '23' THEN 'ACTIVITY_STOPPED'
     ELSE types
END types,
  classs,
  source
FROM data
where package = 'jp.naver.line.android'   and usage_type ='event-log'
ORDER BY lasttimeactive ;
```



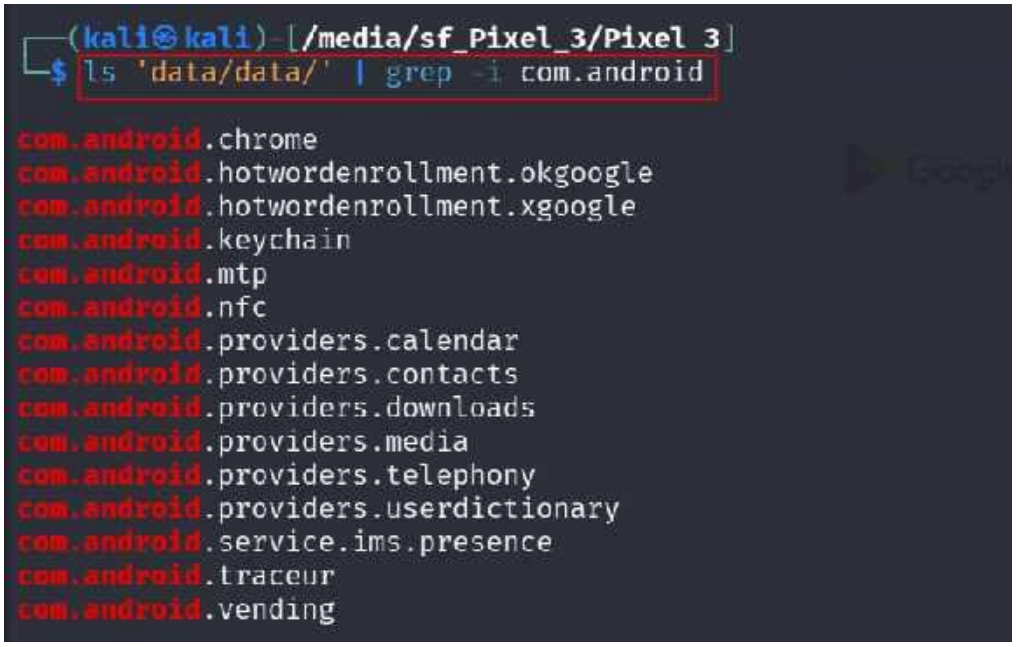

```
sqlite> SELECT
  usage_type,
  lastime,
  datetime(lastime/1000, 'UNIXEPOCH') as lasttimeactive,
  timeactive as time_Active_in_msecs,
  timeactive/1000 as timeactive_in_secs,
  package,
CASE types
     WHEN '1' THEN 'MOVE_TO_FOREGROUND'
     WHEN '2' THEN 'MOVE_TO_BACKGROUND'
     WHEN '5' THEN 'CONFIGURATION_CHANGE'
     WHEN '7' THEN 'USER_INTERACTION'
     WHEN '8' THEN 'SHORTCUT_INVOCATION'
     WHEN '23' THEN 'ACTIVITY_STOPPED'
     ELSE types
END types,
  classs,
  source
FROM data
where usage_type ='event-log' and lastime > 1581271520730
ORDER BY lasttimeactive limit 20;
```



Artefact Findings

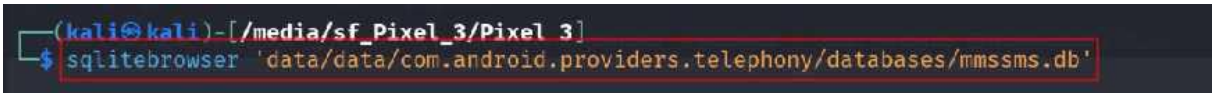
AOSP Communications, Contacts and Calendar

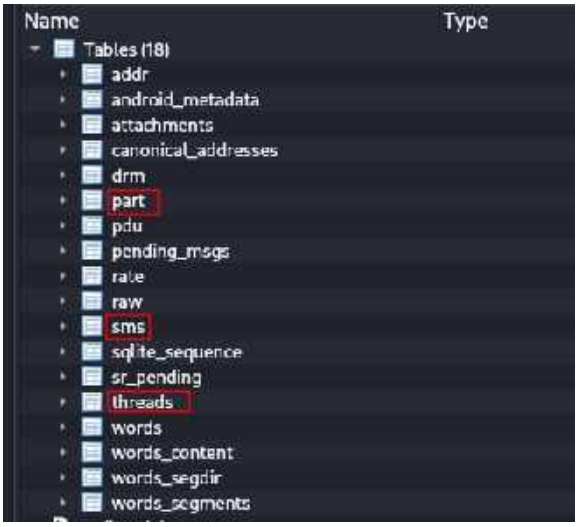
I examined the Telephony Provider (mmssms.db), the Contacts Provider (contacts2.db, calllog.db) and the Calendar Provider (calendar.db). The tables and rows reviewed are shown in the screenshots below. A saved contact name only labels a number. It does not prove who used that number.

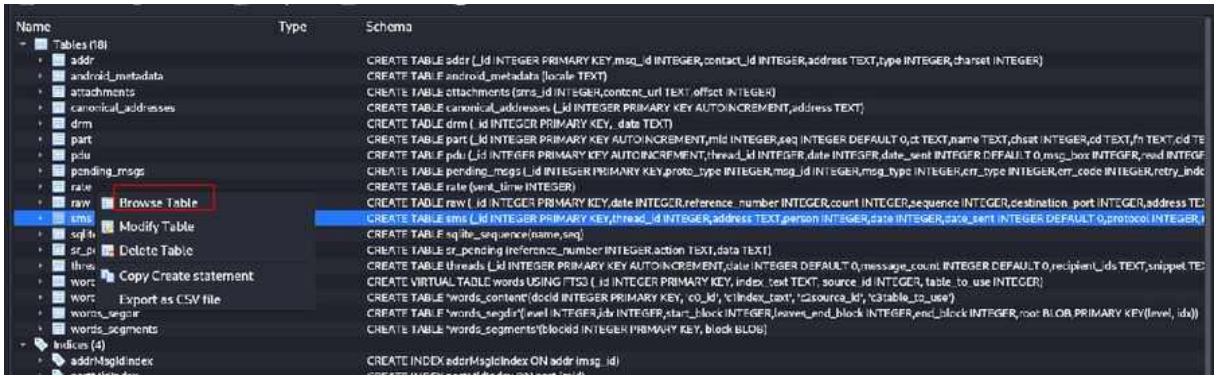


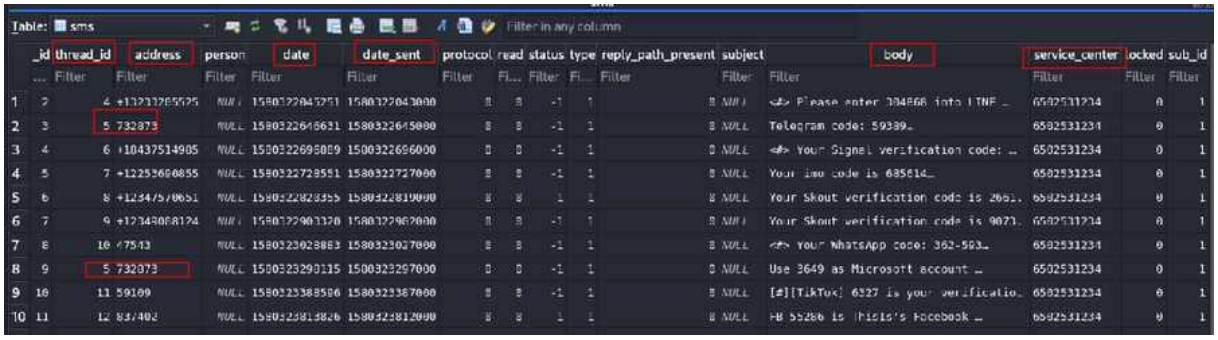













```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d '@1580322045' -u
Wed Jan 29 06:20:45 PM UTC 2020

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

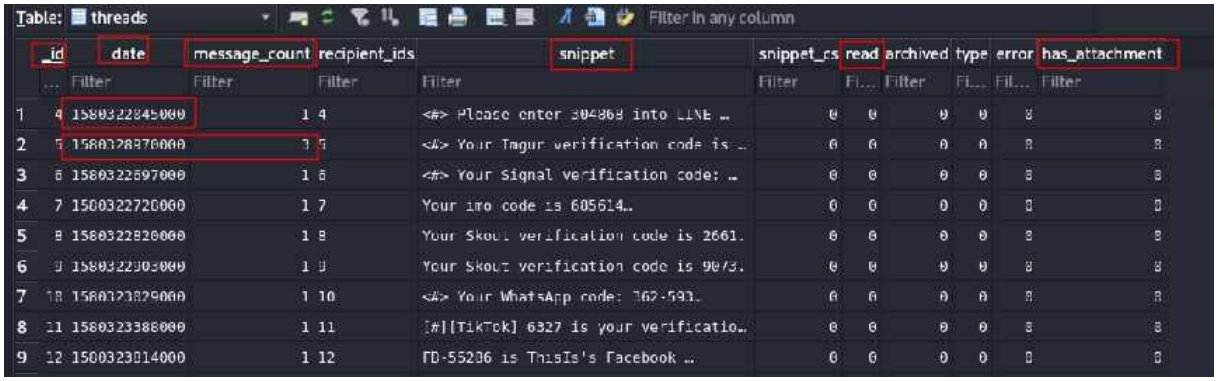


Table: threads

Filter in any column

_id	date	message_count	recipient_ids	snippet	snippet_cs	read	archived	type	error	has_attachment
4	1580322845000	1	4	<#> Please enter 304868 into LINE …	0	0	0	0	0	0
5	1580328870000	1	5	<#> Your Imgur verification code is …	0	0	0	0	0	0
6	1580322697000	1	6	<#> Your Signal verification code: …	0	0	0	0	0	0
7	1580322720000	1	7	Your imo code is 605614..	0	0	0	0	0	0
8	1580322820000	1	8	Your Skout verification code is 2661.	0	0	0	0	0	0
9	1580322902000	1	9	Your Skout verification code is 9073.	0	0	0	0	0	0
10	1580323829000	1	10	<#> Your WhatsApp code: 362-593.	0	0	0	0	0	0
11	1580323388000	1	11	[#][TikTok] 6327 is your verificatio…	0	0	0	0	0	0
12	1580323014000	1	12	FD-55206 is ThisIs's Facebook …	0	0	0	0	0	0
13	1581628344000	8	13	🔔 ThisIs, you have 9 new …	0	0	0	0	0	0
14	1580332533000	1	14	389067 is your verification code for…	0	0	0	0	0	0
15	1580518807000	1	15	Your Wickr Me confirmation code is …	0	1	0	0	0	0
16	1580519353000	1	16	Your Wire code is 172488.…	0	1	0	0	0	0
17	1581605000000	8	17		106	1	0	0	0	0
18	1581692447000	8	18	NULL	0	1	0	0	0	0
19	1581692447000	8	19	NULL	0	1	0	0	0	0
20	1581692457000	8	20	NULL	0	1	0	0	0	0

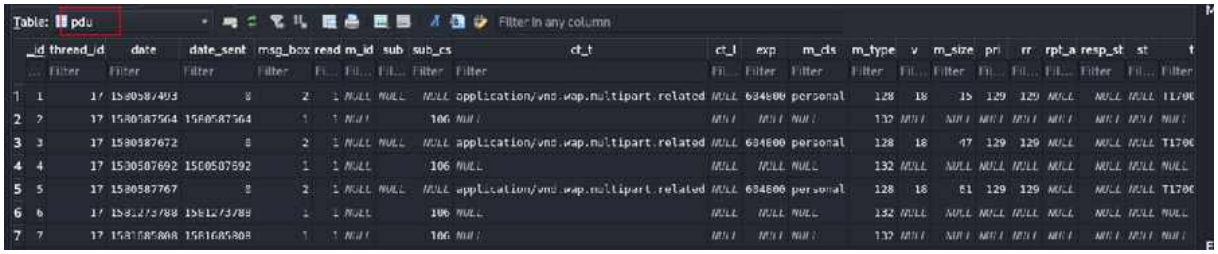


Table: pdu

Filter in any column

_id	thread_id	date	date_sent	msg_box	read	m_id	sub	sub_cs	ct_t	ct_l	exp	m_cls	m_type	v	m_size	pri	rr	rpt_a	resp_st	st	t
1	1	17	1580587493	8	2	1	NULL	NULL	application/vnd.wap.multipart.related	NULL	604800	personal	128	18	15	129	129	NULL	NULL	NULL	T170
2	2	17	1580587564	1580587564	1	1	NULL	106	NULL	NULL	NULL	NULL	132	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
3	3	17	1580587672	8	2	1	NULL	NULL	application/vnd.wap.multipart.related	NULL	604800	personal	128	18	47	129	129	NULL	NULL	NULL	T170
4	4	17	1580587692	1580587592	1	1	NULL	106	NULL	NULL	NULL	NULL	132	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
5	5	17	1580587767	8	2	1	NULL	NULL	application/vnd.wap.multipart.related	NULL	604800	personal	128	18	61	129	129	NULL	NULL	NULL	T170
6	6	17	1581273788	1581273788	1	1	NULL	106	NULL	NULL	NULL	NULL	132	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
7	7	17	1581605808	1581605808	1	1	NULL	106	NULL	NULL	NULL	NULL	132	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

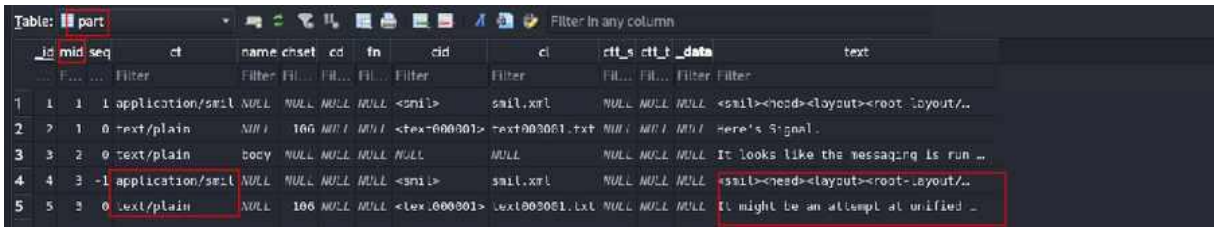


Table: part

Filter in any column

_id	mid	seq	ct	name	chset	cd	fn	cid	ci	ctt_s	ctt_t	_data	text
1	1	1	application/smil	NULL	NULL	NULL	NULL	<smil>	smil.xml	NULL	NULL	NULL	<smil><head><layout><root-layout/..
2	2	1	0 text/plain	NULL	106	NULL	NULL	<text000001>	text000001.txt	NULL	NULL	NULL	Here's Signal.
3	3	2	0 text/plain	body	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	It looks like the messaging is run …
4	4	3	-1 application/smil	NULL	NULL	NULL	NULL	<smil>	smil.xml	NULL	NULL	NULL	<smil><head><layout><root-layout/..
5	5	3	0 text/plain	NULL	106	NULL	NULL	<text000001>	text000001.txt	NULL	NULL	NULL	It might be an attempt at unified …



Table: pdu

Filter in any column

_id	thread_id	date	date_sent	msg_box	read	m_id	sub	sub_cs	ct_t	ct_l	exp	m_cls	m_type	v	m_size	pri	rr	rpt_a	resp_st	st	t
1	1	17	1580587493	8	2	1	NULL	NULL	application/vnd.wap.multipart.related	NULL	604800	personal	128	18	15	129	129	NULL	NULL	NULL	T170



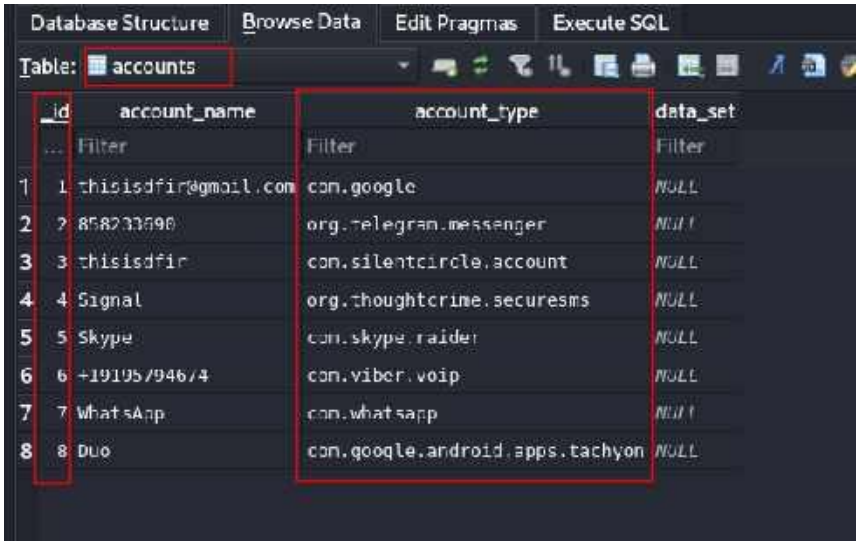
Table: part

Filter in any column

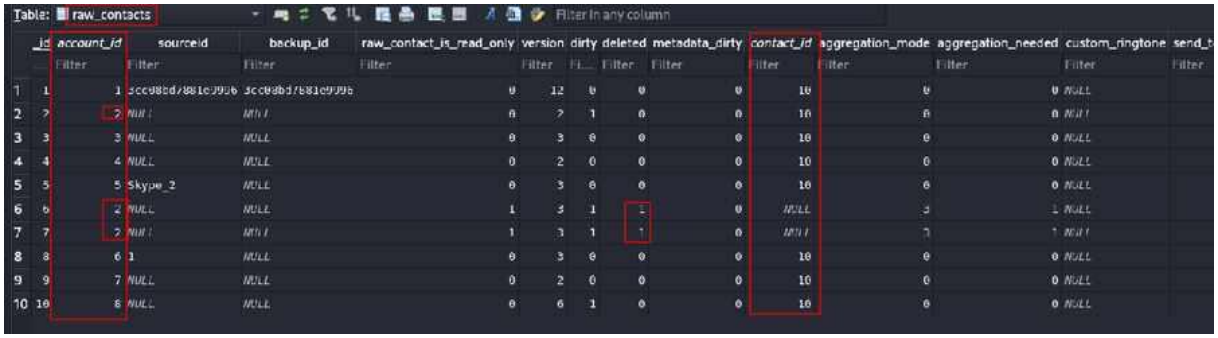
_id	mid	seq	ct	name	chset	cd	fn	cid	ci	ctt_s	ctt_t	_data	text
1	1	1	application/smil	NULL	NULL	NULL	NULL	<smil>	smil.xml	NULL	NULL	NULL	<smil><head><layout><root-layout/..
2	2	1	0 text/plain	NULL	106	NULL	NULL	<text000001>	text000001.txt	NULL	NULL	NULL	Here's Signal.

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/data/data]
└─$ tree -L 2 com.android.providers.contacts
com.android.providers.contacts
├── app_voicemail-data
│   ├── voicemail4257613984156813452
│   └── voicemail6279788559607275412
├── databases
│   ├── calllog.db
│   ├── contacts2.db
│   └── profile.db
└── shared_prefs
    ├── com.android.providers.contacts_preferences.xml
    └── ContactsUpgradeReceiver.xml
```

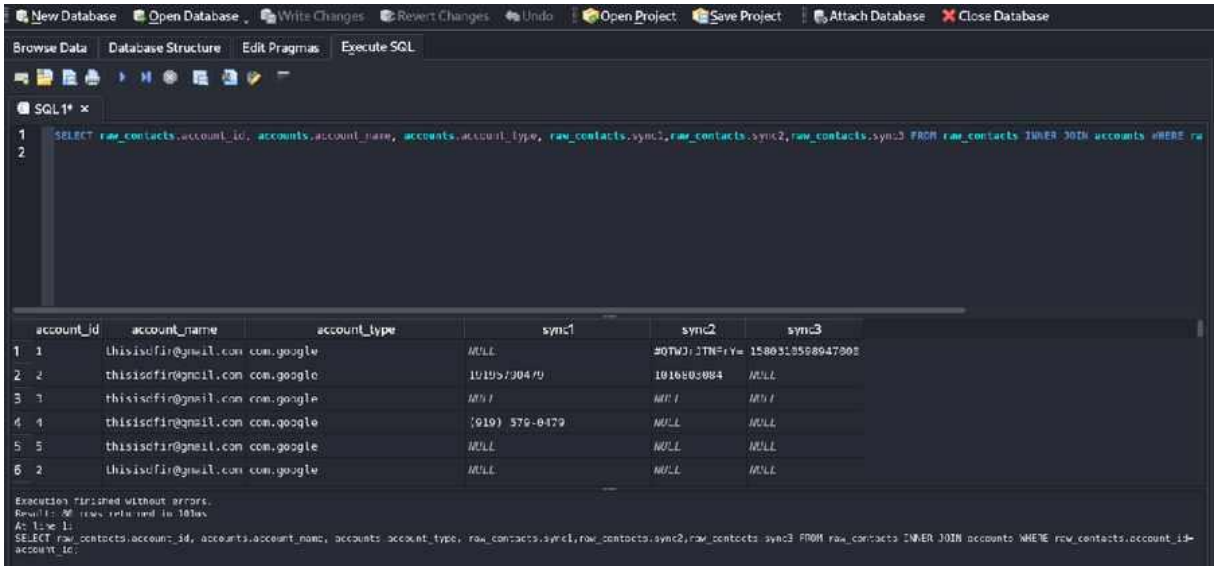
```
┌──(kali㉿kali)-[/media/…/data/data/com.android.providers.contacts/databases]
└─$ sqlite3 contacts2.db
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .tables
_sync_state               phone_lookup              view_data_usage
_sync_state_metadata      photo_files               view_entities
accounts                  pre_authorized_uris       view_groups
agg_exceptions            presence                  view_metadata_sync
agg_presence              properties                view_metadata_sync_state
android_metadata          raw_contacts              view_raw_contacts
contacts                  search_index              view_raw_entities
data                      search_index_content      view_stream_items
data_usage_stat           search_index_docsize      view_v1_contact_methods
default_directory         search_index_segdir       view_v1_extensions
deleted_contacts          search_index_segments     view_v1_group_membership
directories               search_index_stat         view_v1_groups
groups                    settings                  view_v1_organizations
metadata_sync             status_updates            view_v1_people
metadata_sync_state       stream_item_photos        view_v1_phones
mimetypes                 stream_items              view_v1_photos
name_lookup               v1_settings               visible_contacts
nickname_lookup           view_contacts
packages                  view_data
sqlite>
```



_id	account_name	account_type	data_set
	Filter	Filter	Filter
1	1	thisisdfir@gmail.com	com.google	NULL
2	2	8582333698	org.telegram.messenger	NULL
3	3	thisisdfir	com.silentcircle.account	NULL
4	4	Signal	org.thoughtcrime.securesms	NULL
5	5	Skype	com.skype.raider	NULL
6	6	+19195794674	com.viber.voip	NULL
7	7	WhatsApp	com.whatsapp	NULL
8	8	Duo	com.google.android.apps.tachyon	NULL



_id	account_id	sourceid	backup_id	raw_contact_is_read_only	version	dirty	deleted	metadata_dirty	contact_id	aggregation_mode	aggregation_needed	custom_ringtone	send_t
	Filter	Filter	Filter	Filter	Filter	Fi...	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	1	3cc08bd/881e9996	3cc08bd/881e9996	0	12	0	0	0	10	0	0	NULL	
2	2	2	NULL	NULL	0	2	1	0	0	10	0	0	NULL	
3	3	3	NULL	NULL	0	3	0	0	0	10	0	0	NULL	
4	4	4	NULL	NULL	0	2	0	0	0	10	0	0	NULL	
5	5	5	Skype_2	NULL	0	3	0	0	0	10	0	0	NULL	
6	6	2	NULL	NULL	1	3	1	1	0	NULL	3	1	NULL	
7	7	2	NULL	NULL	1	3	1	1	0	NULL	3	1	NULL	
8	8	6	1	NULL	0	3	0	0	0	10	0	0	NULL	
9	9	7	NULL	NULL	0	2	0	0	0	10	0	0	NULL	
10	10	8	NULL	NULL	0	6	1	0	0	10	0	0	NULL	



account_id	account_name	account_type	sync1	sync2	sync3
1	1	thisisdfir@gmail.com	com.google	NULL	#QTWJ:ITN=rY=	1580319508947008
2	2	thisisdfir@gmail.com	com.google	19195794479	1016803084	NULL
3	3	thisisdfir@gmail.com	com.google	NULL	NULL	NULL
4	4	thisisdfir@gmail.com	com.google	(919) 579-0479	NULL	NULL
5	5	thisisdfir@gmail.com	com.google	NULL	NULL	NULL
6	2	thisisdfir@gmail.com	com.google	NULL	NULL	NULL

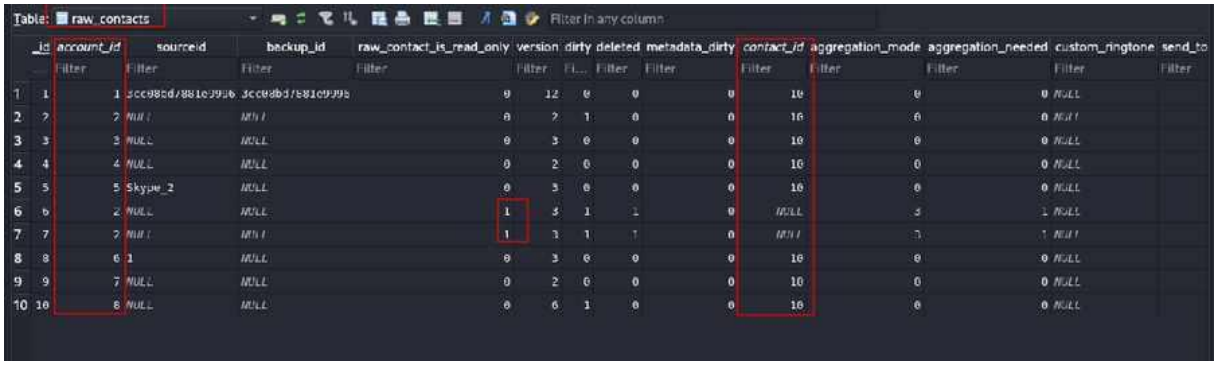


Table: raw_contacts

Filter in any column

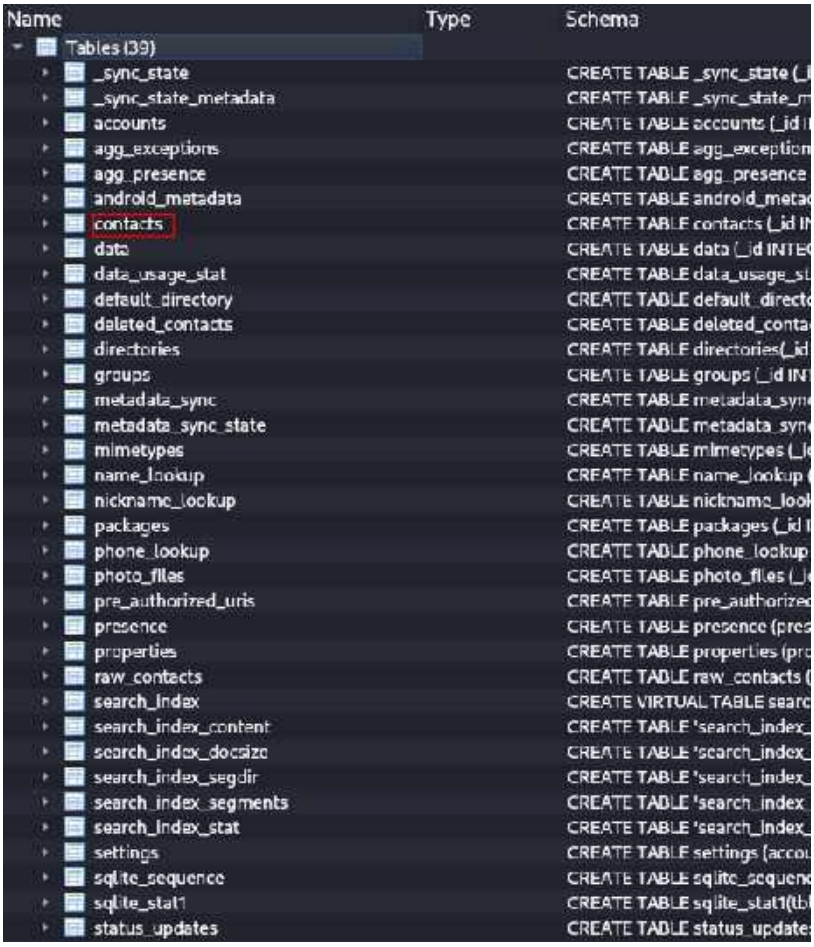
	_id	account_id	sourceid	backup_id	raw_contact_is_read_only	version	dirty	deleted	metadata_dirty	contact_id	aggregation_mode	aggregation_needed	custom_ringtone	send_to
	Filter	Filter	Filter	Filter	Filter	Filter	Fi...	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	1	3cc988d/8810996	3cc98d/8810996		9	12	0	0	0	10	0	0	NULL	
2	2	2	NULL	NULL		0	2	1	0	0	10	0	0	NULL	
3	3	3	NULL	NULL		0	3	0	0	0	10	0	0	NULL	
4	4	4	NULL	NULL		0	2	0	0	0	10	0	0	NULL	
5	5	5	Skype_2	NULL		0	3	0	0	0	10	0	0	NULL	
6	6	2	NULL	NULL		1	3	1	1	0	NULL	3	1	NULL	
7	7	2	NULL	NULL		1	1	1	1	0	NULL	1	1	NULL	
8	8	6	1	NULL		0	3	0	0	0	10	0	0	NULL	
9	9	7	NULL	NULL		0	2	0	0	0	10	0	0	NULL	
10	10	8	NULL	NULL		0	6	1	0	0	10	0	0	NULL	

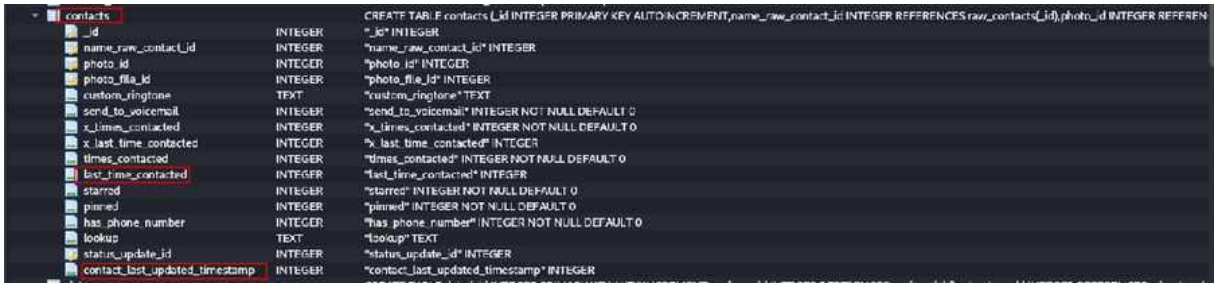


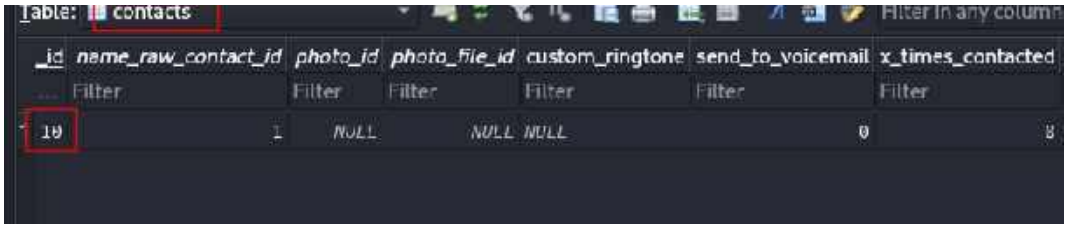
Name	Type	Schema
Tables (39)		
_sync_state		CREATE TABLE _sync_state (_id I
_sync_state_metadata		CREATE TABLE _sync_state_meta
accounts		CREATE TABLE accounts (_id INT
agg_exceptions		CREATE TABLE agg_exceptions (
agg_presence		CREATE TABLE agg_presence (pr
android_metadata		CREATE TABLE android_metadat
contacts		CREATE TABLE contacts (_id INTE
data		CREATE TABLE data (_id INTEGE
data_usage_stat		CREATE TABLE data_usage_stat(
default_directory		CREATE TABLE default_directory
deleted_contacts		CREATE TABLE deleted_contacts
directories		CREATE TABLE directories(_id IN
groups		CREATE TABLE groups (_id INTEG
metadata_sync		CREATE TABLE metadata_sync (_
metadata_sync_state		CREATE TABLE metadata_sync_s
mimetypes		CREATE TABLE mimetypes (_id IN
name_lookup		CREATE TABLE name_lookup (da
nickname_lookup		CREATE TABLE nickname_lookup
packages		CREATE TABLE packages (_id INT
phone_lookup		CREATE TABLE phone_lookup (de
photo_files		CREATE TABLE photo_files (_id IN
pre_authorized_uris		CREATE TABLE pre_authorized_u
presence		CREATE TABLE presence (presen
properties		CREATE TABLE properties (prope
raw_contacts		CREATE TABLE raw_contacts (_id
search_index		CREATE VIRTUAL TABLE search_i
search_index_content		CREATE TABLE 'search_index_co
search_index_docsize		CREATE TABLE 'search_index_do
search_index_segdir		CREATE TABLE 'search_index_se
search_index_segments		CREATE TABLE 'search_index_se
search_index_stat		CREATE TABLE 'search_index_sta
settings		CREATE TABLE settings (account_
sqlite_sequence		CREATE TABLE sqlite_sequence(n
sqlite_stat1		CREATE TABLE sqlite_stat1(tbl,id
status_updates		CREATE TABLE status_updates (s

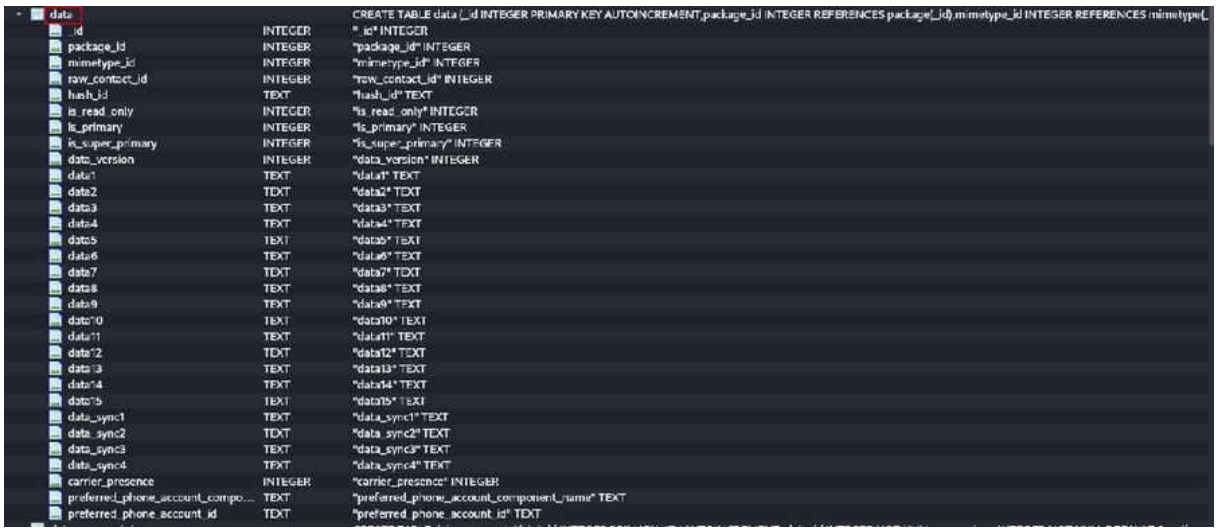
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/data/data]
└─$ sqlite3 'com.android.providers.contacts/databases/contacts2.db'
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> pragma table_info(contacts);
0|_id|INTEGER|0||1
1|name_raw_contact_id|INTEGER|0||0
2|photo_id|INTEGER|0||0
3|photo_file_id|INTEGER|0||0
4|custom_ringtone|TEXT|0||0
5|send_to_voicemail|INTEGER|1|0|0
6|x_times_contacted|INTEGER|1|0|0
7|x_last_time_contacted|INTEGER|0||0
8|times_contacted|INTEGER|1|0|0
9|last_time_contacted|INTEGER|0||0
10|starred|INTEGER|1|0|0
11|pinned|INTEGER|1|0|0
12|has_phone_number|INTEGER|1|0|0
13|lookup|TEXT|0||0
14|status_update_id|INTEGER|0||0
15|contact_last_updated_timestamp|INTEGER|0||0
sqlite>
```

```
sqlite> SELECT * FROM contacts;
10|1|||0|0|0|0||0|0|1|329813cc08bd7801e9996.3750r2-3C464E38383A2E3E422A44.386r3-3C464E38383A2E3E422A44.2399r4-3C464E38383A2E3E422A44.14501Skype 2.102811.3789r9-3C464E38383A2E3E422A44.847r10-3C464E38383A2E3E422A44||1581684660833
sqlite>
```


Name
Type
Schema
Tables (39)
_sync_state
_sync_state_metadata
accounts
agg_exceptions
agg_presence
android_metadata
contacts
data
data_usage_stat
default_directory
deleted_contacts
directories
groups
metadata_sync
metadata_sync_state
mimetypes
name_lookup
nickname_lookup
packages
phone_lookup
photo_files
pre_authorized_uris
presence
properties
raw_contacts
search_index
search_index_content
search_index_docsize
search_index_segdir
search_index_segments
search_index_stat
settings
sqlite_sequence
sqlite_stat1
status_updates


contacts
CREATE TABLE contacts (_id INTEGER PRIMARY KEY AUTOINCREMENT,name_raw_contact_id INTEGER REFERENCES raw_contacts(_id),photo_id INTEGER REFEREN
_id INTEGER "_id" INTEGER
name_raw_contact_id INTEGER "name_raw_contact_id" INTEGER
photo_id INTEGER "photo_id" INTEGER
photo_file_id INTEGER "photo_file_id" INTEGER
custom_ringtone TEXT "custom_ringtone" TEXT
send_to_voicemail INTEGER "send_to_voicemail" INTEGER NOT NULL DEFAULT 0
x_times_contacted INTEGER "x_times_contacted" INTEGER NOT NULL DEFAULT 0
x_last_time_contacted INTEGER "x_last_time_contacted" INTEGER
times_contacted INTEGER "times_contacted" INTEGER NOT NULL DEFAULT 0
last_time_contacted INTEGER "last_time_contacted" INTEGER
starred INTEGER "starred" INTEGER NOT NULL DEFAULT 0
pinned INTEGER "pinned" INTEGER NOT NULL DEFAULT 0
has_phone_number INTEGER "has_phone_number" INTEGER NOT NULL DEFAULT 0
lookup TEXT "lookup" TEXT
status_update_id INTEGER "status_update_id" INTEGER
contact_last_updated_timestamp INTEGER "contact_last_updated_timestamp" INTEGER


Table: contacts
Filter in any column
_id
name_raw_contact_id
photo_id
photo_file_id
custom_ringtone
send_to_voicemail
x_times_contacted
Filter
10
1
NULL
NULL
NULL
0
8



data		CREATE TABLE data (_id INTEGER PRIMARY KEY AUTOINCREMENT,package_id INTEGER REFERENCES package(_id),mimetype_id INTEGER REFERENCES mimetype(_
_id	INTEGER	"_id" INTEGER
package_id	INTEGER	"package_id" INTEGER
mimetype_id	INTEGER	"mimetype_id" INTEGER
raw_contact_id	INTEGER	"raw_contact_id" INTEGER
hash_id	TEXT	"hash_id" TEXT
is_read_only	INTEGER	"is_read_only" INTEGER
is_primary	INTEGER	"is_primary" INTEGER
is_super_primary	INTEGER	"is_super_primary" INTEGER
data_version	INTEGER	"data_version" INTEGER
data1	TEXT	"data1" TEXT
data2	TEXT	"data2" TEXT
data3	TEXT	"data3" TEXT
data4	TEXT	"data4" TEXT
data5	TEXT	"data5" TEXT
data6	TEXT	"data6" TEXT
data7	TEXT	"data7" TEXT
data8	TEXT	"data8" TEXT
data9	TEXT	"data9" TEXT
data10	TEXT	"data10" TEXT
data11	TEXT	"data11" TEXT
data12	TEXT	"data12" TEXT
data13	TEXT	"data13" TEXT
data14	TEXT	"data14" TEXT
data15	TEXT	"data15" TEXT
data_sync1	TEXT	"data_sync1" TEXT
data_sync2	TEXT	"data_sync2" TEXT
data_sync3	TEXT	"data_sync3" TEXT
data_sync4	TEXT	"data_sync4" TEXT
carrier_presence	INTEGER	"carrier_presence" INTEGER
preferred_phone_account_compo...	TEXT	"preferred_phone_account_component_name" TEXT
preferred_phone_account_id	TEXT	"preferred_phone_account_id" TEXT

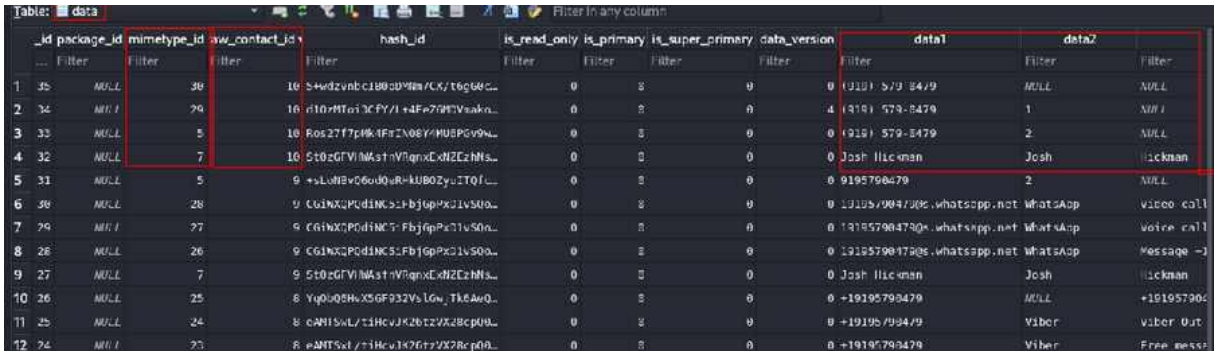
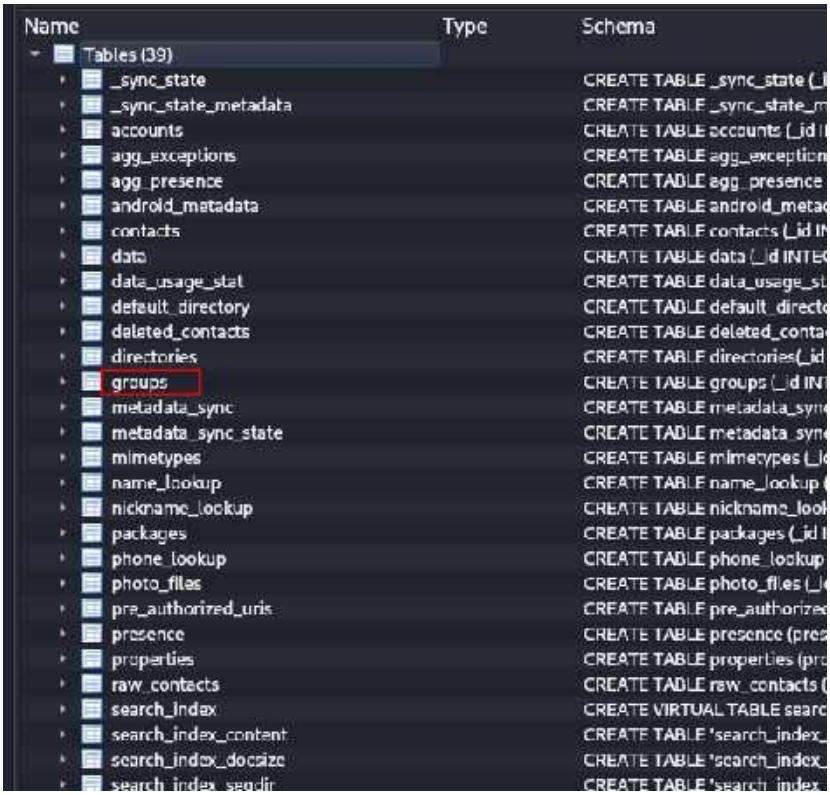


Table: data

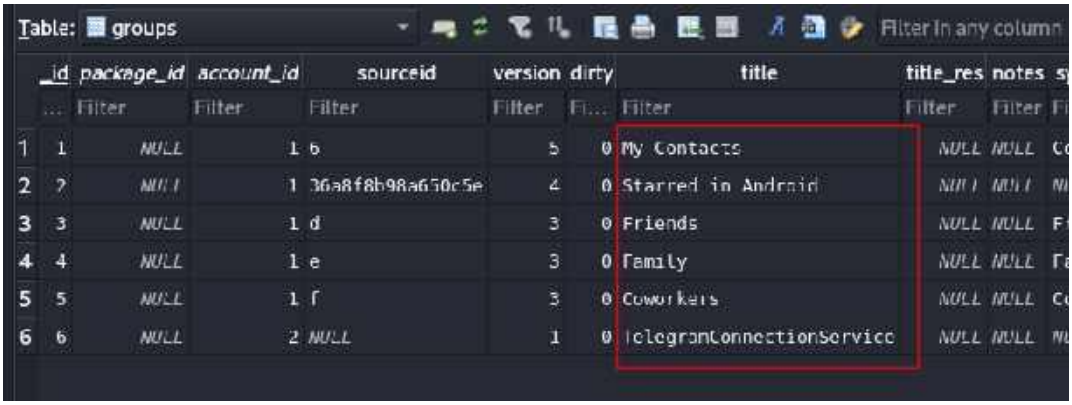
Filter in any column

	_id	package_id	mimetype_id	raw_contact_id	hash_id	is_read_only	is_primary	is_super_primary	data_version	data1	data2	
	...	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	35	NULL	30	10	S+wdzvnbc1B0bDYWm/CX/t6g6Ec...	0	0	0	0	(919) 579-8479	NULL	NULL
2	34	NULL	29	10	d10rMTOi3CfY/1+4Fe7GMIVsako...	0	0	0	4	(919) 579-8479	1	NULL
3	33	NULL	5	10	Ros27f7pMk4FrIN08Y/MU8PGv9w...	0	0	0	0	(919) 579-8479	2	NULL
4	32	NULL	7	10	St0zGfVlNAstnVRqnxExNZEzhNs...	0	0	0	0	Josh Hickman	Josh	Hickman
5	31	NULL	5	9	+sLuNBvQ6od0uR+kUBOZyuITQf...	0	0	0	0	9195790479	2	NULL
6	30	NULL	28	9	CGiNXQPQdiNC5:FbjGpPxJ1vSOo...	0	0	0	0	19195790479@s.whatsapp.net	WhatsApp	video call
7	29	NULL	27	9	CGiNXQPQdiNC5:FbjGpPxJ1vSOo...	0	0	0	0	19195790479@s.whatsapp.net	WhatsApp	voice call
8	28	NULL	26	9	CGiNXQPQdiNC5:FbjGpPxJ1vSOo...	0	0	0	0	19195790479@s.whatsapp.net	WhatsApp	Message -1
9	27	NULL	7	9	St0zGfVlNAstnVRqnxExNZEzhNs...	0	0	0	0	Josh Hickman	Josh	Hickman
10	26	NULL	25	8	YqQbQ5HwX5GF932VslGw/Tk6AwQ...	0	0	0	0	+19195790479	NULL	+19195790
11	25	NULL	24	8	cAMI5wL/tiHcvJK2btzVX2BcpU0...	0	0	0	0	+19195790479	Viber	Viber Out
12	24	NULL	23	8	cAMI5wL/tiHcvJK2GtzVX2RcpU0...	0	0	0	0	+19195790479	Viber	Free mess


Table: mimetypes
_id
mimetype
Filter
1 1 vnd.android.cursor.item/email_v2
2 2 vnd.android.cursor.item/im
3 3 vnd.android.cursor.item/nickname
4 4 vnd.android.cursor.item/organization
5 5 vnd.android.cursor.item/phone_v2
6 6 vnd.android.cursor.item/sip_address
7 7 vnd.android.cursor.item/name
8 8 vnd.android.cursor.item/postal-…
9 9 vnd.android.cursor.item/identity
10 10 vnd.android.cursor.item/photo
11 11 vnd.android.cursor.item/…
12 12 vnd.android.cursor.item/note
13 13 vnd.android.cursor.item/contact_event
14 14 vnd.android.cursor.item/website
15 15 vnd.android.cursor.item/relation
16 16 vnd.com.google.cursor.item/…
17 17 vnd.android.cursor.item/…
18 18 vnd.android.cursor.item/…
19 19 vnd.android.cursor.item/…
20 20 vnd.android.cursor.item/…
21 21 vnd.android.cursor.item/…
22 22 vnd.android.cursor.item/…
1 - 23 of 30


Name
Type
Schema
Tables (39)
_sync_state
_sync_state_metadata
accounts
agg_exceptions
agg_presence
android_metadata
contacts
data
data_usage_stat
default_directory
deleted_contacts
directories
groups
metadata_sync
metadata_sync_state
mimetypes
name_lookup
nickname_lookup
packages
phone_lookup
photo_files
pre_authorized_uris
presence
properties
raw_contacts
search_index
search_index_content
search_index_docsize
search_index_segdir


groups
CREATE TABLE groups (_id INTEGER PRIMARY KEY AUTOINCREMENT,package_id INTEGER REFERENCES package(_id),account_id INTEGER REFERENCES accounts(_i
_id INTEGER "_id" INTEGER
package_id INTEGER "package_id" INTEGER
account_id INTEGER "account_id" INTEGER
sourceid TEXT "sourceid" TEXT
version INTEGER "version" INTEGER NOT NULL DEFAULT 1
dirty INTEGER "dirty" INTEGER NOT NULL DEFAULT 0
title TEXT "title" TEXT
title_res INTEGER "title_res" INTEGER
notes TEXT "notes" TEXT
system_id TEXT "system_id" TEXT
deleted INTEGER "deleted" INTEGER NOT NULL DEFAULT 0
group_visible INTEGER "group_visible" INTEGER NOT NULL DEFAULT 0
should_sync INTEGER "should_sync" INTEGER NOT NULL DEFAULT 1
auto_add INTEGER "auto_add" INTEGER NOT NULL DEFAULT 0
favorites INTEGER "favorites" INTEGER NOT NULL DEFAULT 0
group_is_read_only INTEGER "group_is_read_only" INTEGER NOT NULL DEFAULT 0
sync1 TEXT "sync1" TEXT
sync2 TEXT "sync2" TEXT
sync3 TEXT "sync3" TEXT
sync4 TEXT "sync4" TEXT

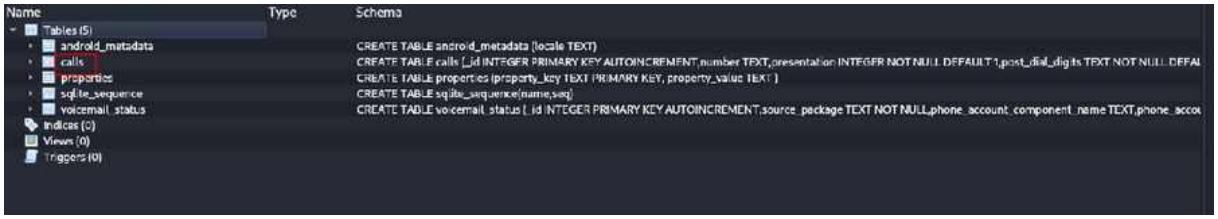

Table: groups
Filter in any column
_id package_id account_id sourceid version dirty title title_res notes
1 1 NULL 1 6 5 0 My Contacts NULL NULL
2 2 NULL 1 36a8f8b98a650c5e 4 0 Starred in Android NULL NULL
3 3 NULL 1 d 3 0 Friends NULL NULL
4 4 NULL 1 e 3 0 Family NULL NULL
5 5 NULL 1 f 3 0 Coworkers NULL NULL
6 6 NULL 2 NULL 1 0 TelegramConnectionService NULL NULL

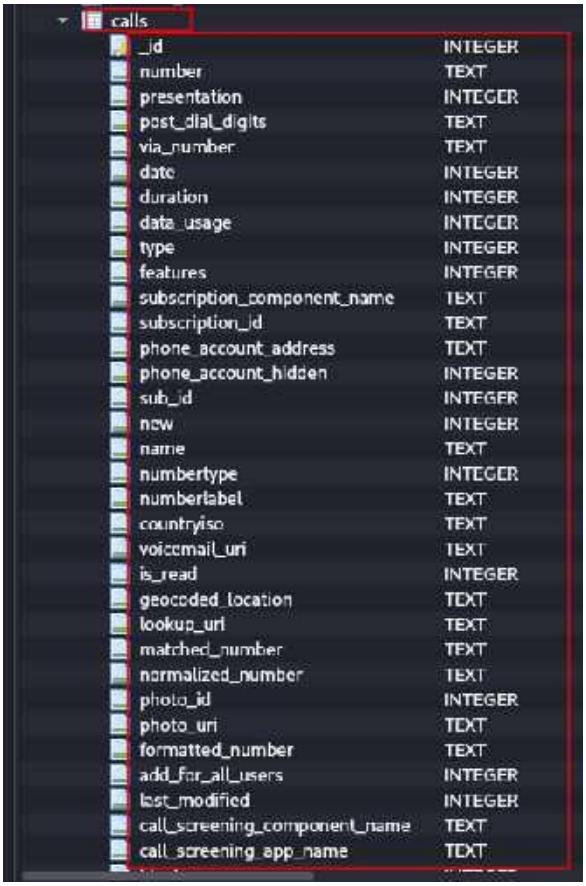
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/data/data]
└─$ tree 'com.android.providers.contacts/databases'
com.android.providers.contacts/databases
├── calllog.db
├── contacts2.db
├── contacts2.db-shm
└── profile.db

1 directory, 4 files

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/data/data]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/data/data]
└─$ sqlitebrowser 'com.android.providers.contacts/databases/calllog.db'
```





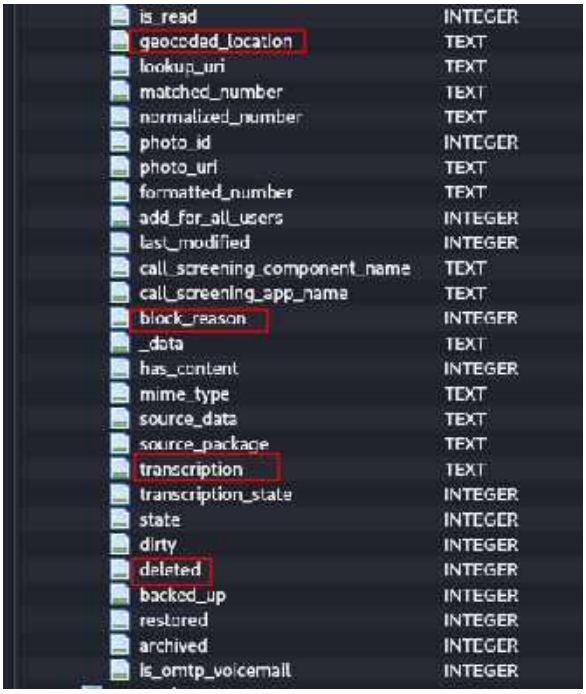


Table: calls

	_id	number	presentation	post_dial_digits	via_number	date	duration	data_usage	type	features	subscription_component_name	subscription_id	phone_account_
1	1		2			1580483796807	0	NULL	3	0	com.android.phone/...	8901260971148578693F	19195794674
2	2		2			1580484070810	0	NULL	3	0	com.android.phone/...	8901260971148578693F	19195794674
3	3		2			1580562115252	0	NULL	3	0	com.android.phone/...	8901260971148578693F	19195794674
4	4		2			1580565200414	0	NULL	3	0	com.android.phone/...	8901260971148578693F	19195794674
5	5		2			1580758342425	0	NULL	3	0	com.android.phone/...	8901260971148578693F	19195794674
6	6	+33269243087	1			1581152896305	0	NULL	3	0	com.android.phone/...	8901260971148578693F	19195794674
7	7	+19195121837	1			1581521849339	0	NULL	3	0	com.android.phone/...	8901233004056883642	9832684955
8	8	+19195121837	1			1581521901008	3	NULL	4	0	NULL	NULL	NULL
9	9	+15755863247	1			1581555164008	0	NULL	3	0	com.android.phone/...	8901233004056883642	9832684955
10	10	+15755863247	1			1581555213008	3	NULL	4	0	NULL	NULL	NULL
11	11	3195794674	1			1581643741051	11	NULL	2	0	com.android.phone/...	8901233004056883642	9832684955

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/data/data]
└─$ date -d '@1581555164' -u
Thu Feb 13 12:52:44 AM UTC 2020
```

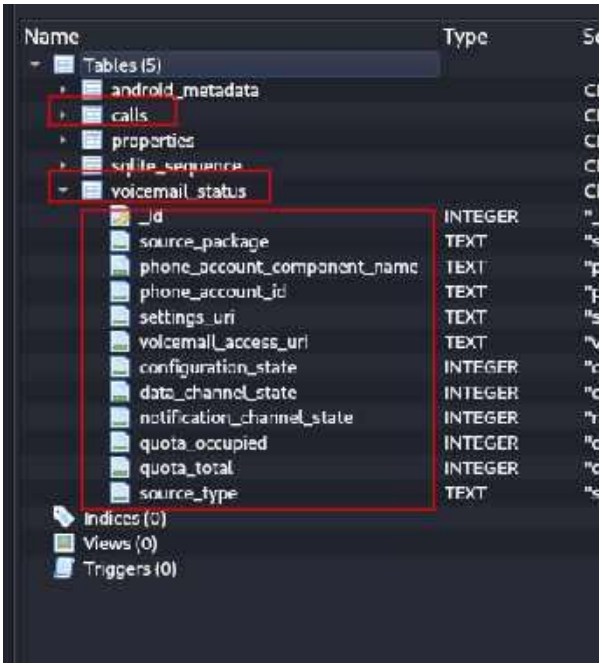
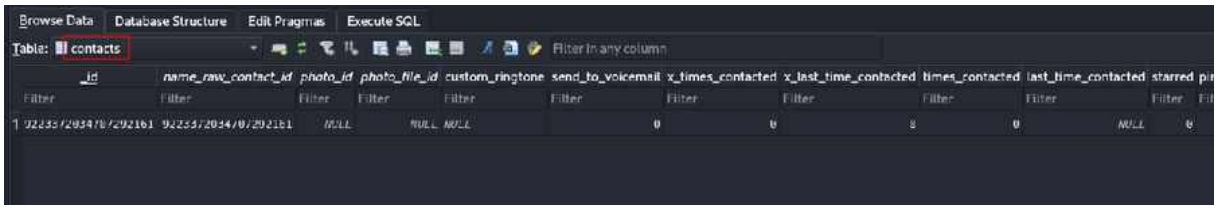
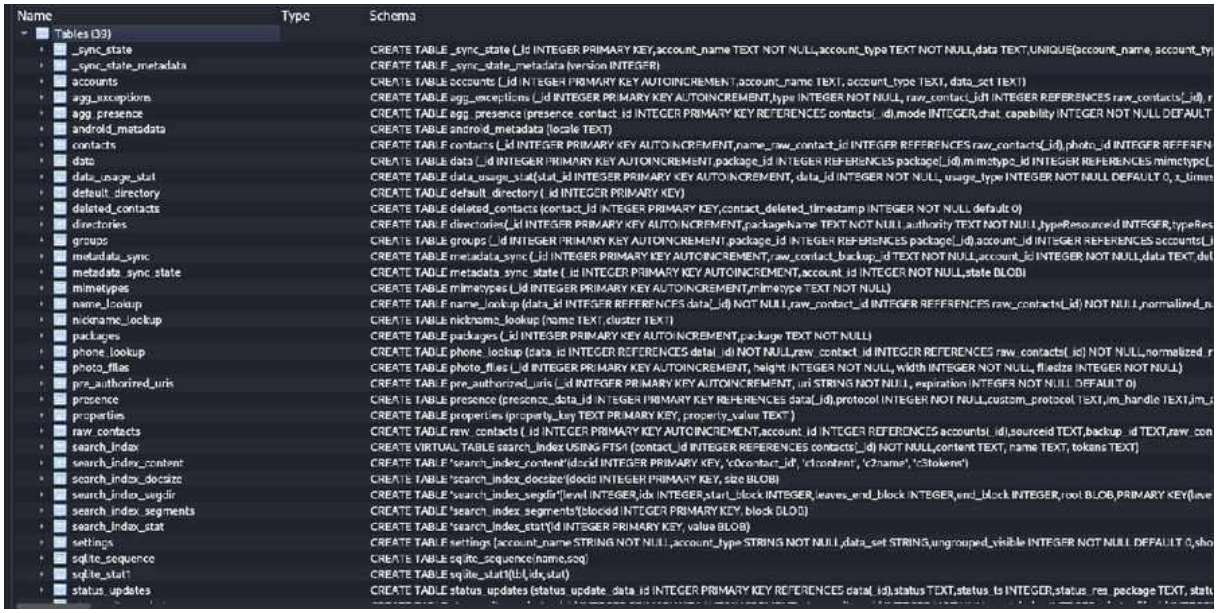

Name
Type
Tables (5)
android_metadata
calls
properties
voicemail_status
_id
source_package
phone_account_component_name
phone_account_id
settings_uri
voicemail_access_uri
configuration_state
data_channel_state
notification_channel_state
quota_occupied
quota_total
source_type
INTEGER
TEXT
TEXT
TEXT
TEXT
TEXT
INTEGER
INTEGER
INTEGER
INTEGER
INTEGER
TEXT
Indices (0)
Views (0)
Triggers (0)


Table: voicemail_status
Filter in any column
_id
source_package
phone_account_component_name
phone_account_id
settings_uri
voicemail_access_uri
configuration_state
data_channel_state
notification_chann
1 com.google.android.dialer com.android.phone/... NULL NULL 1 1
2 com.google.android.apps.tycho NULL NULL NULL 0 0

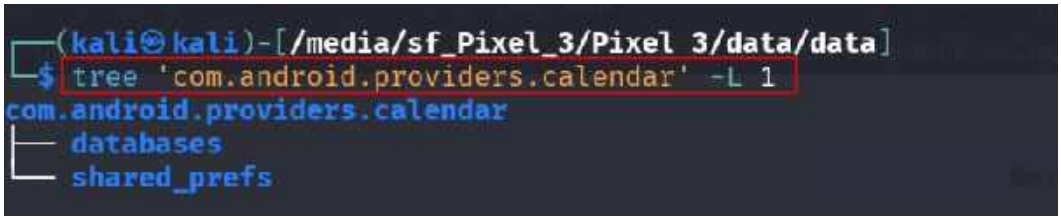

(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/data/data]
$ tree 'com.android.providers.contacts/databases'
com.android.providers.contacts/databases
calllog.db
contacts2.db
contacts2.db-shm
profile.db


(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/data/data]
$ sqlitebrowser 'com.android.providers.contacts/databases/profile.db'

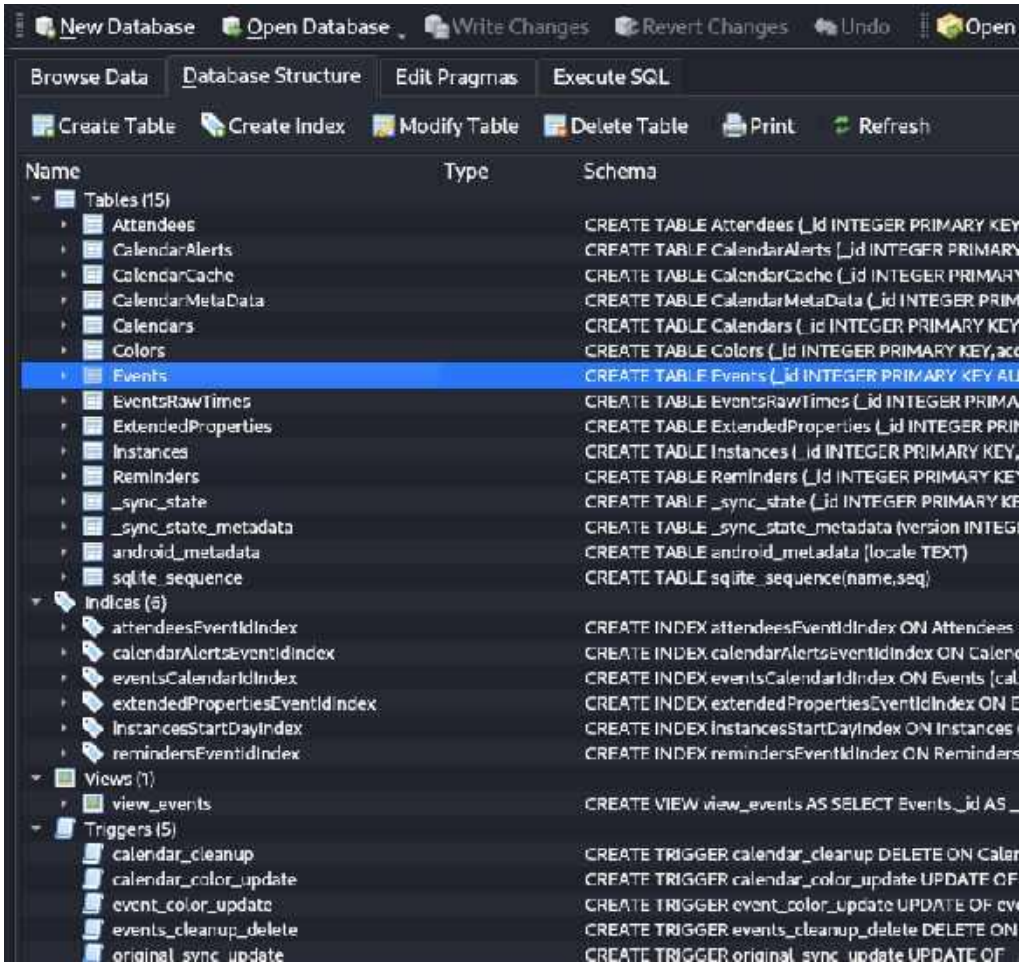

Browse Data
Database Structure
Edit Pragmas
Execute SQL
Table: contacts
Filter in any column
_id
name_raw_contact_id
photo_id
photo_file_id
custom_ringtone
send_to_voicemail
x_times_contacted
x_last_time_contacted
times_contacted
last_time_contacted
starred
1 922337203478... 922337203478... NULL NULL NULL 0 0 8 0 NULL 0


Name
Type
Schema
Tables (33)
_sync_state
_sync_state_metadata
accounts
agg_exceptions
agg_presence
android_metadata
contacts
data
data_usage_stat
default_directory
deleted_contacts
directories
groups
metadata_sync
metadata_sync_state
mimetypes
name_lookup
nickname_lookup
packages
phone_lookup
photo_files
pre_authorized_uris
presence
properties
raw_contacts
search_index
search_index_content
search_index_docsize
search_index_segdir
search_index_segments
search_index_stat
settings
sqlite_sequence
sqlite_stat1
status_updates


accounts
_id
INTEGER
account_name
TEXT
account_type
TEXT
data_set
TEXT


(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/data/data]
$ tree 'com.android.providers.calendar' -L 1
com.android.providers.calendar
databases
shared_prefs


(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3/data/data]
$ tree 'com.android.providers.calendar/databases'
com.android.providers.calendar/databases
calendar.db



Google Applications (Messages, Dialer, Maps, Photos)

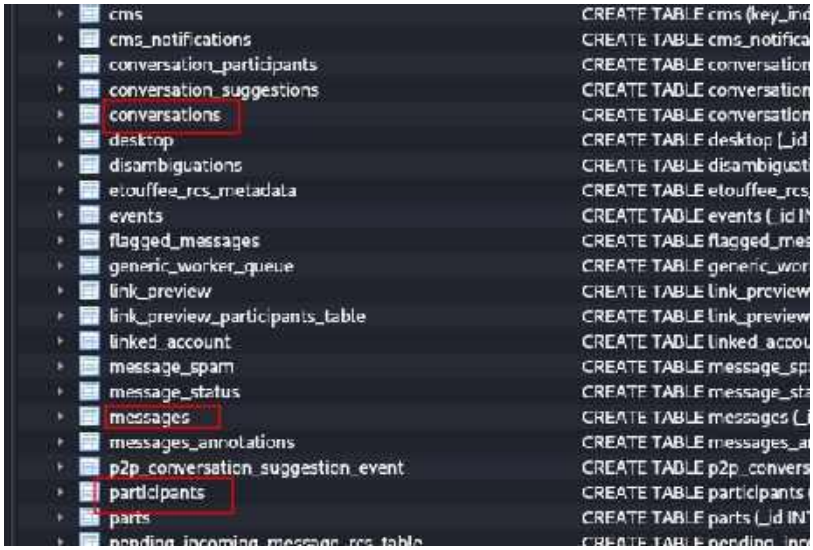
I checked Google Messages (bugle_db), Phone/Dialer, Maps and Photos at the paths listed in Appendix A of the brief. In Photos, backup_complete_times.xml holds account:0:backupComplete = 1581647595518 (2020-02-14 02:33:15 UTC). That is when a backup finished, not when a photo was taken. Maps sync_item stores coordinates in E6 form (latitude 35734602 = 35.734602). Coordinates are reported as stored and are not taken as proof of anyone's location.

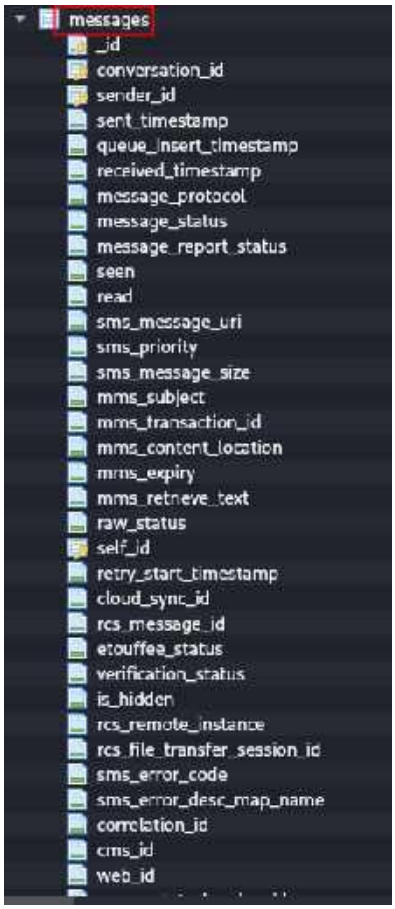
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ls 'data/data/' | grep -i com.google

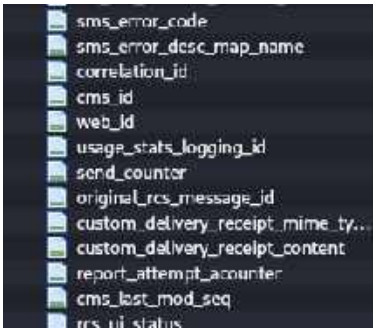
com.google.android.apps.chromecast.app
com.google.android.apps.docs
com.google.android.apps.docs.editors.docs
com.google.android.apps.dreamliner
com.google.android.apps.enterprise.dmagent
com.google.android.apps.gcs
com.google.android.apps.helprtc
com.google.android.apps.magazines
com.google.android.apps.maps
com.google.android.apps.messaging
com.google.android.apps.nexuslauncher
com.google.android.apps.photos
com.google.android.apps.pixelmigrate
com.google.android.apps.safetyhub
com.google.android.apps.tachyon
com.google.android.apps.tips
com.google.android.apps.turbo
com.google.android.apps.tycho
com.google.android.apps.walletnfcrel
com.google.android.apps.wallpaper
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.google.android.apps.messaging' -L 1
data/data/com.google.android.apps.messaging
├── app_logs
├── cache
├── databases
├── files
├── no_backup
└── shared_prefs
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.google.android.apps.messaging/databases'
data/data/com.google.android.apps.messaging/databases
├── androidx.work.workdb
├── androidx.work.workdb-shm
├── androidx.work.workdb-wal
├── bugle_db
├── bugle_db-shm
├── bugle_db-wal
├── google_app_measurement_local.db
├── growthkit.db
├── imsft.db
└── undelivered_messages
```


cms
cms_notifications
conversation_participants
conversation_suggestions
conversations
desktop
disambiguations
etouffee_rcs_metadata
events
flagged_messages
generic_worker_queue
link_preview
link_preview_participants_table
linked_account
message_spam
message_status
messages
messages_annotations
p2p_conversation_suggestion_event
participants
parts


messages
_id
conversation_id
sender_id
sent_timestamp
queue_insert_timestamp
received_timestamp
message_protocol
message_status
message_report_status
seen
read
sms_message_uri
sms_priority
sms_message_size
mms_subject
mms_transaction_id
mms_content_location
mms_expiry
mms_retrieve_text
raw_status
self_id
retry_start_timestamp
cloud_sync_id
rcs_message_id
etouffee_status
verification_status
is_hidden
rcs_remote_instance
rcs_file_transfer_session_id
sms_error_code
sms_error_desc_map_name
correlation_id
cms_id
web_id



eceived_timestamp	message_protocol	message_status	message_report_status	seen	read	sms_message_ur
ilter	Filter	Filter	Filter	Fi...	Fi...	Filter
1580322845251	0	108	0	1	0	content://sms/2
1580322646631	0	108	0	1	0	content://sms/3
1580322846632	5	218	0	1	1	NULL
1580322596009	0	108	0	1	0	content://sms/4
1580322728551	0	108	0	1	0	content://sms/5
1580322820355	0	108	0	1	0	content://sms/6
1580322903320	0	108	0	1	0	content://sms/7
1580323028883	0	108	0	1	0	content://sms/8
1580323020004	5	218	0	1	1	NULL
1580323298115	0	108	0	1	0	content://sms/9
1580323386596	0	108	0	1	0	content://sms/1
1580323813826	0	108	0	1	0	content://sms/1

Table: messages — Filter in any column

_id	conversation_id	sender_id	sent_timestamp	queue_insert_timestamp	received_timestamp	message_protocol	message_status	message_report_status	seen	read	sms_message_uri	sms_prio
1	2	4	1580322845000	1580322845550	1580322845251	0	108	0	1	0	content://sms/2	
2	3	5	1580322646000	1580322646908	1580322646631	0	108	0	1	0	content://sms/3	
3	3	2	1580322846632	1580322647511	1580322846632	5	218	0	1	1	NULL	
4	4	6	1500322596000	1580322697118	1500322596009	0	108	0	1	0	content://sms/4	
5	5	7	1580322727000	1580322728715	1580322728551	0	108	0	1	0	content://sms/5	
6	6	8	1580322810000	1580322820523	1580322820355	0	108	0	1	0	content://sms/6	
7	7	9	1580322902000	1580322903587	1580322903320	0	108	0	1	0	content://sms/7	
8	8	10	1580323027000	1580323029128	1580323028883	0	108	0	1	0	content://sms/8	
9	9	8	1500323020004	1580323029064	1500323020004	5	218	0	1	1	NULL	

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d '@1580322045' -u
Wed Jan 29 06:20:45 PM UTC 2020
```

Browse Data | Database Structure | Edit Pragmas | Execute SQL

Table: messages — Filter in any column

	_id	conversation_id	sender_id	sent_timestamp	queue_insert_timestamp	received_timestamp	message_protocol	message_status	message_report_status	seen	read	sms_message_uri	sms_priori
	...	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Fi...	Fi...	Filter	Filter
1	1	2	4	1580322845000	1580322845550	1580322845251	0	108	0	1	0	content://sms/2	1
2	2	3	5	1580322646000	1580322646908	1580322646631	0	108	0	1	0	content://sms/3	1
3	3	3	2	1580322846632	1580322647511	1580322846632	5	218	0	1	1	NULL	1
4	4	4	6	1500322596000	1580322697118	1500322596009	0	108	0	1	0	content://sms/4	1
5	5	5	7	1580322727000	1580322728715	1580322728551	0	108	0	1	0	content://sms/5	1
6	6	6	8	1580322810000	1580322820523	1580322820355	0	108	0	1	0	content://sms/6	1
7	7	7	9	1580322902000	1580322903587	1580322903320	0	108	0	1	0	content://sms/7	1

Table: conversations

_id	sms_thread_id	name	name_is_automatic	latest_message_id	snippet_text
1	2	4 (323) 320 5525	1	1	<#> Please enter 304868 into LINE …
2	3	5 732873	1	15	<#> Your Imgur verification code is …
3	4	6 (843) 751-4985	1	4	<#> Your Signal verification code: …
4	5	7 (225) 369-0055	1	5	Your imo code is 585614…
5	6	8 (234) 757-0651	1	6	Your Skout verification code is 2661.
6	7	9 (234) 900 8124	1	7	Your Skout verification code is 9873.
7	8	10 47543	1	9	Link previews are on. Learn mor…

Table: messages

_id	conversation_id	sender_id	sent_timestamp	queue_insert_timestamp	received_timestamp	message_protocol	message_status	message_report_status	seen	read	sms_message_uri	sms_prior
1	2	4	1580322845000	1580322845553	1580322845251	0	108	0	1	0	content://sms/2	1
2	3	5	1580322646000	1580322646908	1580322646611	0	108	0	1	0	content://sms/3	1
3	3	2	1580322646632	1580322647511	1580322646632	5	218	0	1	1	NULL	1
4	4	6	1500322596000	1500322697116	1500322596009	0	108	0	1	0	content://sms/4	1
5	5	7	1580322727000	1580322728715	1580322728551	0	108	0	1	0	content://sms/5	1
6	6	8	1580322810000	1580322820620	1580322820355	0	108	0	1	0	content://sms/6	1
7	7	9	1580322802000	1580322803587	1580322803320	0	108	0	1	0	content://sms/7	1
8	8	10	1580323827000	1580323829128	1580323828883	0	108	0	1	0	content://sms/8	1

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d '@1580322043' -u
Wed Jan 29 06:20:43 PM UTC 2020
```

Table: messages

_id	conversation_id	sender_id	sent_timestamp	queue_insert_timestamp	received_timestamp	message_protocol	message_status	message_report_status	seen	read	sms_message_uri	sms_prior
1	2	4	1580322845000	1580322845553	1580322845251	0	108	0	1	0	content://sms/2	1
2	3	5	1580322646000	1580322646908	1580322646611	0	108	0	1	0	content://sms/3	1
3	3	2	1580322646632	1580322647511	1580322646632	5	218	0	1	1	NULL	1
4	4	6	1500322596000	1500322697116	1500322596009	0	108	0	1	0	content://sms/4	1
5	5	7	1580322727000	1580322728715	1580322728551	0	108	0	1	0	content://sms/5	1
6	6	8	1580322810000	1580322820620	1580322820355	0	108	0	1	0	content://sms/6	1
7	7	9	1580322802000	1580322803587	1580322803320	0	108	0	1	0	content://sms/7	1

Table: participants

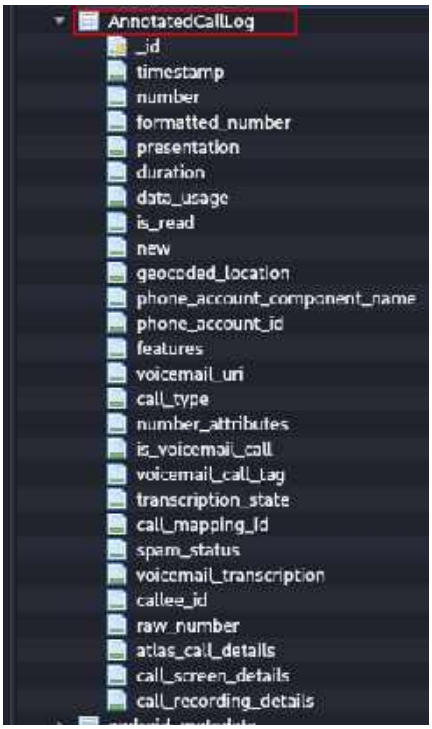
_id	sub_id	sim_slot_id	normalized_destination	send_destination	display_destination	full_name	first_name	profile_photo_uri	contact_id	lookup_key
1	1	0	+19195794674	NULL	(919) 579 4674	This is DFIR	This is	NULL	92233720347072327161	profile
2	2	-1				This Is DFIR	This Is	NULL	92233720347072327161	profile
3	3	-2	-1 22000	22000	22000	NULL	NULL	NULL	-2	NULL
4	4	-2	-1 13233205525	+13233205525	(323) 320-5525	NULL	NULL	NULL	-2	NULL
5	5	-2	-1 732873	732873	732873	NULL	NULL	NULL	-2	NULL
6	6	2	1 +18437514985	+18437514985	(843) 751 4985	NULL	NULL	NULL	2	NULL
7	7	-2	-1 +12255690855	+12255690855	(225) 369-0855	NULL	NULL	NULL	-2	NULL
8	8	-2	-1 +12347570651	+12347570651	(234) 757-0651	NULL	NULL	NULL	-2	NULL
9	9	-2	-1 +12349000124	+12349000124	(234) 900-0124	NULL	NULL	NULL	-2	NULL

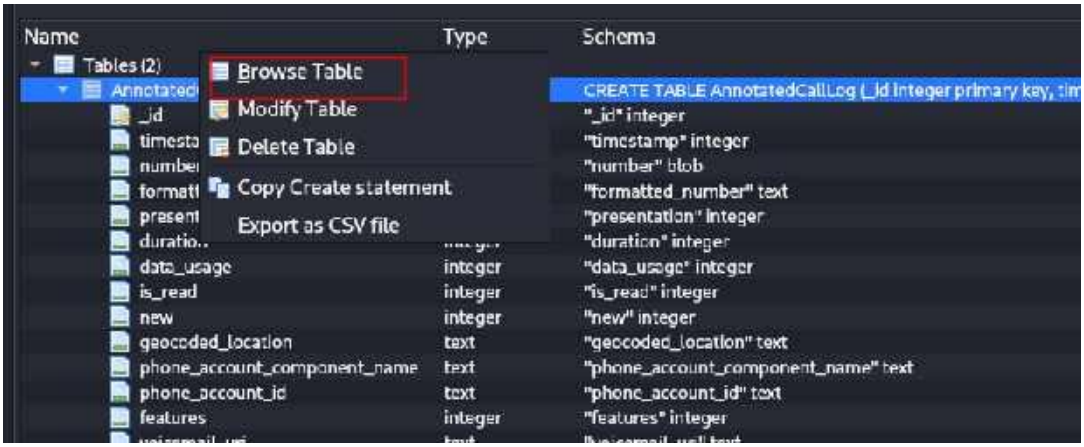
Table: messages

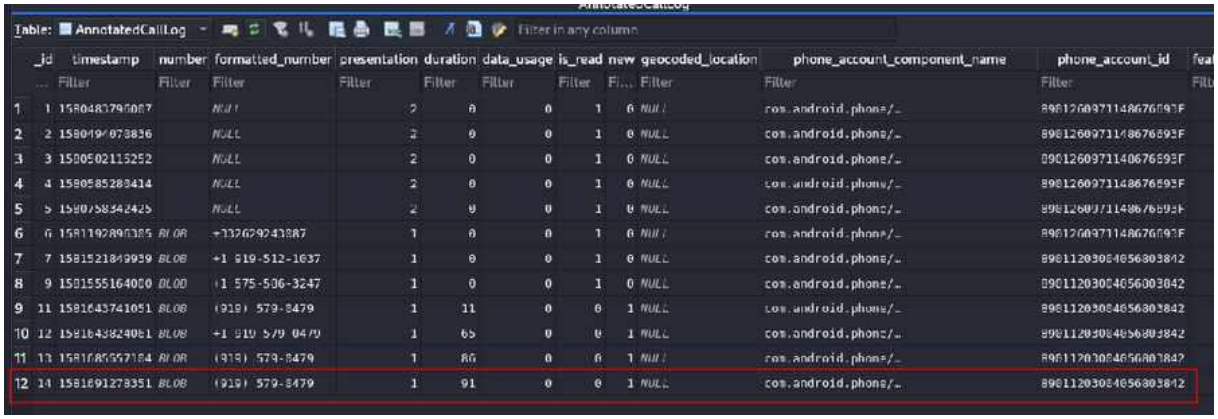
_id	conversation_id	sender_id	sent_timestamp	queue_insert_timestamp	received_timestamp	message_protocol	message_status	message_report_status	seen	read	sms_message_uri	sms_prior
1	2	4	1580322845000	1580322845553	1580322845251	0	108	0	1	0	content://sms/2	1
2	3	5	1580322646000	1580322646908	1580322646611	0	108	0	1	0	content://sms/3	1
3	3	2	1580322646632	1580322647511	1580322646632	5	218	0	1	1	NULL	1
4	4	6	1500322596000	1500322697116	1500322596009	0	108	0	1	0	content://sms/4	1
5	5	7	1580322727000	1580322728715	1580322728551	0	108	0	1	0	content://sms/5	1

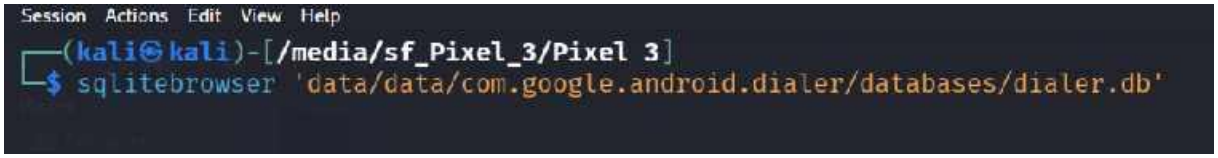
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ll 'data/data/com.google.android.dialer/databases/'
total 8164
-rwxrwx--- 1 root vboxsf   24576 Feb 16 22:52 annotated_call_log.db
-rwxrwx--- 1 root vboxsf    4096 Feb 16 22:52 callscreen_transcripts
-rwxrwx--- 1 root vboxsf   32768 Feb 16 22:52 callscreen_transcripts-shm
-rwxrwx--- 1 root vboxsf   37112 Feb 16 22:52 callscreen_transcripts-wal
-rwxrwx--- 1 root vboxsf   16384 Feb 16 22:52 CPSpeedDialEntry
-rwxrwx--- 1 root vboxsf   77824 Feb 16 22:52 dialer.db
-rwxrwx--- 1 root vboxsf   94208 Feb 16 22:52 growthkit.db
-rwxrwx--- 1 root vboxsf 7348224 Feb 16 22:52 normalized_spam.db
-rwxrwx--- 1 root vboxsf   24576 Feb 16 22:52 phone_lookup_history.db
-rwxrwx--- 1 root vboxsf   16384 Feb 16 22:52 rtt_transcript.db
-rwxrwx--- 1 root vboxsf    4096 Feb 16 22:52 spam_call_database
-rwxrwx--- 1 root vboxsf   32768 Feb 16 22:52 spam_call_database-shm
-rwxrwx--- 1 root vboxsf   74192 Feb 16 22:52 spam_call_database-wal
-rwxrwx--- 1 root vboxsf   24576 Feb 16 22:52 suggest_contact_database
-rwxrwx--- 1 root vboxsf   32768 Feb 16 22:52 suggest_contact_database-shm
-rwxrwx--- 1 root vboxsf  416152 Feb 16 22:52 suggest_contact_database-wal
-rwxrwx--- 1 root vboxsf   65536 Feb 16 22:52 superpacks.db
-rwxrwx--- 1 root vboxsf   24576 Feb 16 22:52 voicemail.db
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.google.android.dialer/databases/annotated_call_log.db'
```

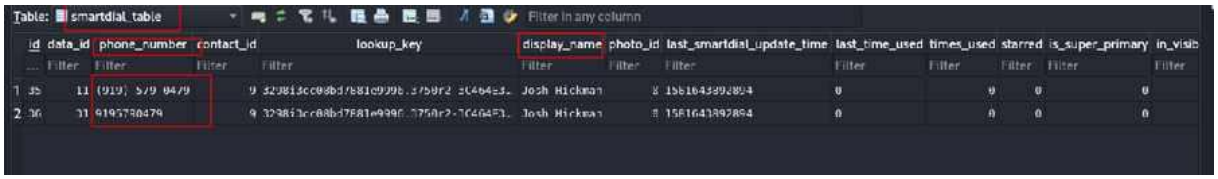



Name
Type
Schema
Tables (2)
Browse Table
Modify Table
Delete Table
Copy Create statement
Export as CSV file
CREATE TABLE AnnotatedCallLog (_id integer primary key, tim
"_id" integer
"timestamp" integer
"number" blob
"formatted_number" text
"presentation" integer
"duration" integer
"data_usage" integer
"is_read" integer
"new" integer
"geocoded_location" text
"phone_account_component_name" text
"phone_account_id" text
"features" integer
data_usage
integer
is_read
integer
new
integer
geocoded_location
text
phone_account_component_name
text
phone_account_id
text
features
integer


Table: AnnotatedCallLog
Filter in any column
_id timestamp number formatted_number presentation duration data_usage is_read new geocoded_location phone_account_component_name phone_account_id
1 1 1580483790887 NULL 2 0 0 1 0 NULL com.android.phone/... 8901260971148676893F
2 2 1580497073836 NULL 2 0 0 1 0 NULL com.android.phone/... 8901260971148676893F
3 3 1580502115252 NULL 2 0 0 1 0 NULL com.android.phone/... 8901260971140676593F
4 4 1580585283414 NULL 2 0 0 1 0 NULL com.android.phone/... 8901260971148676893F
5 5 1580758342425 NULL 2 0 0 1 0 NULL com.android.phone/... 8901260971148676893F
6 6 1581192893385 BLOB +13262924387 1 0 0 1 0 NULL com.android.phone/... 8901260971148676893F
7 7 1581521849939 BLOB +1 919-512-1637 1 0 0 1 0 NULL com.android.phone/... 8901120304056303842
8 9 1581555164020 BLOB (1 575-536-3247 1 0 0 1 0 NULL com.android.phone/... 8901120304056303842
9 11 1581643741051 BLOB (919) 579-8479 1 11 0 0 1 NULL com.android.phone/... 8901120304056303842
10 12 1581643824061 BLOB +1 919 579 8479 1 65 0 0 1 NULL com.android.phone/... 8901120304056303842
11 13 1581685557184 BLOB (919) 579-8479 1 86 0 0 1 NULL com.android.phone/... 8901120304056303842
12 14 1581691278351 BLOB (919) 579-8479 1 91 0 0 1 NULL com.android.phone/... 8901120304056303842

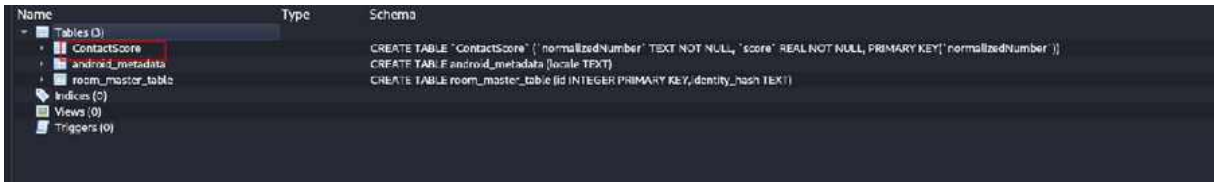

Session Actions Edit View Help
(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$ sqlitebrowser 'data/data/com.google.android.dialer/databases/dialer.db'


Name
Type
Tables (8)
android_metadata
cached_number_contacts
filtered_numbers_table
prefix_table
properties
smartdial_table
sqlite_sequence
sqlite_stat1
Indices (5)
cached_number_index
nameprefix_contact_id_index


Table: smartdial_table
Filter in any column
id data_id phone_number contact_id lookup_key display_name photo_id last_smartdial_update_time last_time_used times_used starred is_super_primary in_visib
1 35 11 (919) 579 8479 9 329813cc08bd7881c9998...3750r2-3C464e3... Josh Hickman 8 1581643892894 0 0 0 0
2 36 11 9195798479 9 329813cc08bd7881c9998...3750r2-3C464e3... Josh Hickman 8 1581643892894 0 0 0 0

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.google.android.dialer/databases'
data/data/com.google.android.dialer/databases
├── annotated_call_log.db
├── callscreen_transcripts
├── callscreen_transcripts-shm
├── callscreen_transcripts-wal
├── CPSpeedDialEntry
├── dialer.db
├── growthkit.db
├── normalized_spam.db
├── phone_lookup_history.db
├── rtt_transcript.db
├── spam_call_database
├── spam_call_database-shm
├── spam_call_database-wal
├── suggest_contact_database
├── suggest_contact_database-shm
├── suggest_contact_database-wal
├── superpacks.db
└── voicemail.db
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.google.android.dialer/databases/suggest_contact_database'
```

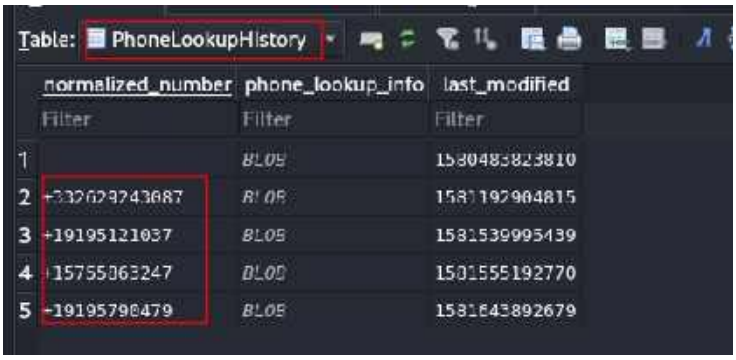




```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.google.android.dialer/databases'
data/data/com.google.android.dialer/databases
├── annotated_call_log.db
├── callscreen_transcripts
├── callscreen_transcripts-shm
├── callscreen_transcripts-wal
├── CPSpeedDialEntry
├── dialer.db
├── growthkit.db
├── normalized_spam.db
├── phone_lookup_history.db
├── rtt_transcript.db
├── spam_call_database
├── spam_call_database-shm
├── spam_call_database-wal
├── suggest_contact_database
├── suggest_contact_database-shm
├── superpacks.db
└── voicemail.db
```

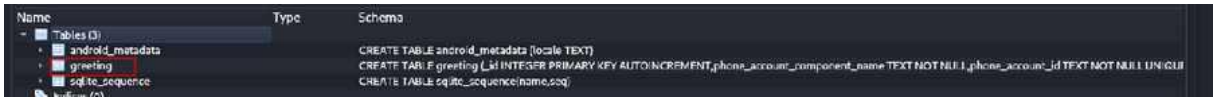
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.google.android.dialer/databases/phone_lookup_history.db'
```

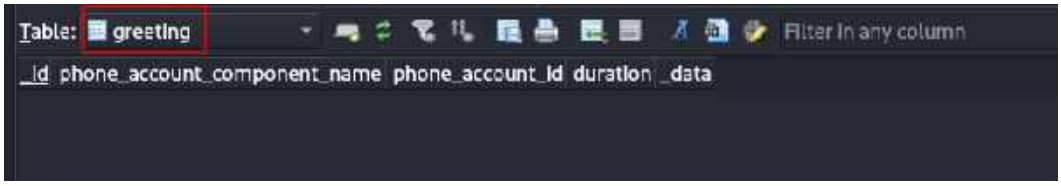




```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.google.android.dialer/databases'
data/data/com.google.android.dialer/databases
├── annotated_call_log.db
├── callscreen_transcripts
├── callscreen_transcripts-shm
├── callscreen_transcripts-wal
├── CPSpeedDialEntry
├── dialer.db
├── growthkit.db
├── normalized_spam.db
├── phone_lookup_history.db
├── rtt_transcript.db
├── spam_call_database
├── spam_call_database-shm
├── spam_call_database-wal
├── suggest_contact_database
├── suggest_contact_database-shm
├── superpacks.db
└── voicemail.db
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.google.android.dialer/databases/voicemail.db'
```





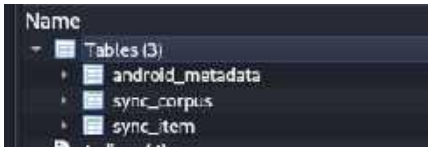
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.google.android.dialer/databases'

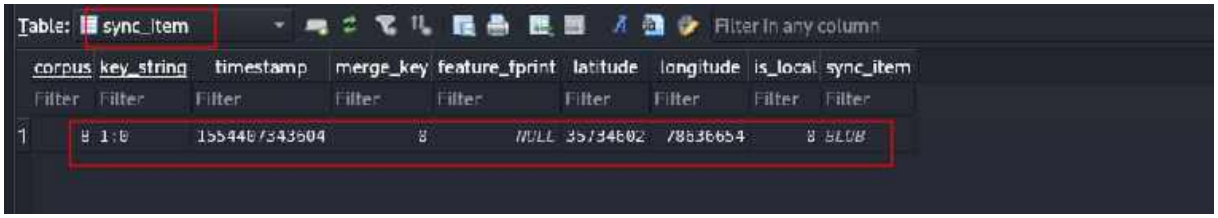
data/data/com.google.android.dialer/databases
├── annotated_call_log.db
├── callscreen_transcripts
├── callscreen_transcripts-shm
├── callscreen_transcripts-wal
├── CPSpeedDialEntry
├── dialer.db
├── growthkit.db
├── normalized_spam.db
├── phone_lookup_history.db
├── rtt_transcript.db
├── spam_call_database
├── spam_call_database-shm
├── spam_call_database-wal
├── suggest_contact_database
├── suggest_contact_database-shm
├── superpacks.db
└── voicemail.db
```

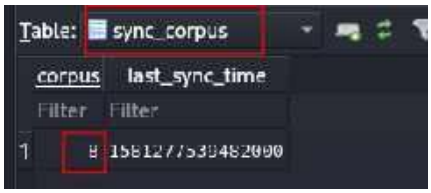
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ll 'data/data/com.google.android.apps.maps'
total 0
drwxrwx--- 1 root vboxsf 0 Feb 16 22:51 app_
drwxrwx--- 1 root vboxsf 0 Feb 16 22:51 app_tts-cache
drwxrwx--- 1 root vboxsf 0 Feb 16 22:51 app_tts-temp
drwxrwx--- 1 root vboxsf 0 Feb 16 22:51 cache
drwxrwx--- 1 root vboxsf 0 Feb 16 22:51 databases
drwxrwx--- 1 root vboxsf 0 Feb 16 22:51 files
drwxrwx--- 1 root vboxsf 0 Feb 16 22:51 no_backup
drwxrwx--- 1 root vboxsf 0 Feb 16 22:51 shared_prefs
```

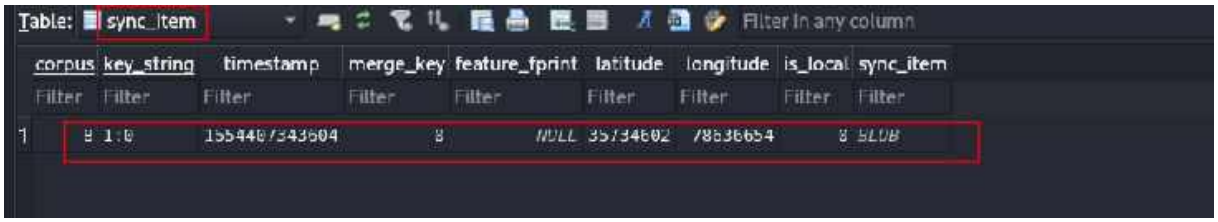
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ll 'data/data/com.google.android.apps.maps/databases'
total 248
-rwxrwx--- 1 root vboxsf 40960 Feb 16 22:51 gmm_myplaces.db
-rwxrwx--- 1 root vboxsf 94208 Feb 16 22:51 gmm_storage.db
-rwxrwx--- 1 root vboxsf 45056 Feb 16 22:51 gmm_sync.db
-rwxrwx--- 1 root vboxsf 20480 Feb 16 22:51 inbox_notifications.db
-rwxrwx--- 1 root vboxsf 24576 Feb 16 22:51 ue3.db
-rwxrwx--- 1 root vboxsf 28672 Feb 16 22:51 ugc_photos_location_data.db
```

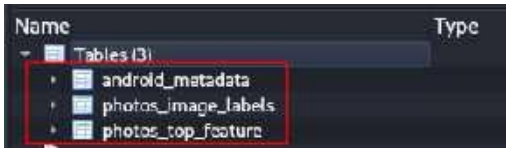
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.google.android.apps.maps/databases/gmm_myplaces.db'
```

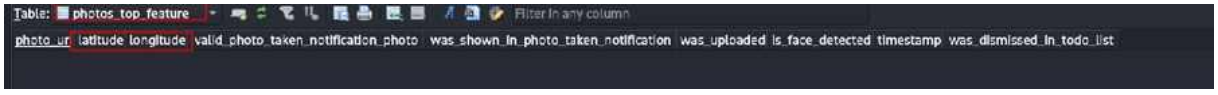













```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.google.android.apps.maps/databases/ugc_photos_location_data.db'
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ll 'data/data/com.google.android.apps.maps/databases'
total 248
-rwxrwx--- 1 root vboxsf 40960 Feb 16 22:51 gmm_myplaces.db
-rwxrwx--- 1 root vboxsf 94208 Feb 16 22:51 gmm_storage.db
-rwxrwx--- 1 root vboxsf 45056 Feb 16 22:51 gmm_sync.db
-rwxrwx--- 1 root vboxsf 20480 Feb 16 22:51 inbox_notifications.db
-rwxrwx--- 1 root vboxsf 24576 Feb 16 22:51 ue3.db
-rwxrwx--- 1 root vboxsf 28672 Feb 16 22:51 ugc_photos_location_data.db

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ll 'data/data/com.google.android.apps.maps/databases'
total 248
-rwxrwx--- 1 root vboxsf 40960 Feb 16 22:51 gmm_myplaces.db
-rwxrwx--- 1 root vboxsf 94208 Feb 16 22:51 gmm_storage.db
-rwxrwx--- 1 root vboxsf 45056 Feb 16 22:51 gmm_sync.db
-rwxrwx--- 1 root vboxsf 20480 Feb 16 22:51 inbox_notifications.db
-rwxrwx--- 1 root vboxsf 24576 Feb 16 22:51 ue3.db
-rwxrwx--- 1 root vboxsf 28672 Feb 16 22:51 ugc_photos_location_data.db

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.google.android.apps.maps/databases/gmm_storage.db'
```

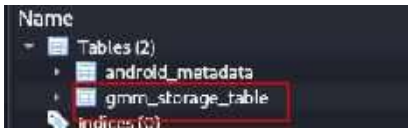


Table: gmm_storage_table

	_key_pri	_key_sec	_data
	Filter	Filter	Filter
1	uri	0	BLOB
2	uri	1	BLOB
3	CompletedNavigation	0	BLOB
4	CommuteSetupNotificationModel	1	BLOB

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.google.android.apps.maps/databases/gmm_sync.db'
```

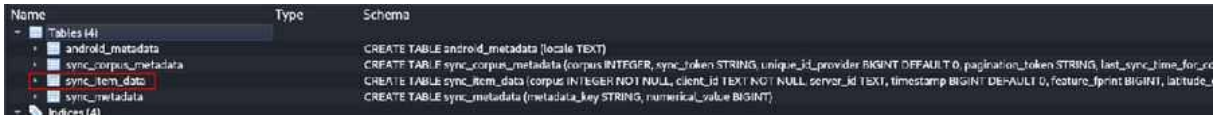


Table: sync_item_data

	corpus	client_id	server_id	timestamp	feature_fprint	latitude_e6	longitude_e6	numerical_index	string_index	sync_state	item_proto
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	11	4:0	4:0	0	14/12911058619583	35659996	-78872848	NULL	NULL	0	BLOB
2	11	5:0	5:0	0	811588047063364497	40784374	-74065336	NULL	NULL	0	BLOB
3	11	1:0	1:0	1554107343604	6121181384959223625	35734002	-78636554	NULL	NULL	0	BLOB
4	13	1581273852322_175453746935554	NULL	0	NULL	0	0	NULL	1581273852322_175453746935554	0	BLOB
5	13	1581273852322_175453747018211	NULL	0	NULL	0	0	NULL	1581273852322_175453747018211	0	BLOB
6	13	1581273852322_175453747049450	NULL	0	NULL	0	0	NULL	1581273852322_175453747049450	0	BLOB
7	6	User Parameters	User Parameters	0	NULL	0	0	NULL	NULL	0	BLOB

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ll 'data/data/com.google.android.apps.maps'
total 4
drwxrwx--- 1 root vboxsf    0 Feb 16 22:51 app_
drwxrwx--- 1 root vboxsf    0 Feb 16 22:51 app_tts-cache
drwxrwx--- 1 root vboxsf    0 Feb 16 22:51 app_tts-temp
drwxrwx--- 1 root vboxsf    0 Feb 16 22:51 cache
drwxrwx--- 1 root vboxsf 4096 Feb 16 22:51 databases
drwxrwx--- 1 root vboxsf    0 Feb 16 22:51 files
drwxrwx--- 1 root vboxsf    0 Feb 16 22:51 no_backup
drwxrwx--- 1 root vboxsf    0 Feb 16 22:51 shared_prefs
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ll 'data/data/com.google.android.apps.maps/files'
total 142
-rwxrwx--- 1 root vboxsf  2000 Feb 16 22:51 DATA_Preferences
-rwxrwx--- 1 root vboxsf    96 Feb 16 22:51 DATA_ShortTermStorage_consumed_subscriber_state
-rwxrwx--- 1 root vboxsf     8 Feb 16 22:51 DATA_ShortTermStorage_consumed_subscriber_state_Version
-rwxrwx--- 1 root vboxsf   107 Feb 16 22:51 DATA_ShortTermStorage_gmm_notification_status_active
-rwxrwx--- 1 root vboxsf     8 Feb 16 22:51 DATA_ShortTermStorage_gmm_notification_status_active_Version
-rwxrwx--- 1 root vboxsf     4 Feb 16 22:51 DATA_ShortTermStorage_nearby_alert_state
-rwxrwx--- 1 root vboxsf     8 Feb 16 22:51 DATA_ShortTermStorage_nearby_alert_state_Version
-rwxrwx--- 1 root vboxsf   192 Feb 16 22:51 DATA_ShortTermStorage_nearby_places_maintainer_state
-rwxrwx--- 1 root vboxsf     8 Feb 16 22:51 DATA_ShortTermStorage_nearby_places_maintainer_state_Version
-rwxrwx--- 1 root vboxsf  3529 Feb 16 22:51 DATA_st_epoch_resources_498
-rwxrwx--- 1 root vboxsf  3636 Feb 16 22:51 DATA_st_epoch_resources_499
-rwxrwx--- 1 root vboxsf     8 Feb 16 22:51 location_uploader_persistence.cs
-rwxrwx--- 1 root vboxsf     8 Feb 16 22:51 ls_state_cache.pb.cs
-rwxrwx--- 1 root vboxsf   356 Feb 16 22:51 new_recent_history_cache_navigated.cs
-rwxrwx--- 1 root vboxsf  3506 Feb 16 22:51 odelay_cache.cs
drwxrwx--- 1 root vboxsf     0 Feb 16 22:51 offlinedatabase
-rwxrwx--- 1 root vboxsf 14061 Feb 16 22:51 offline_saved_directions.data.cs
-rwxrwx--- 1 root vboxsf 89840 Feb 16 22:51 SavedClientParameters.data.cs
-rwxrwx--- 1 root vboxsf 14061 Feb 16 22:51 saved_directions.data.cs
drwxrwx--- 1 root vboxsf     0 Feb 16 22:51 users
```

```
TETurn left onto West Elm Avenue, then Turn left onto North Main Street (•  c2••••  (•• Turn left onto West Elm Avenue••••••••••  ••••••••••
•  0•(
•J7••A@!••k•[•S••
                •G••A@!8[••W•S•/

Turn z

leftz

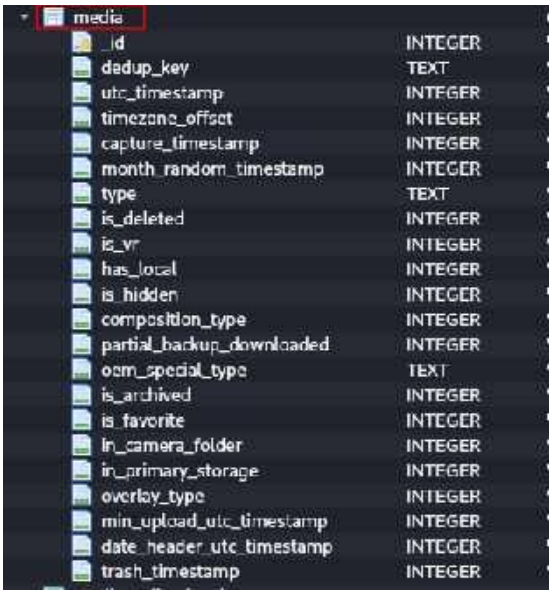
onto z
      N Main St2,0ahUKEwjwy8mik8XnAhVudt8KHeeHCR8Q9iQICSgCMAA8@Hb;    N Main St•North Main Street2North Main Street:en8USz
d(@
    ]eb
T
S
TTake the next left onto North Main Street, then Your destination will be on the left (J••••Y
T
S
KTurn left onto North Main Street, then Your destination will be on the left (•  c2••••  (••" Turn left onto North Main Street•8  ••••••••••(  ••••••••••
B•(
•1•A@!••Z•S•c•••        •A@!••••Z•S•R•Destination will be on the leftz• no_id_set•F
@//maps.gstatic.com/mapfiles/traffic_report/svg/ic_qu_info_v2.svg(•K
E//maps.gstatic.com/mapfiles/traffic_report/svg_dark/ic_qu_info_v2.svg(•2,0ahUKEwjwy8mik8XnAhVudt8KHeeHCR8Q9iQICigDMAA8b242 S Main St Suite 118•2 42 S Main S
t Suite 1 18ir Walter Coffee + Kitchenz <J••••%
 Your destination is on the left (•••*,0ahUKEwjwy8mik8XnAhVudt8KHeeHCR8Q7FcIBigAMAA ***America/New_York2EST
•
ir Walter Coffee + Kitchen
0242 S Main St Suite 118, Holly Springs, NC 27540ir Walter Coffee + Kitchen0242 S Main St Suite 118, Holly Springs, NC 27540"242 S Main St Suite 118•
```

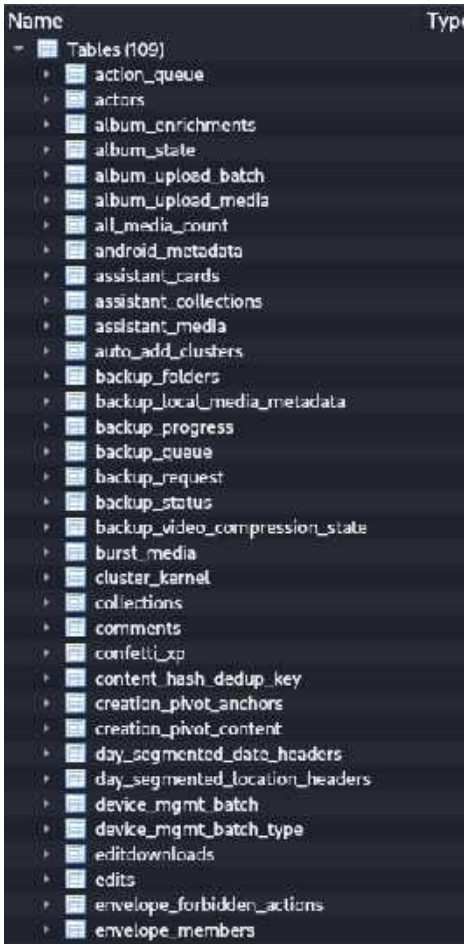
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/data/com.google.android.apps.maps/files/offline_saved_directions.data.cs'
F•W~
```

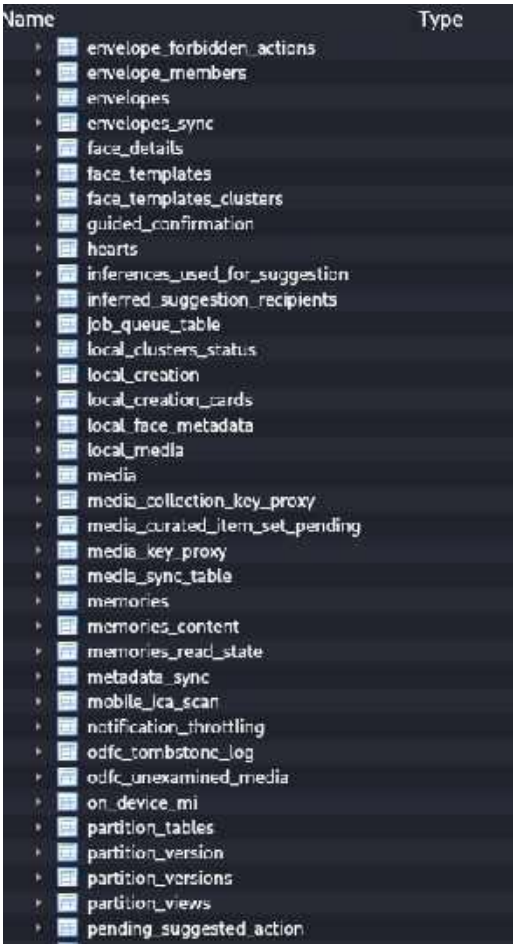
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.google.android.apps.photos' -L 1
data/data/com.google.android.apps.photos
├── app_null
├── app_phenotype_file
├── cache
├── databases
├── files
├── no_backup
└── shared_prefs
```

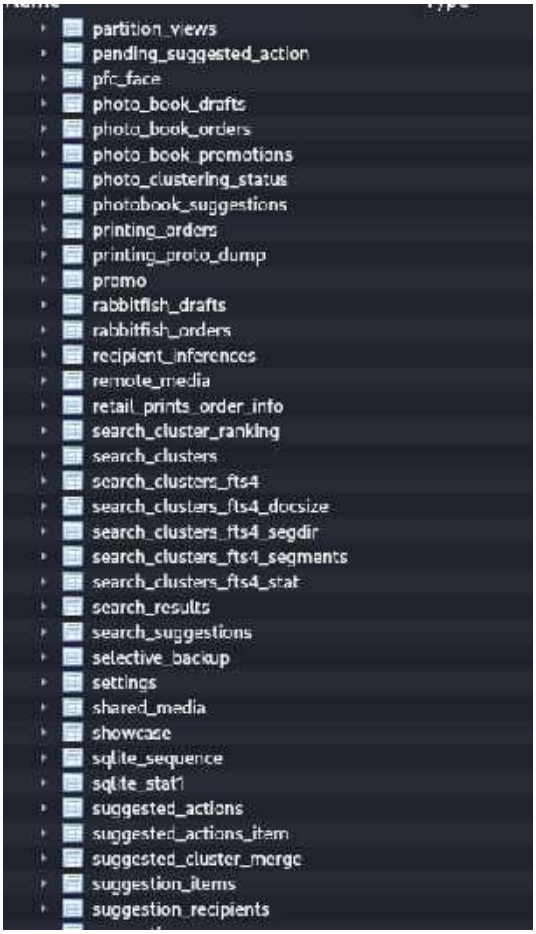
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ll 'data/data/com.google.android.apps.photos/databases'
total 5820
-rwxrwx--- 1 root vboxsf  172032 Feb 16 22:51 disk_cache
-rwxrwx--- 1 root vboxsf   32768 Feb 16 22:51 disk_cache-shm
-rwxrwx--- 1 root vboxsf  469712 Feb 16 22:51 disk_cache-wal
-rwxrwx--- 1 root vboxsf   16384 Feb 16 22:51 google_app_measurement_local.db
-rwxrwx--- 1 root vboxsf 2191360 Feb 16 22:51 gphotos0.db
-rwxrwx--- 1 root vboxsf   32768 Feb 16 22:51 gphotos0.db-shm
-rwxrwx--- 1 root vboxsf  524288 Feb 16 22:51 gphotos0.db-wal
-rwxrwx--- 1 root vboxsf 1232896 Feb 16 22:51 gphotos-1.db
-rwxrwx--- 1 root vboxsf   32768 Feb 16 22:51 gphotos-1.db-shm
-rwxrwx--- 1 root vboxsf  519152 Feb 16 22:51 gphotos-1.db-wal
-rwxrwx--- 1 root vboxsf   28672 Feb 16 22:51 local_trash.db
-rwxrwx--- 1 root vboxsf  176128 Feb 16 22:51 media_store_extras
-rwxrwx--- 1 root vboxsf   32768 Feb 16 22:51 media_store_extras-shm
-rwxrwx--- 1 root vboxsf  457352 Feb 16 22:51 media_store_extras-wal
-rwxrwx--- 1 root vboxsf   36864 Feb 16 22:51 NotificationsDB0
```

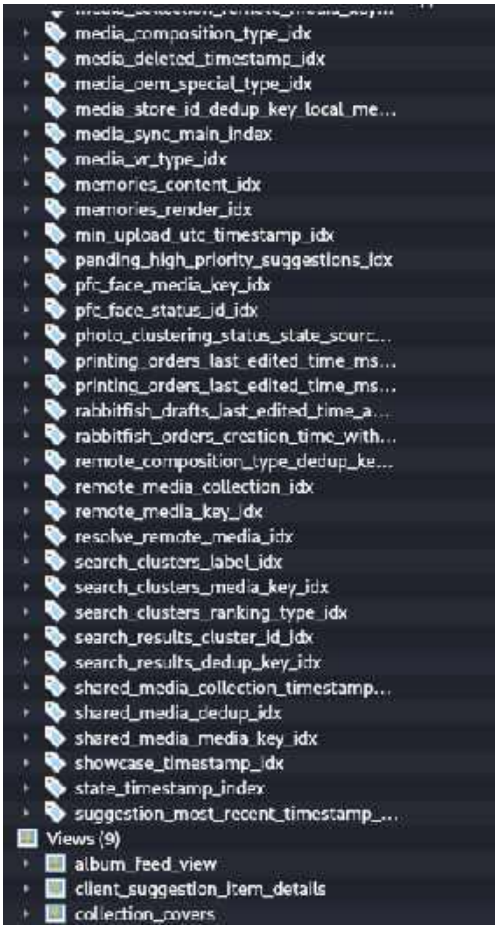
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser  'data/data/com.google.android.apps.photos/databases/gphotos-1.db'
```

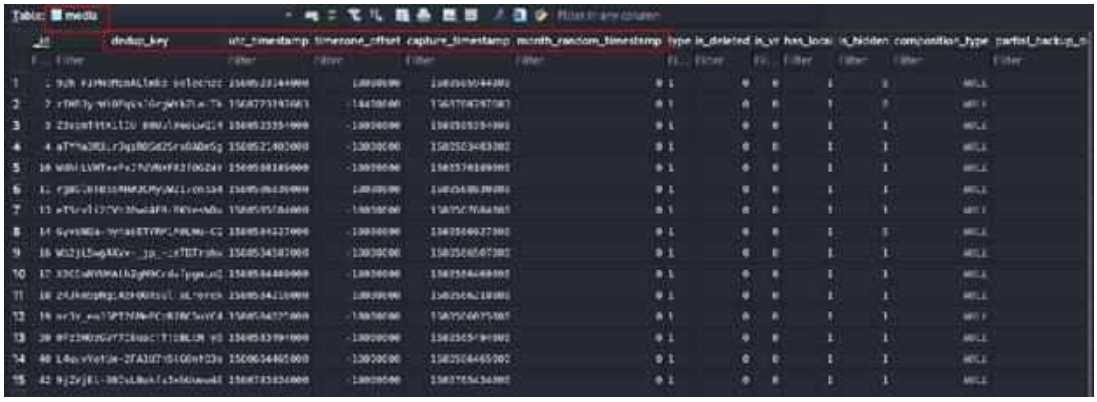


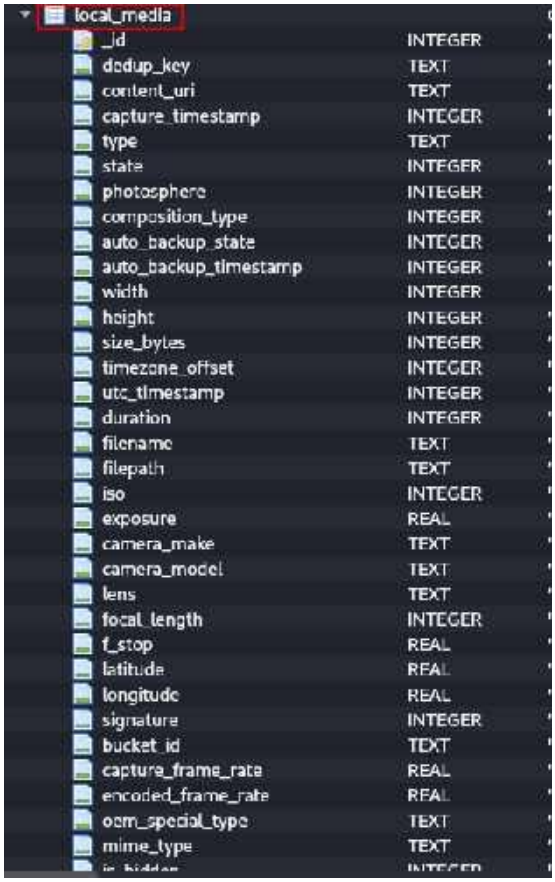

Name
Typ
Tables (109)
action_queue
actors
album_enrichments
album_state
album_upload_batch
album_upload_media
all_media_count
android_metadata
assistant_cards
assistant_collections
assistant_media
auto_add_clusters
backup_folders
backup_local_media_metadata
backup_progress
backup_queue
backup_request
backup_status
backup_video_compression_state
burst_media
cluster_kernel
collections
comments
confetti_xp
content_hash_dedup_key
creation_pivot_anchors
creation_pivot_content
day_segmented_date_headers
day_segmented_location_headers
device_mgmt_batch
device_mgmt_batch_type
editdownloads
edits
envelope_forbidden_actions
envelope_members


Name
Type
envelope_forbidden_actions
envelope_members
envelopes
envelopes_sync
face_details
face_templates
face_templates_clusters
guided_confirmation
hearts
inferences_used_for_suggestion
inferred_suggestion_recipients
job_queue_table
local_clusters_status
local_creation
local_creation_cards
local_face_metadata
local_media
media
media_collection_key_proxy
media_curated_item_set_pending
media_key_proxy
media_sync_table
memories
memories_content
memories_read_state
metadata_sync
mobile_lca_scan
notification_throttling
odfc_tombstone_log
odfc_unexamined_media
on_device_ml
partition_tables
partition_version
partition_versions
partition_views
pending_suggested_action


partition_views
pending_suggested_action
pfc_face
photo_book_drafts
photo_book_orders
photo_book_promotions
photo_clustering_status
photobook_suggestions
printing_orders
printing_proto_dump
promo
rabbitfish_drafts
rabbitfish_orders
recipient_inferences
remote_media
retail_prints_order_info
search_cluster_ranking
search_clusters
search_clusters_fts4
search_clusters_fts4_docsize
search_clusters_fts4_segdir
search_clusters_fts4_segments
search_clusters_fts4_stat
search_results
search_suggestions
selective_backup
settings
shared_media
showcase
sqlite_sequence
sqlite_stat1
suggested_actions
suggested_actions_item
suggested_cluster_merge
suggestion_items
suggestion_recipients


media_composition_type_idx
media_deleted_timestamp_idx
media_oem_special_type_idx
media_store_id_dedup_key_local_me...
media_sync_main_index
media_vr_type_idx
memories_content_idx
memories_render_idx
min_upload_utc_timestamp_idx
pending_high_priority_suggestions_idx
pfc_face_media_key_idx
pfc_face_status_id_idx
photo_clustering_status_state_sourc...
printing_orders_last_edited_time_ms...
printing_orders_last_edited_time_ms...
rabbitfish_drafts_last_edited_time_a...
rabbitfish_orders_creation_time_with...
remote_composition_type_dedup_ke...
remote_media_collection_idx
remote_media_key_idx
resolve_remote_media_idx
search_clusters_label_idx
search_clusters_media_key_idx
search_clusters_ranking_type_idx
search_results_cluster_id_idx
search_results_dedup_key_idx
shared_media_collection_timestamp...
shared_media_dedup_idx
shared_media_media_key_idx
showcase_timestamp_idx
state_timestamp_index
suggestion_most_recent_timestamp_...
Views (9)
album_feed_view
client_suggestion_item_details
collection_covers


Table: media
_id
dedup_key
utc_timestamp
timezone_offset
capture_timestamp
month_random_timestamp
type
is_deleted
is_vr
has_local
is_hidden
composition_type
partial_backup_s


local_media
_id INTEGER
dedup_key TEXT
content_uri TEXT
capture_timestamp INTEGER
type TEXT
state INTEGER
photosphere INTEGER
composition_type INTEGER
auto_backup_state INTEGER
auto_backup_timestamp INTEGER
width INTEGER
height INTEGER
size_bytes INTEGER
timezone_offset INTEGER
utc_timestamp INTEGER
duration INTEGER
filename TEXT
filepath TEXT
iso INTEGER
exposure REAL
camera_make TEXT
camera_model TEXT
lens TEXT
focal_length INTEGER
f_stop REAL
latitude REAL
longitude REAL
signature INTEGER
bucket_id TEXT
capture_frame_rate REAL
encoded_frame_rate REAL
osm_special_type TEXT
mime_type TEXT

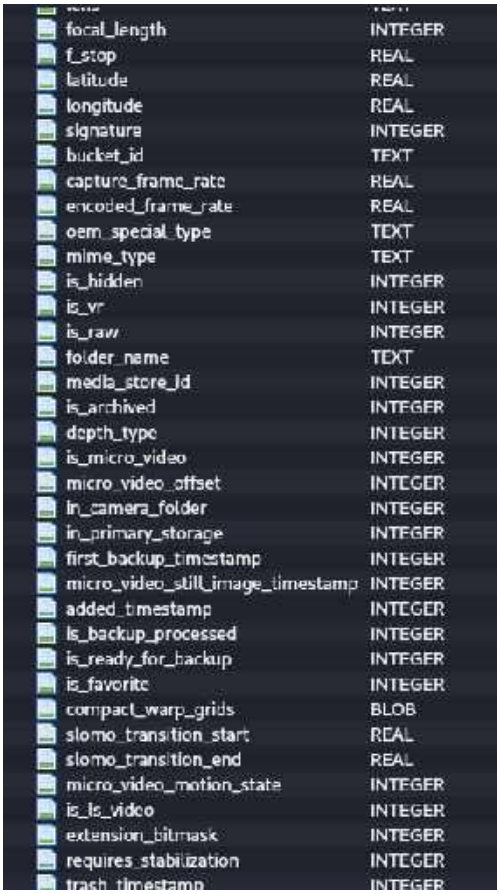


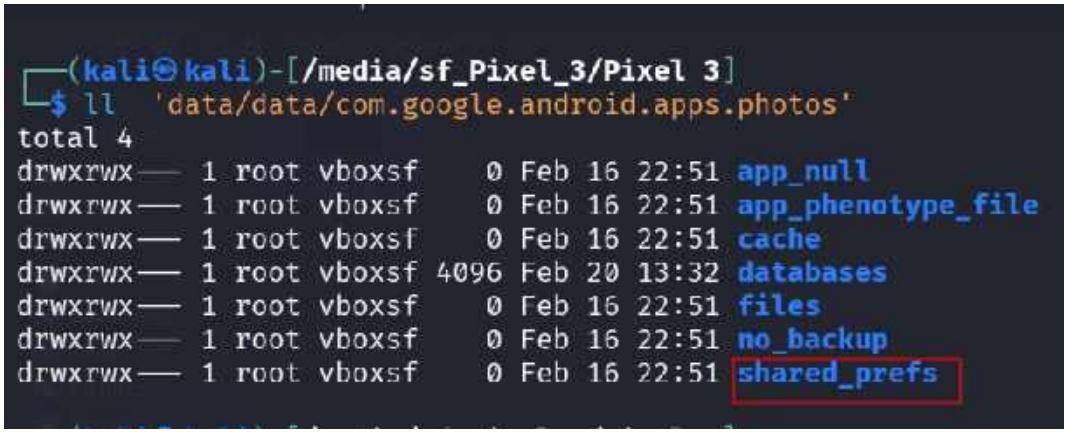
Table: local_media

Filter in any column

	utc_timestamp	duration	filename	filepath	iso
	Filter	Filter	Filter	Filter	Fil...
1	1568723197584	0	IMG_20190917_082637.jpg	/storage/emulated/0/DCIM/Restored/...	NULL
2	1580521403008	0	image0.jpg	/storage/emulated/0/Download/...	NULL
3	1580523354008	0	Nintendo-Famicom-Disk-System.jpg	/storage/emulated/0/Download/...	NULL
4	1580523944008	0	2ECCD1DB-39FD-454D-0D63-...	/storage/emulated/0/DCIM/Restored/...	NULL
5	1580583494008	0	received_517951378846562.jpeg	/storage/emulated/0/Pictures/...	NULL
6	1580584025008	0	t5MbUp3.jpg	/storage/emulated/0/Download/Imgur/...	NULL
7	1580584218008	0	aHTO2nX.jpg	/storage/emulated/0/Download/Imgur/...	NULL

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d '@1580521403' -u
Sat Feb  1 01:43:23 AM UTC 2020

┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$
```









Third-Party Applications

Kik (kik.android): installed 1580317722000 (2020-01-29 17:08:42 UTC), app version 15.19.0.22104. kikCoreDatabase.db › CoreTable holds the username thisisdfir. KikPreferences gives the profile name ThisIs DFIR and the email thisisdfir@gmail.com. kik.logintime is 1580322563292 (2020-01-29 18:29:23 UTC). messagesTable contains six rows with partner joshuahickman1_ulc@talk.kik.com. Two of them link through content_id to KIKContentTable

gallery media, and the matching thumbnails are in files/staging/thumbs (for example 59c3951061454f1df4663ff0720223c0, a 48x48 PNG). The online content URL returned 404.

TextNow (com.enflick.android.TextNow): the preferences give userinfo_phone 9842352054 and username thisisdfir. textnow_data.db holds contacts (including Josh Hickman) and one conversation with +19842032223. Its messages are in both directions, with text and media links, dated around 2020-02-08 20:14 UTC onward (for example raw 1581192840000).

WhatsApp (com.whatsapp): registration.RegisterPhone.xml gives phone 9195794674. msgstore.db › jid lists 16505434800 and 19195790479. Joining messages to jid and chat recovers "Hi there! I switched over.", "Awesome!" and "The Osbourne!" (around 2020-02-08 20:55 UTC). Two location messages captioned "Here I am!!" carry latitude ≈ 35.8609. wa.db › wa_contacts names 19195790479 as Josh Hickman. A public reverse lookup places that area code in Sanford, NC. That is background context only, not a location fix.

Snapchat (com.snapchat.android): identity_persistent_store.xml records INSTALL_ON_DEVICE 1580317793035 (2020-01-29 17:09:53 UTC), FIRST_LOGGED_IN 1580322609553 (18:30:09 UTC) and LAST_LOGGED_IN_USERNAME thisisdfir. user_session_shared_pref.xml adds the display name This Is DFIR and the email thisisdfir@gmail.com. Message content was not recovered.

```
└─$ ls 'data/data/' | grep -Eiv 'com.google|com.android'
com.discord
com.enflick.android.TextNow
com.facebook.orca
com.imgur.mobile
com.imo.android.imous
com.instagram.android
com.instagram.threadsapp
com.mewe
com.mywickr.wickr2
com.qti.ltebc
com.radicalapps.cyberdust
com.silentcircle.silentphone
com.skout.android
com.skype.raider
com.snapchat.android
com.spotify.music
com.thinkyeah.galleryvault
com.topjohnwu.magisk
com.twitter.android
com.venmo
com.viber.voip
com.whatsapp
com.wire
com.zhiliaoapp.musically
jp.naver.line.android
kik.android
org.telegram.messenger
org.thoughtcrime.securesms
org.torproject.torbrowser
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree -L 1 'data/data/kik.android'
data/data/kik.android
├── app_webview
├── cache
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0
├── databases
├── files
├── no_backup
└── shared_prefs
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree -L 1 'data/data/kik.android/databases'
data/data/kik.android/databases
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.abTesting.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.alternatesTable
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.assetEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.contactProfileEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.convoProfileEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikAndroidFileDatabase.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikDatabase.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikImageDatabase.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikItemDatabase.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikOfferTransactionEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kinProductTransactionEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.p2pTransactionEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.smileyTable
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.themeCache.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.userRosterEntries.db
├── google_app_measurement_local.db
└── kikCoreDatabase.db
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree -L 1 'data/data/kik.android'
data/data/kik.android
├── app_webview
├── cache
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0
├── databases
├── files
├── no_backup
└── shared_prefs
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree -L 1 'data/data/kik.android/shared_prefs'
data/data/kik.android/shared_prefs
├── admob.xml
├── BNC_Server_Request_Queue.xml
├── branch_referral_shared_pref.xml
├── com.crashlytics.prefs.xml
├── com.crashlytics.sdk.android&#58;answers&#58;settings.xml
├── com.google.android.gms.appid.xml
├── com.google.android.gms.measurement.prefs.xml
├── com.mopub.privacy.xml
├── com.mopub.settings.identifier.xml
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.com.google.android.gms.measurement.prefs.xml
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.com.kik.android.registerSharedPrefs.xml
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.KikPreferences.joshuahickman1_ulc@talk.kik.com.xml
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.KikPreferences.xml
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.Kik.Storage.ContactImageCache.pref.xml
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.Kik.Storage.ContentImageCache.pref.xml
├── io.fabric.sdk.android&#58;fabric&#58;io.fabric.sdk.android.Onboarding.xml
├── kik.android_preferences.xml
├── KikUltraPersistence.xml
├── kinecosystem_account_manager.xml
├── kinecosystem_blockchain_source.xml
├── kinecosystem_sign_in_pref.xml
├── KinKeyStore_kinecosystem_store.xml
├── mopubSettings.xml
├── TwitterAdvertisingInfoPreferences.xml
└── WebViewChromiumPrefs.xml
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ ll 'data/data/kik.android/shared_prefs'
total 60
-rwxrwx--- 1 root vboxsf   257 Feb 16 22:56 admob.xml
-rwxrwx--- 1 root vboxsf   123 Feb 16 22:56 BNC_Server_Request_Queue.xml
-rwxrwx--- 1 root vboxsf  2049 Feb 16 22:56 branch_referral_shared_pref.xml
-rwxrwx--- 1 root vboxsf   251 Feb 16 22:56 com.crashlytics.prefs.xml
-rwxrwx--- 1 root vboxsf   125 Feb 16 22:56 'com.crashlytics.sdk.android&#58;answers&#58;settings.xml'
-rwxrwx--- 1 root vboxsf  2514 Feb 16 22:56 com.google.android.gms.appid.xml
-rwxrwx--- 1 root vboxsf  1001 Feb 16 22:56 com.google.android.gms.measurement.prefs.xml
-rwxrwx--- 1 root vboxsf   932 Feb 16 22:56 com.mopub.privacy.xml
-rwxrwx--- 1 root vboxsf   377 Feb 16 22:56 com.mopub.settings.identifier.xml
-rwxrwx--- 1 root vboxsf   441 Feb 16 22:56 d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.com.google.android.gms.measurement.prefs.xml
-rwxrwx--- 1 root vboxsf    65 Feb 16 22:56 d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.com.kik.android.registerSharedPrefs.xml
-rwxrwx--- 1 root vboxsf   160 Feb 16 22:56 d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.KikPreferences.joshuahickman1_ulc@talk.kik.com.xml
-rwxrwx--- 1 root vboxsf 18943 Feb 16 22:56 d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.KikPreferences.xml
-rwxrwx--- 1 root vboxsf   140 Feb 16 22:56 d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.Kik.Storage.ContactImageCache.pref.xml
-rwxrwx--- 1 root vboxsf   140 Feb 16 22:56 d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.Kik.Storage.ContentImageCache.pref.xml
-rwxrwx--- 1 root vboxsf   168 Feb 16 22:56 'io.fabric.sdk.android&#58;fabric&#58;io.fabric.sdk.android.Onboarding.xml'
-rwxrwx--- 1 root vboxsf  2661 Feb 16 22:56 kik.android_preferences.xml
-rwxrwx--- 1 root vboxsf  2598 Feb 16 22:56 KikUltraPersistence.xml
-rwxrwx--- 1 root vboxsf   113 Feb 16 22:56 kinecosystem_account_manager.xml
-rwxrwx--- 1 root vboxsf   348 Feb 16 22:56 kinecosystem_blockchain_source.xml
-rwxrwx--- 1 root vboxsf   714 Feb 16 22:56 kinecosystem_sign_in_pref.xml
-rwxrwx--- 1 root vboxsf   359 Feb 16 22:56 KinKeyStore_kinecosystem_store.xml
-rwxrwx--- 1 root vboxsf   187 Feb 16 22:56 mopubSettings.xml
-rwxrwx--- 1 root vboxsf   213 Feb 16 22:56 TwitterAdvertisingInfoPreferences.xml
-rwxrwx--- 1 root vboxsf   127 Feb 16 22:56 WebViewChromiumPrefs.xml
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/data/kik.android/shared_prefs/com.google.android.gms.measurement.prefs.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <long name="time_active" value="16946" />
    <long name="last_upload" value="1580322534556" />
    <long name="first_open_time" value="1580322534533" />
    <string name="gmp_app_id">1:7203525089:android:c7f7cd15833ee205</string>
    <string name="app_instance_id">3e215f4b36b0d6baf2483569077efbc3</string>
    <boolean name="app_backgrounded" value="true" />
    <long name="health_monitor:start" value="1581692483339" />
    <boolean name="use_service" value="true" />
    <boolean name="deferred_analytics_collection" value="false" />
    <long name="app_install_time" value="1580317722000" />
    <string name="previous_os_version">10</string>
    <boolean name="has_been_opened" value="true" />
    <boolean name="use_dynamite_api" value="false" />
    <boolean name="allow_remote_dynamite" value="false" />
    <long name="last_pause_time" value="1581270799473" />
    <boolean name="start_new_session" value="false" />
</map>
```

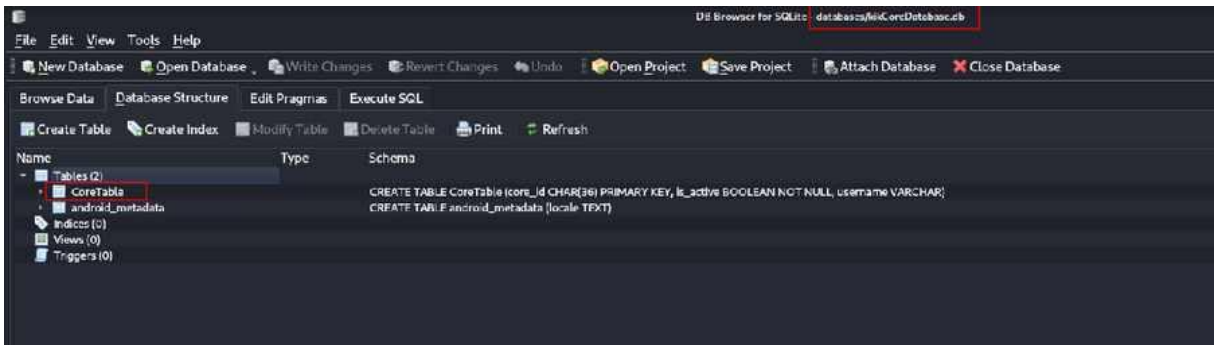
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ date -d '@1580317722' -u
Wed Jan 29 05:08:42 PM UTC 2020
```

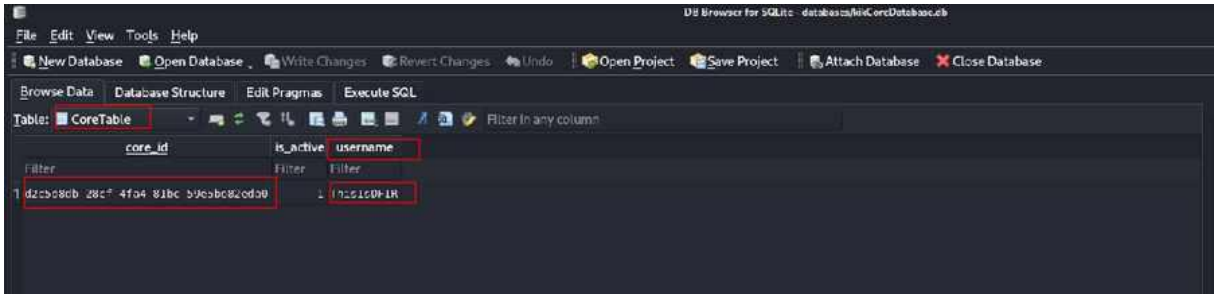
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cat 'data/data/kik.android/shared_prefs/branch_referral_shared_pref.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <string name="bnc_push_identifier">bnc_no_value</string>
    <string name="bnc_branch_analytical_data"></string>
    <string name="bnc_google_play_install_referrer_extras">bnc_no_value</string>
    <boolean name="bnc_is_full_app_conversion" value="false" />
    <string name="bnc_device_fingerprint_id">751134039444391423</string>
    <string name="bnc_link_click_id">bnc_no_value</string>
    <string name="bnc_external_intent_extra">bnc_no_value</string>
    <long name="bnc_last_known_update_time" value="1580317727506" />
    <long name="bnc_branch_strong_match_time" value="1580322535593" />
    <string name="bnc_user_url">https://kik.co/a/key_live_fpxk8WuTwagFbmshrrtx3lijCvnqVRgC?%24identity_id=751134039482145721</string>
    <boolean name="bnc_triggered_by_fb_app_link" value="false" />
    <string name="bnc_session_params">bnc_no_value</string>
    <int name="bnc_is_referrable" value="1" />
    <long name="bnc_original_install_time" value="1580317727506" />
    <string name="bnc_branch_key">key_live_fpxk8WuTwagFbmshrrtx3lijCvnqVRgC</string>
    <string name="bnc_session_id">755111266351332525</string>
    <string name="bnc_link_click_identifier">bnc_no_value</string>
    <string name="skip_url_format_key">{&quot;version&quot;:1,&quot;uri_skip_list&quot;:[&quot;^fb\\d+:&quot;,&quot;^li\\d+:&quot;,&quo
;^pdk\\d+:&quot;,&quot;^twitterkit-.*:&quot;,&quot;^com\\.googleusercontent\\.apps\\.\\d+-.*:\\/oauth&quot;,&quot;^(?i).+:.*[?].*\\b(
assword|o?auth|o?auth.?token|access|access.?token)\\b&quot;]}</string>
    <string name="bnc_identity_id">751134039482145721</string>
    <string name="bnc_app_link">bnc_no_value</string>
    <string name="bnc_app_version">15.19.0.22104</string>
    <string name="bnc_google_search_install_identifier">bnc_no_value</string>
    <long name="bnc_previous_update_time" value="1580317727506" />
    <string name="bnc_external_intent_uri">bnc_no_value</string>
    <string name="bnc_install_referrer">bnc_no_value</string>
</map>
```

```
┌──(kali㉿kali)-[/media/…/Pixel 3/data/data/kik.android]
└─$ grep -rI "version_name" .
./files/.Fabric/com.crashlytics.sdk.android.crashlytics-core/SE46C25903D1-0001-298A-1760C076DC36SessionApp.json:{"version_name":"15.19.0.22104","
api_key":"05f490868c9518194f9555a7d8664f006da667ac","unity_version":"","version_code":"30000539","app_identifier":"kik.android","install_uuid":"3
7f2de3f38524f30bd595b03d845093a","delivery_mechanism":4}
```

```
┌──(kali㉿kali)-[/media/…/Pixel 3/data/data/kik.android]
└─$ sqlitebrowser 'databases/kikCoreDatabase.db'
```






```
┌──(kali㉿kali)-[/media/…/Pixel 3/data/data/kik.android]
└─$ grep -E -i 'firstname|lastname' 'shared_prefs/d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikPreferences.xml'

    <string name="user_profile_firstName">ThisIs</string>
    <string name="user_profile_lastName">DFIR</string>
```

```
┌──(kali㉿kali)-[/media/…/Pixel 3/data/data/kik.android]
└─$ grep -i 'email' 'shared_prefs/d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.KikPreferences.xml'

    <string name="user_profile_email_emailConfirmed">false</string>
    <string name="user_profile_email">thisisdfir@gmail.com</string>
    <string name="CredentialData.email_passkey">23d2332a980840379dcc0cf93bbbe4fa</string>
```

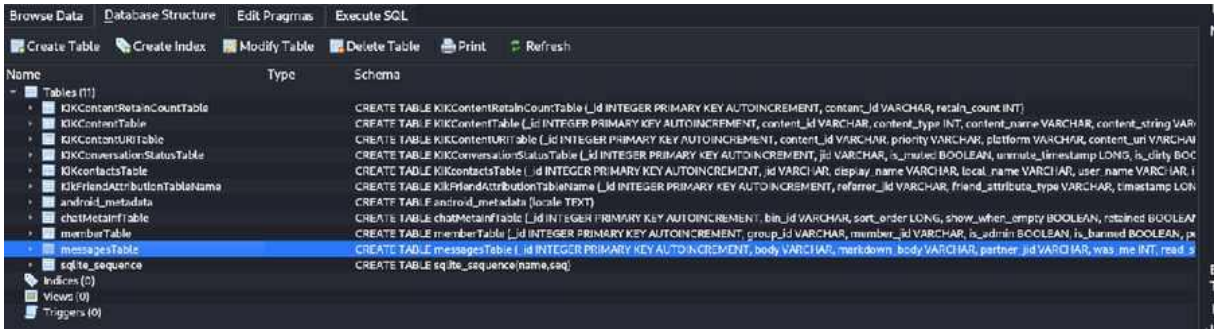
```
┌──(kali㉿kali)-[/media/…/Pixel 3/data/data/kik.android]
└─$ grep -r "logintime" .
./shared_prefs/d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.KikPreferences.xml:    <long name="kik.logintime" value="1580322563292" />
```

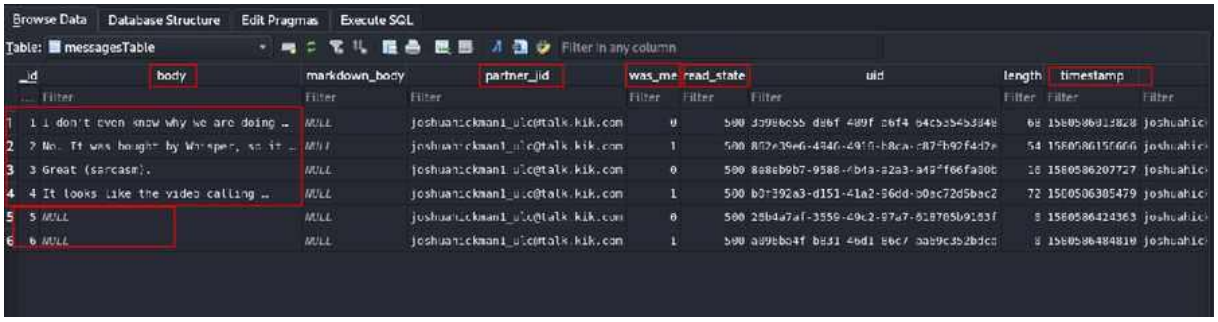
```
┌──(kali㉿kali)-[/media/…/Pixel 3/data/data/kik.android]
└─$ date -d '@1580322563' -u
Wed Jan 29 06:29:23 PM UTC 2020
```

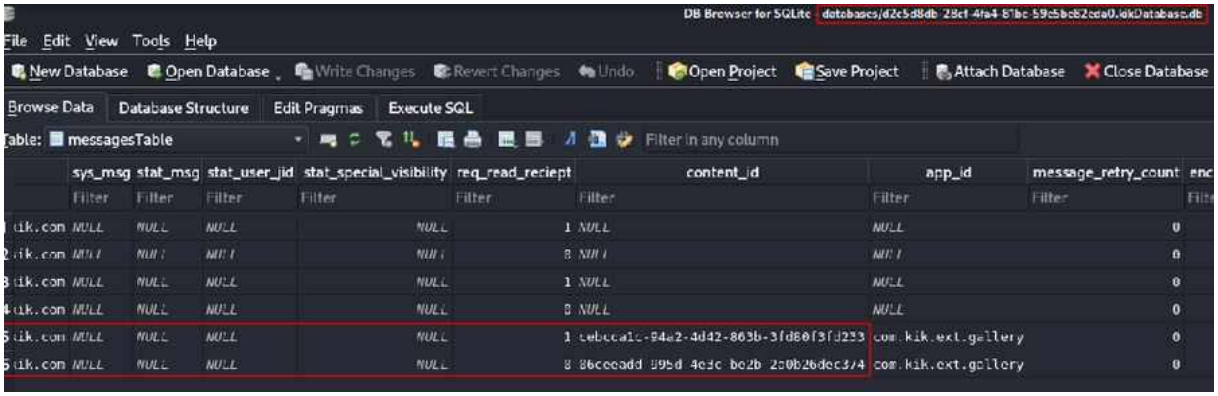
```
┌──(kali㉿kali)-[/media/…/Pixel 3/data/data/kik.android]
└─$ tree -L 1
.
├── app_webview
├── cache
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0
├── databases
├── files
├── no_backup
└── shared_prefs
```

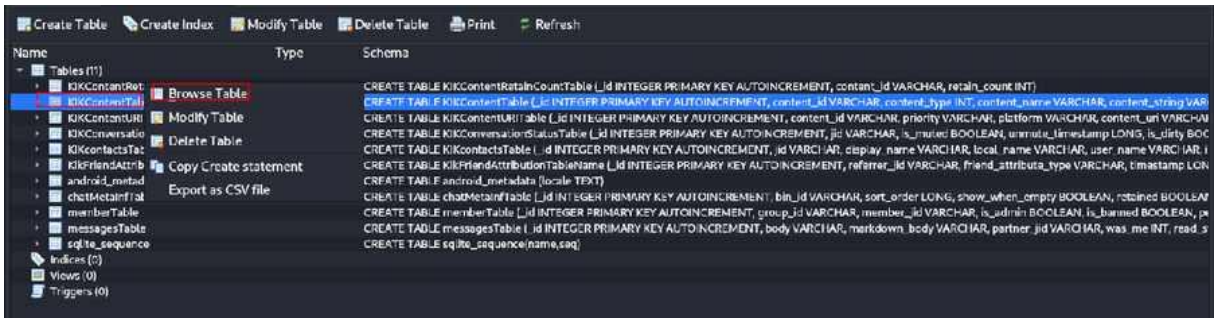
```
┌──(kali㉿kali)-[/media/…/Pixel 3/data/data/kik.android]
└─$ tree 'databases'
databases
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.abTesting.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.alternatesTable
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.assetEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.contactProfileEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.convoProfileEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikAndroidFileDatabase.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikDatabase.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikImageDatabase.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikItemDatabase.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikOfferTransactionEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kinProductTransactionEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.p2pTransactionEntries.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.smileyTable
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.themeCache.db
├── d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.userRosterEntries.db
├── google_app_measurement_local.db
└── kikCoreDatabase.db
```

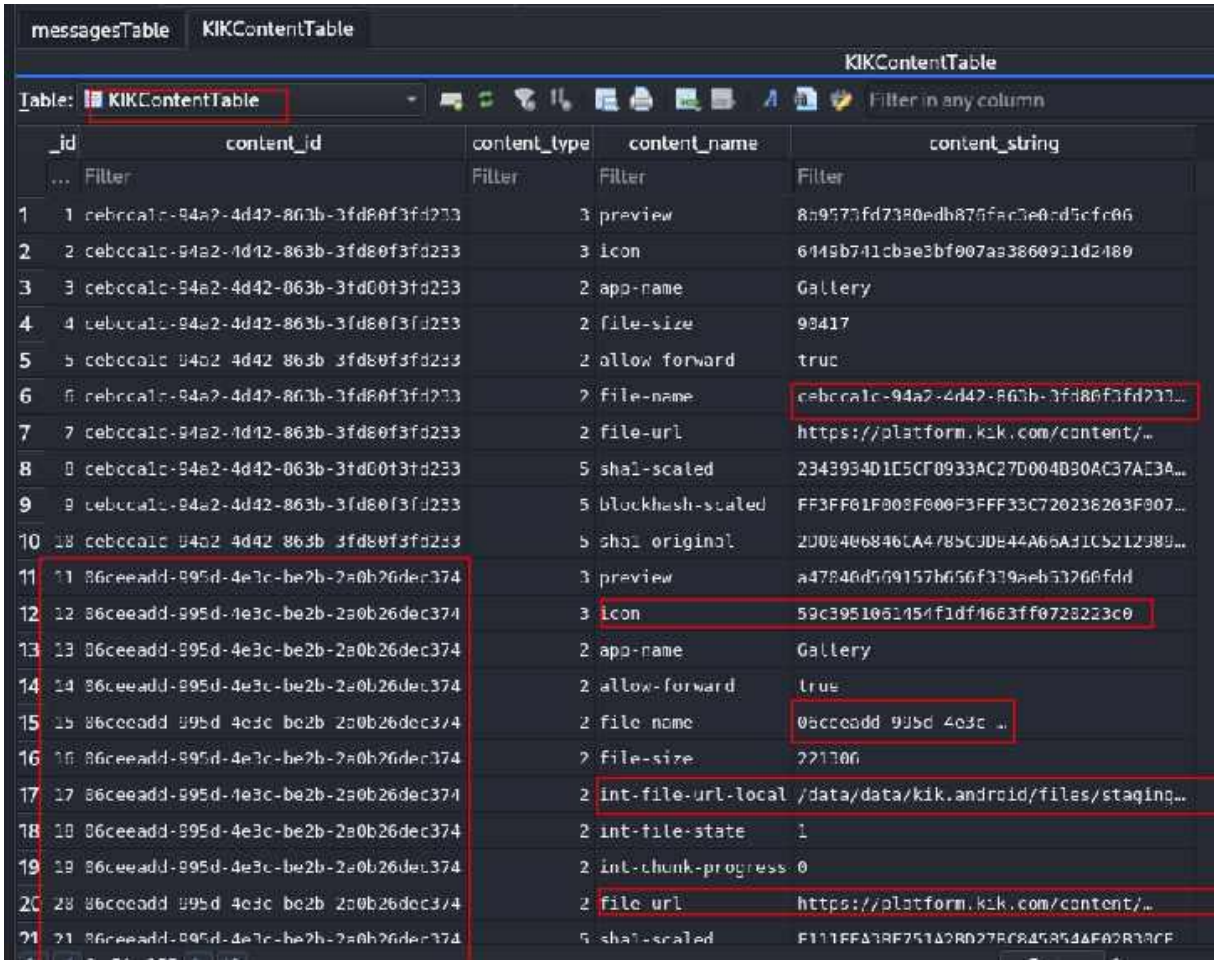
```
┌──(kali㉿kali)-[/media/…/Pixel 3/data/data/kik.android]
└─$ sqlitebrowser 'databases/d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikDatabase.db'
```




Browse Data
Database Structure
Edit Pragmas
Execute SQL
Table: messagesTable
Filter in any column
_id
body
markdown_body
partner_jid
was_me
read_state
uid
length
timestamp
1 1 don't even know why we are doing …
2 2 No. It was bought by Whisper, so it …
3 3 Great (sarcasm).
4 4 It looks like the video calling …
5 5 NULL
6 6 NULL
joshuahickman1_ulc@talk.kik.com


DB Browser for SQLite
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Browse Data
Database Structure
Edit Pragmas
Execute SQL
Table: messagesTable
Filter in any column
sys_msg
stat_msg
stat_user_jid
stat_special_visibility
req_read_reciept
content_id
app_id
message_retry_count
NULL
com.kik.ext.gallery


Create Table
Create Index
Modify Table
Delete Table
Print
Refresh
Name
Type
Schema
Tables (11)
Browse Table
Modify Table
Delete Table
Copy Create statement
Export as CSV file
messagesTable
sqlite_sequence
Indices (0)
Views (0)
Triggers (0)



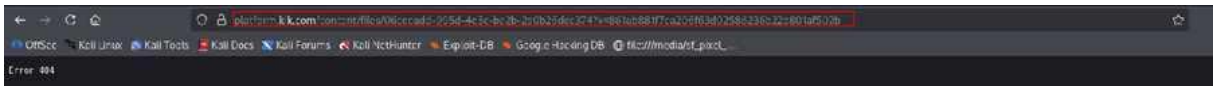
messagesTable | KIKContentTable

KIKContentTable

Table: KIKContentTable

Filter in any column

	_id	content_id	content_type	content_name	content_string
	...	Filter	Filter	Filter	Filter
1	1	cebccalc-94a2-4d42-863b-3fd80f3fd233	3	preview	8b9573fd7380edb876fac3e0cd5cfc06
2	2	cebccalc-94a2-4d42-863b-3fd80f3fd233	3	icon	6449b741cbae3bf007aa3860911d2480
3	3	cebccalc-94a2-4d42-063b-3fd80f3fd233	2	app-name	Gallery
4	4	cebccalc-94a2-4d42-863b-3fd80f3fd233	2	file-size	98417
5	5	cebccalc 94a2 4d42 863b 3fd80f3fd233	2	allow-forward	true
6	6	cebccalc-94a2-4d42-863b-3fd80f3fd233	2	file-name	cebccalc-94a2-4d42-863b-3fd80f3fd233...
7	7	cebccalc-94a2-4d42-863b-3fd80f3fd233	2	file-url	https://platform.kik.com/content/...
8	8	cebccalc-94a2-4d42-063b-3fd80f3fd233	5	sha1-scaled	2343934D1E5CF0933AC27D004B30AC37AE3A...
9	9	cebccalc-94a2-4d42-863b-3fd80f3fd233	5	blockhash-scaled	FF3FF01F003F000F3FFF33C720238203F007...
10	10	cebccalc 94a2 4d42 863b 3fd80f3fd233	5	sha1-original	2D08406846CA4785C9DE44A66AD1C5212989...
11	11	86ceeadd-995d-4e3c-be2b-2a0b26dec374	3	preview	a47040d569157b656f339aeb53260fdd
12	12	86ceeadd-995d-4e3c-be2b-2a0b26dec374	3	icon	59c3951061454f1df4663ff0720223c0
13	13	86ceeadd-995d-4e3c-be2b-2a0b26dec374	2	app-name	Gallery
14	14	86ceeadd-995d-4e3c-be2b-2a0b26dec374	2	allow-forward	true
15	15	86ceeadd 995d 4e3c be2b 2a0b26dec374	2	file-name	86ceeadd-995d-4e3c-...
16	16	86ceeadd-995d-4e3c-be2b-2a0b26dec374	2	file-size	221306
17	17	86ceeadd-995d-4e3c-be2b-2a0b26dec374	2	int-file-url-local	/data/data/kik.android/files/staging...
18	18	86ceeadd-995d-4e3c-be2b-2a0b26dec374	2	int-file-state	1
19	19	86ceeadd-995d-4e3c-be2b-2a0b26dec374	2	int-chunk-progress	0
20	20	86ceeadd 995d 4e3c be2b 2a0b26dec374	2	file-url	https://platform.kik.com/content/...
21	21	86ceeadd-995d-4e3c-be2b-2a0b26dec374	5	sha1-scaled	F111EEA38F751A2BD27BC845A544E02B30CE



platform.kik.com/content/files/86ceeadd-995d-4e3c-be2b-2a0b26dec374?v=8b9573fd7380edb876fac3e0cd5cfc06

Error 404

```
┌──(kali㉿kali)-[/media/…/Pixel 3/data/data/kik.android]
└─$ cd 'files/staging/thumbs'

┌──(kali㉿kali)-[/media/…/kik.android/files/staging/thumbs]
└─$ ll
total 44
-rwxrwx--- 1 root vboxsf   895 Feb 16 22:56 59c3951061454f1df4663ff0720223c0
-rwxrwx--- 1 root vboxsf  1260 Feb 16 22:56 6449b741cbae3bf007aa3860911d2480
-rwxrwx--- 1 root vboxsf 14914 Feb 16 22:56 8b9573fd7380edb876fac3e0cd5cfc06
-rwxrwx--- 1 root vboxsf 18661 Feb 16 22:56 a47040d569157b656f339aeb53260fdd

┌──(kali㉿kali)-[/media/…/kik.android/files/staging/thumbs]
└─$ file 59c3951061454f1df4663ff0720223c0
59c3951061454f1df4663ff0720223c0: PNG image data, 48 x 48, 8-bit/color RGBA, non-interlaced

┌──(kali㉿kali)-[/media/…/kik.android/files/staging/thumbs]
└─$ display 59c3951061454f1df4663ff0720223c0
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.enflick.android.TextNow' -L 1

data/data/com.enflick.android.TextNow
├── app_Leanplum_Documents
├── app_webview
├── cache
├── databases
├── files
├── no_backup
└── shared_prefs
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cd 'data/data/com.enflick.android.TextNow/shared_prefs'

┌──(kali㉿kali)-[/media/…/data/data/com.enflick.android.TextNow/shared_prefs]
└─$ find -type f -size +1000c
./admob.xml
./com.amazonaws.android.auth.xml
./com.applovin.sdk.1.xml
./com.enflick.android.TextNow_preferences.xml
./com.facebook.internal.preferences.APP_GATEKEEPERS.xml
./com.google.android.gms.measurement.prefs.xml
./LeanplumVariables.xml
./Mediation_Shared_Preferences.xml
./mopubCustomEventSettings.xml
./profig.xml
./SafeDKEvents.com.enflick.android.TextNow&#58;sync.xml
./SafeDKEvents.com.enflick.android.TextNow.xml
./SafeDKToggles.xml
./TNSettingsInfo.xml
./__leanplum__.xml
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree 'data/data/com.enflick.android.TextNow/databases'

data/data/com.enflick.android.TextNow/databases
├── ads.db
├── androidx.work.workdb
├── androidx.work.workdb-shm
├── androidx.work.workdb-wal
├── com_textnow_android_events_TNEventStreaming_buffer_db
├── com_textnow_android_events_TNEventStreaming_buffer_db-shm
├── com_textnow_android_events_TNEventStreaming_buffer_db-wal
├── EmogiSDK.db
├── google_app_measurement_local.db
├── __leanplum.db
├── supersonic_sdk.db
└── textnow_data.db
```

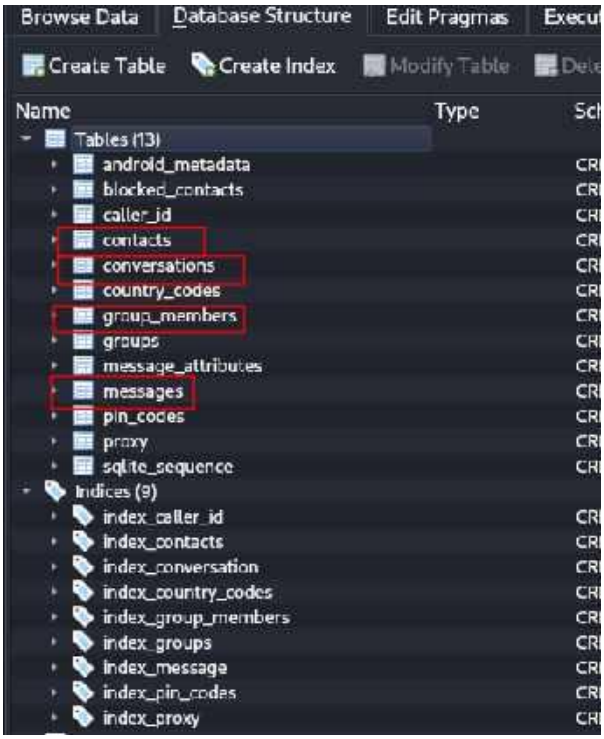
```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep 'phone'  'data/data/com.enflick.android.TextNow/shared_prefs/com.enflick.android.TextNow_preferences.xml'

    <string name="userinfo_phone">9842352054</string>
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ grep 'username'  'data/data/com.enflick.android.TextNow/shared_prefs/com.enflick.android.TextNow_preferences.xml'

    <string name="userinfo_username">thisisdfir</string>
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.enflick.android.TextNow/databases/textnow_data.db'
```



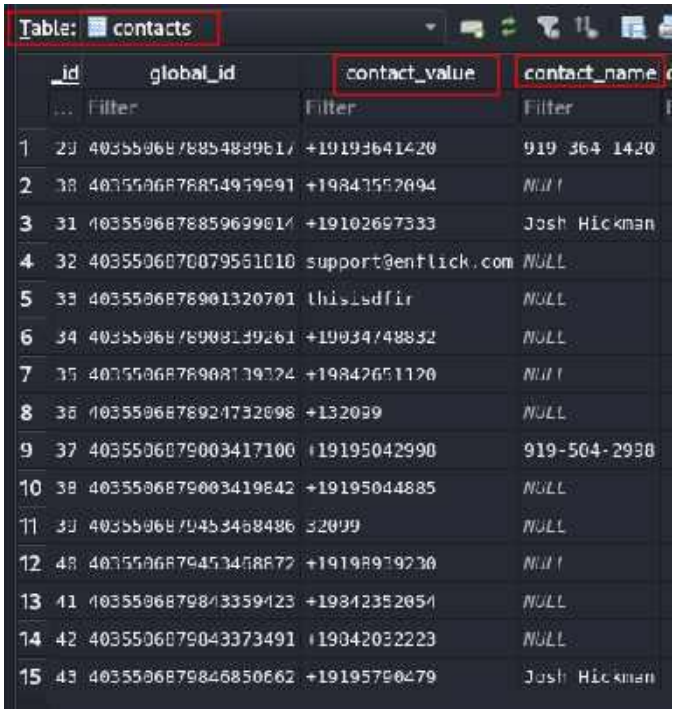
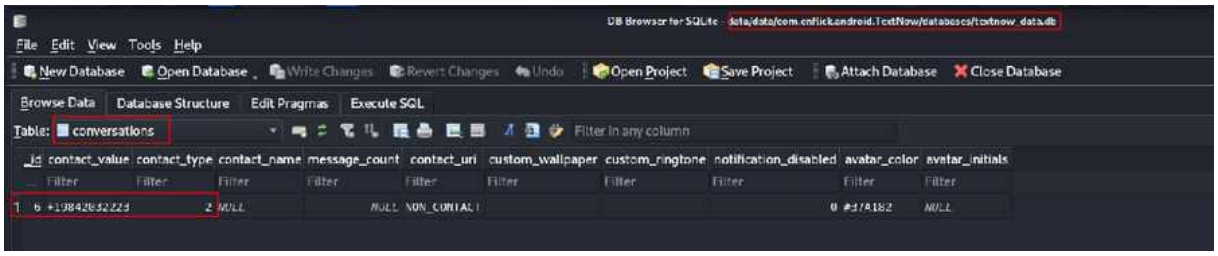


Table: contacts

	_id	global_id	contact_value	contact_name
	... Filter	Filter	Filter	Filter
1	29	4035506879885489617	+19193641420	919 364 1420
2	38	4035506878854959991	+19843552094	NULL
3	31	4035506878859699014	+19102697333	Josh Hickman
4	32	4035506870079551010	support@enflick.com	NULL
5	33	4035506878901320701	thisisdfir	NULL
6	34	4035506879008139261	+19034748832	NULL
7	35	4035506879008139324	+19842651120	NULL
8	36	4035506879024732098	+132099	NULL
9	37	4035506879003417100	+19195042990	919-504-2990
10	39	4035506879003419842	+19195044885	NULL
11	30	4035506879453468486	32099	NULL
12	48	4035506879453468672	+19198919230	NULL
13	41	4035506879843359423	+19842352054	NULL
14	42	4035506879843373491	+19842032223	NULL
15	43	4035506879846850662	+19195790479	Josh Hickman



DB Browser for SQLite - /data/data/com.enflick.android.TextNow/databases/textnow_data.db

File Edit View Tools Help

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Database

Browse Data | Database Structure | Edit Pragmas | Execute SQL

Table: conversations | Filter in any column

_id	contact_value	contact_type	contact_name	message_count	contact_uri	custom_wallpaper	custom_ringtone	notification_disabled	avatar_color	avatar_initials
... Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	6 +19842032223	2	NULL		NULL	NON_CONTACT			0 #37A1B2	NULL

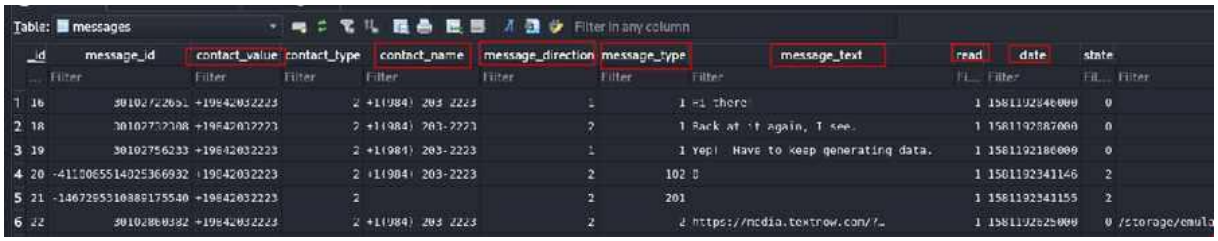
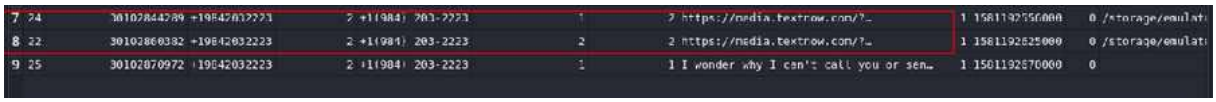


Table: messages | Filter in any column

_id	message_id	contact_value	contact_type	contact_name	message_direction	message_type	message_text	read	date	state
... Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Fi...	Filter	Fil... Filter
1 16	30102722651	+19842032223	2	+1(984) 203-2223	1	1	Hi there	1	1581192840000	0
2 18	30102732308	+19842032223	2	+1(984) 203-2223	2	1	Back at it again, I see.	1	1581192887000	0
3 19	30102756233	+19842032223	2	+1(984) 203-2223	1	1	Yep! Have to keep generating data.	1	1581192186000	0
4 20	-4118065514325366932	+19842032223	2	+1(984) 203-2223	2	102	0	1	1581192341146	2
5 21	-1467285510388175540	+19842032223	2		2	201		1	1581192341155	2
6 22	30102860382	+19842032223	2	+1(984) 203-2223	2	2	https://mcdia.textrow.com/?…	1	1581192825000	0 /storage/emula



7 24	30102844789	+19842032223	2	+1(984) 203-2223	1	2	https://mcdia.textrow.com/?…	1	1581192750000	0 /storage/emulat
8 22	30102860382	+19842032223	2	+1(984) 203-2223	2	2	https://mcdia.textrow.com/?…	1	1581192825000	0 /storage/emulat
9 25	30102870972	+19842032223	2	+1(984) 203-2223	1	1	I wonder why I can't call you or sen…	1	1581192870000	0



```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree -L 1 'data/data/com.whatsapp/'

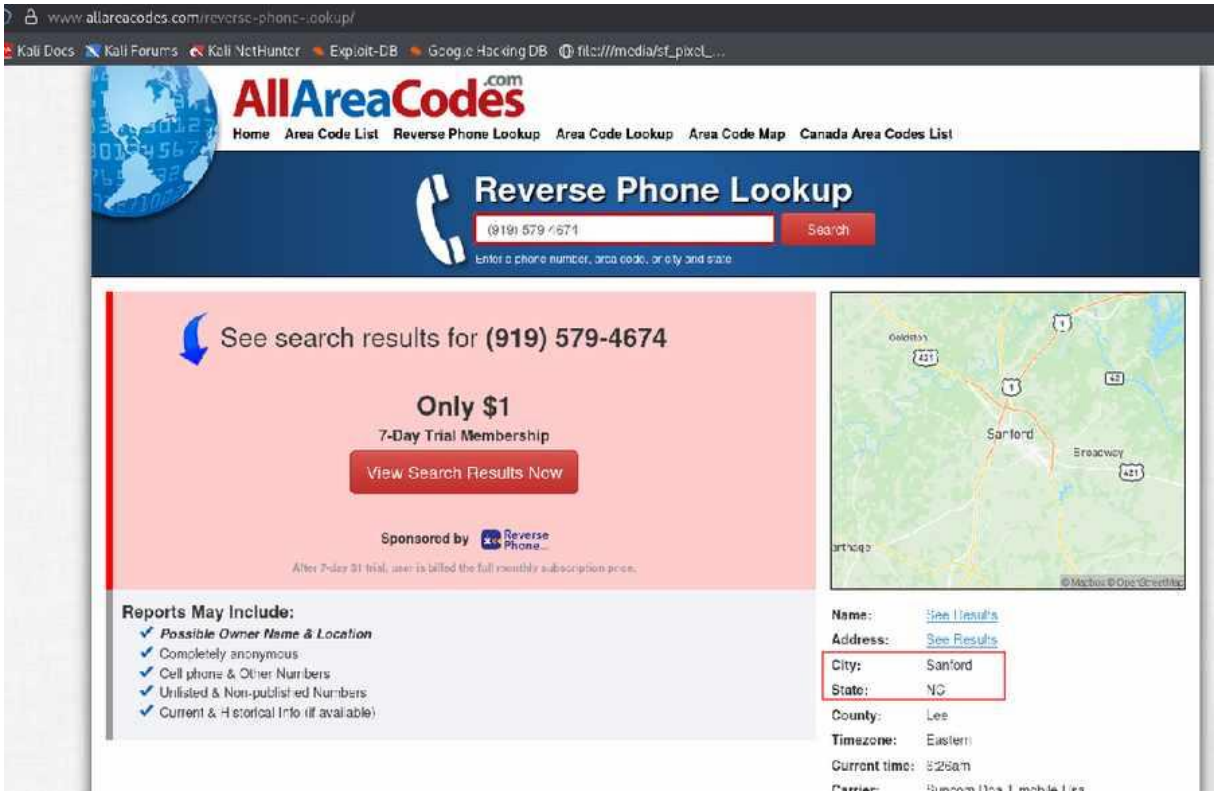
data/data/com.whatsapp/
├── cache
├── databases
├── files
├── no_backup
└── shared_prefs
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ cd 'data/data/com.whatsapp/shared_prefs'

┌──(kali㉿kali)-[/media/…/data/data/com.whatsapp/shared_prefs]
└─$ tree -L 1 .
.
├── ab-props.xml
├── chatCounts.xml
├── com.google.android.gms.appid.xml
├── com.google.maps.api.android.lib6.drd.PREFERENCES_FILE.xml
├── com.whatsapp_payment_preferences.xml
├── com.whatsapp_preferences_light.xml
├── com.whatsapp_preferences.xml
├── contact_sync_prefs.xml
├── field-stats-events-sampling.xml
├── _has_set_default_values.xml
├── keystore.xml
├── MapviewInitializerPreferences.xml
├── media_bandwidth_shared_preferences_v2.xml
├── media_daily_usage_preferences_v1.xml
├── registration.RegisterPhone.xml
├── registration.VerifySms.xml
├── time_spent_prefs.xml
└── voip_prefs.xml
```

```
┌──(kali㉿kali)-[/media/…/data/data/com.whatsapp/shared_prefs]
└─$ cat 'registration.RegisterPhone.xml'

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <int name="com.whatsapp.registration.RegisterPhone.country_code_position" value="-1" />
    <int name="com.whatsapp.registration.RegisterPhone.phone_number_position" value="10" />
    <string name="com.whatsapp.registration.RegisterPhone.input_country_code">1</string>
    <string name="com.whatsapp.registration.RegisterPhone.country_code">1</string>
    <string name="com.whatsapp.registration.RegisterPhone.phone_number">9195794674</string>
    <int name="com.whatsapp.registration.RegisterPhone.verification_state" value="0" />
    <string name="com.whatsapp.registration.RegisterPhone.input_phone_number">(919) 579-4674</string>
</map>
```





```
┌──(kali㉿kali)-[/media/…/data/data/com.whatsapp/shared_prefs]
└─$ cat 'com.whatsapp_preferences.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <string name="TestKeySet">proxy_gif_external_services;mms_vcache_aggregation_enabled;mms_hot_content_timespan_in_seconds</string>
    <boolean name="watls_early_data" value="true" />
    <int name="wam_buffer_count" value="3" />
    <int name="status_video_max_duration" value="30" />
    <int name="status_image_max_edge" value="1280" />
    <boolean name="server_props:one_time_unlocked" value="false" />
    <int name="media_limit_auto_download_mb" value="32" />
    <boolean name="consumer_content_provider" value="true" />
    <int name="subject_length_limit" value="25" />
```



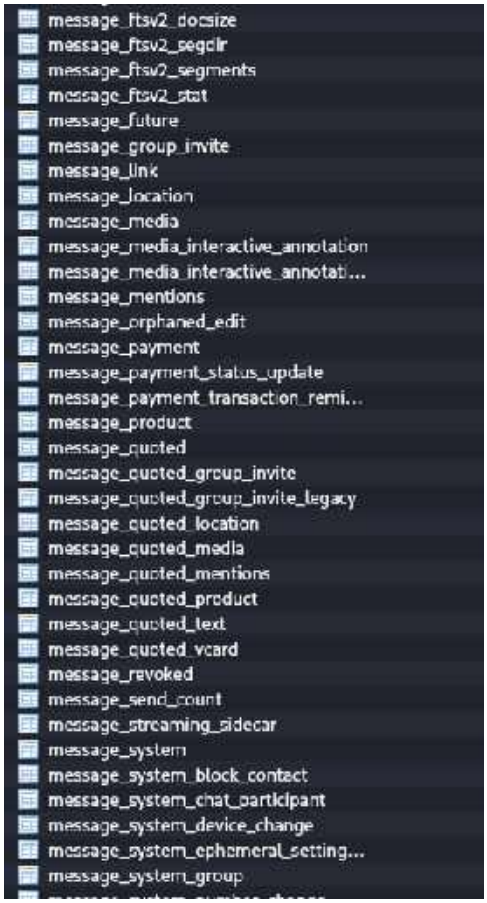
```
┌──(kali㉿kali)-[/media/…/data/data/com.whatsapp/shared_prefs]
└─$ cat 'com.whatsapp_preferences.xml' | wc -l
94
```

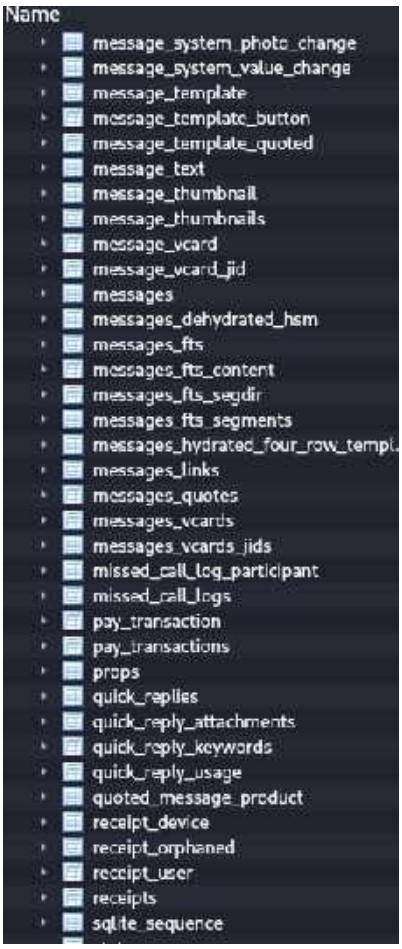
```
┌──(kali㉿kali)-[/media/…/Pixel 3/data/data/com.whatsapp]
└─$ cd databases

┌──(kali㉿kali)-[/media/…/data/data/com.whatsapp/databases]
└─$ ll
total 3448
-rwxrwx--- 1 root vboxsf 233472 Feb 16 22:55 axolotl.db
-rwxrwx--- 1 root vboxsf  32768 Feb 16 22:55 axolotl.db-shm
-rwxrwx--- 1 root vboxsf 432632 Feb 16 22:55 axolotl.db-wal
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:55 chatsettings.db
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:55 companion_devices.db
-rwxrwx--- 1 root vboxsf 425984 Feb 16 22:55 emojidictionary.db
-rwxrwx--- 1 root vboxsf  24576 Feb 16 22:55 hsmpacks.db
-rwxrwx--- 1 root vboxsf  16384 Feb 16 22:55 _jobqueue-WhatsAppJobManager
-rwxrwx--- 1 root vboxsf   4096 Feb 16 22:55 location.db
-rwxrwx--- 1 root vboxsf  32768 Feb 16 22:55 location.db-shm
-rwxrwx--- 1 root vboxsf 123632 Feb 16 22:55 location.db-wal
-rwxrwx--- 1 root vboxsf   4096 Feb 16 22:55 media.db
-rwxrwx--- 1 root vboxsf  32768 Feb 16 22:55 media.db-shm
-rwxrwx--- 1 root vboxsf 164832 Feb 16 22:55 media.db-wal
-rwxrwx--- 1 root vboxsf 823296 Feb 16 22:55 msgstore.db
-rwxrwx--- 1 root vboxsf  32768 Feb 16 22:55 msgstore.db-shm
-rwxrwx--- 1 root vboxsf 449112 Feb 16 22:55 msgstore.db-wal
-rwxrwx--- 1 root vboxsf   4096 Feb 16 22:55 payments.db
-rwxrwx--- 1 root vboxsf  32768 Feb 16 22:55 payments.db-shm
-rwxrwx--- 1 root vboxsf  82432 Feb 16 22:55 payments.db-wal
-rwxrwx--- 1 root vboxsf  32768 Feb 16 22:55 statusranking.db
-rwxrwx--- 1 root vboxsf 118784 Feb 16 22:55 stickers.db
-rwxrwx--- 1 root vboxsf   4096 Feb 16 22:55 wa.db
-rwxrwx--- 1 root vboxsf  32768 Feb 16 22:55 wa.db-shm
-rwxrwx--- 1 root vboxsf 325512 Feb 16 22:55 wa.db-wal
```

```
┌──(kali㉿kali)-[/media/…/data/data/com.whatsapp/databases]
└─$ sqlitebrowser 'msgstore.db'
```


Name
Type
Tables (106)
away_messages
away_messages_exemptions
call_log
call_log_participant_v2
chat
chat_list
conversion_tuples
deleted_chat_job
frequent
frequents
group_participant_device
group_participant_user
group_participants
group_participants_history
jid
keywords
labeled_jid
labeled_jids
labeled_messages
labeled_messages_fts
labeled_messages_fts_content
labeled_messages_fts_segdir
labeled_messages_fts_segments
labels
media_hash_thumbnail
media_refs
message
message_ephemeral
message_forwarded
message_ftsv2
message_ftsv2_content
message_ftsv2_docsize
message_ftsv2_segdir
message_ftsv2_segments
message_ftsv2_stat

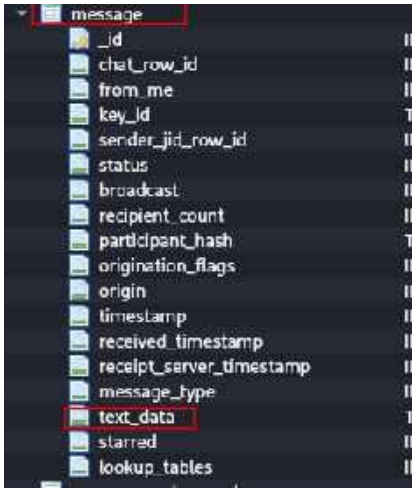

message_ftsv2_docsize
message_ftsv2_segdir
message_ftsv2_segments
message_ftsv2_stat
message_future
message_group_invite
message_link
message_location
message_media
message_media_interactive_annotation
message_media_interactive_annotati...
message_mentions
message_orphaned_edit
message_payment
message_payment_status_update
message_payment_transaction_remi...
message_product
message_quoted
message_quoted_group_invite
message_quoted_group_invite_legacy
message_quoted_location
message_quoted_media
message_quoted_mentions
message_quoted_product
message_quoted_text
message_quoted_vcard
message_revoked
message_send_count
message_streaming_sidecar
message_system
message_system_block_contact
message_system_chat_participant
message_system_device_change
message_system_ephemeral_setting...
message_system_group

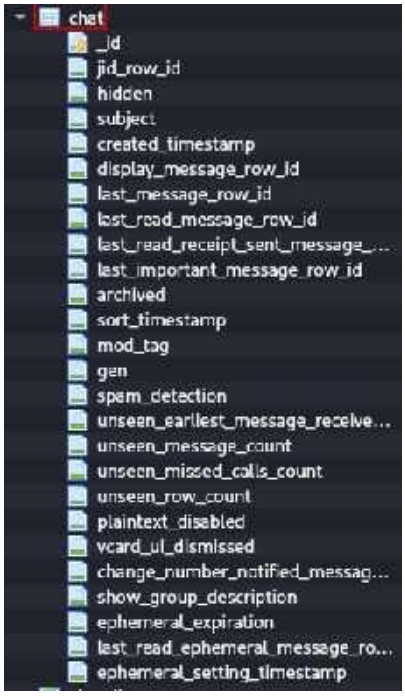


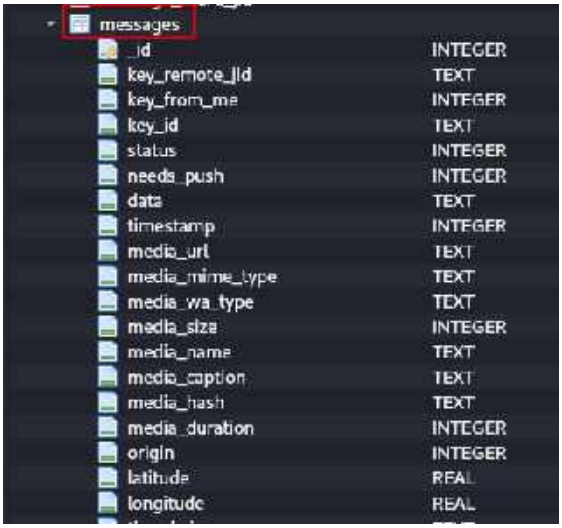

```
┌──(kali㉿kali)-[/media/…/data/data/com.whatsapp/databases]
└─$ sqlite3 'msgstore.db'
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> PRAGMA table_info(jid)
   ...> ;
0|_id|INTEGER|0||1
1|user|TEXT|1||0
2|server|TEXT|1||0
3|agent|INTEGER|0||0
4|device|INTEGER|0||0
5|type|INTEGER|0||0
6|raw_string|TEXT|0||0
sqlite> SELECT * FROM jid;
1|status|broadcast|0|0|5|status@broadcast
2|0|s.whatsapp.net|0|0|7|0@s.whatsapp.net
3|16505434800|s.whatsapp.net|0|0|0|16505434800@s.whatsapp.net
4|19195790479|s.whatsapp.net|0|0|0|19195790479@s.whatsapp.net
5|19195790479|s.whatsapp.net|0|0|17|19195790479.0:0@s.whatsapp.net
sqlite>
```

Table: jid

	_id	user	server	agent	device	type	raw_string
	...	Filter	Filter	Filter	Filter	Fi...	Filter
1	1	status	broadcast	0	0	5	status@broadcast
2	2	0	s.whatsapp.net	0	0	7	0@s.whatsapp.net
3	3	16505434800	s.whatsapp.net	0	0	0	16505434800@s.whatsapp.net
4	4	19195790479	s.whatsapp.net	0	0	0	19195790479@s.whatsapp.net
5	5	19195790479	s.whatsapp.net	0	0	17	19195790479.0:0@s.whatsapp.net







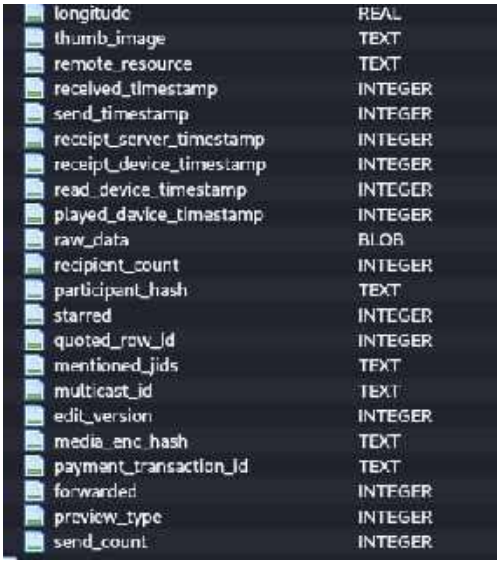


Table: messages

Filter in any column

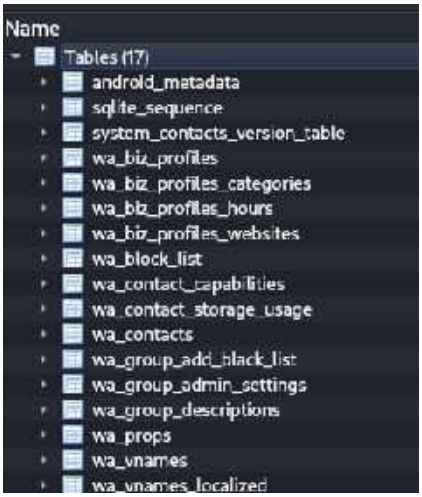
	_id	key_remote_jid	key_from_me	key_id	status	needs_push	data	timestamp	media_url
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	1	0	1	1	0	NULL	0	NULL
2	2	status@broadcast	0	PSAS1487100001	0	0	NULL	1487100001000	https://mmg-fna.whatsapp.net/d/f/…
3	3	status@broadcast	0	PSAS1487100002	0	0	NULL	1487100002000	https://mmg-fna.whatsapp.net/d/f/…
4	4	status@broadcast	0	PSAS1407100003	0	0	NULL	1407100003000	https://mmg-fna.whatsapp.net/d/f/…
5	5	status@broadcast	0	PSAS1487100004	0	0	NULL	1487100004000	https://mmg-fna.whatsapp.net/d/f/…
6	6	status@broadcast	0	PSAS1487100005	0	0	NULL	1487100005000	https://mmg-fna.whatsapp.net/d/f/…
7	7	status@broadcast	0	PSAS1487100006	0	0	NULL	1487100006000	https://mmg-fna.whatsapp.net/d/f/…
8	8	16505434800@s.whatsapp.net	1	3B848CBC6E3E805A2BFC36260300B088	6	0	NULL	1500323811221	NULL
9	9	16505434800@s.whatsapp.net	0	AED229F4D12307A3C1	0	0	NULL	1500323810000	NULL
10	10	16505434800@s.whatsapp.net	1	8A1C679E98A94DB71E392F4485320332	6	0	NULL	1500323811285	NULL
11	11	19195799479@s.whatsapp.net	1	81125A27B65EA6B8108410A51BBC48A	6	0	NULL	1581195314711	NULL
12	12	19195799479@s.whatsapp.net	1	C3FFB4CF2001F9F2F6C80DA901FCD1FF	13	0	Hi there - I switched over…	1581195314692	NULL
13	13	19195799479@s.whatsapp.net	0	A957774C3DB020FFCFCA6DA387909A21	0	0	Awesome!	1581195365000	NULL
14	14	19195799479@s.whatsapp.net	0	9D753662950E0C4139741CE3A3FC26D	13	0	NULL	1581195425000	https://mmg-fna.whatsapp.net/d/f/…
15	15	19195799479@s.whatsapp.net	1	ED3568DDE1CB11F3FDE685A11982679	13	0	The Osbourne!	1581195469433	NULL
16	16	19195799479@s.whatsapp.net	1	24E60A3E0ADBCD3385D44FBEFFE1D2F	13	0	NULL	1581195492603	https://mmg-fna.whatsapp.net/d/f/…
17	17	19195799479@s.whatsapp.net	1	6D1AC9AC9032CC05F2BD9C551F3BFC07	13	0	NULL	1581271214137	NULL
18	18	19195799479@s.whatsapp.net	1	0DCD9C73F8A5A831C5CCA829FF2AF268	13	0	NULL	1581271233018	NULL
19	19	19195799479@s.whatsapp.net	1	8B5001D727C017D0D72540C5DF223258	13	0	NULL	1581271502090	NULL

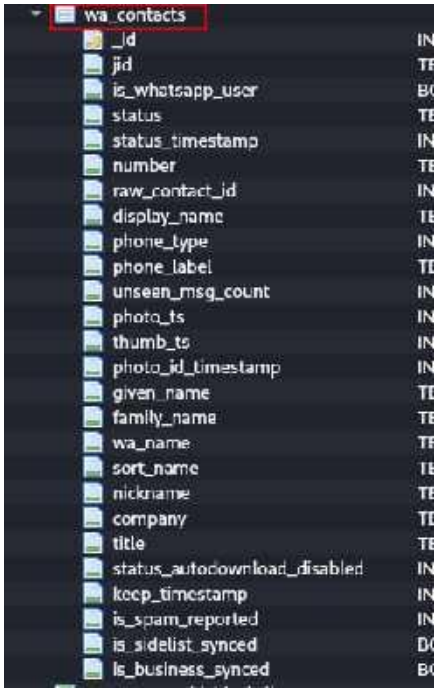


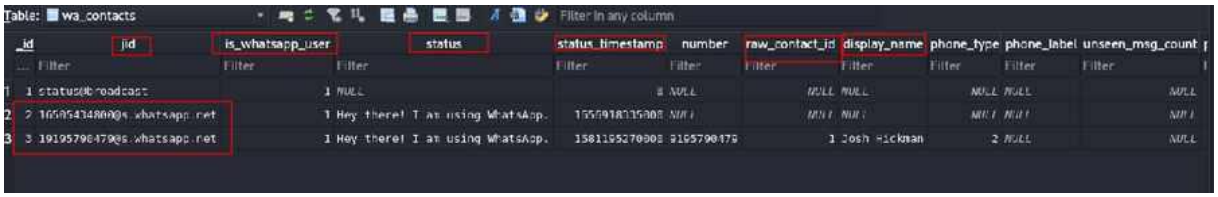
media_wa_type	media_size	media_name	media_caption	media_hash	media_duration	origin	atitude	l
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Fil
1	1	NULL	NULL	NULL	0	0	0.0	
1	91858	NULL	NULL	VAgR95+nono4s98BPS+p5FRWFf004hVkkl3U…	0	0	0.0	
13	274412	NULL	NULL	qdo73F5-68EpDC/…	30	0	0.0	
1	97706	NULL	NULL	OtD90OsTUAX0dll23ha4NLBKgsxO14D52NvSY…	0	0	0.0	
1	72973	NULL	NULL	bPpJaE6ik6EgL+rgAHI2LlNnDUr9PLCvxTu0…	0	0	0.0	
1	118435	NULL	NULL	YAixUY8x2iXPn/c+uOsVuQ/545z+bgMg5hfz…	0	0	0.0	
1	84270	NULL	NULL	nHhlpcnRvrVeolkab/…	0	0	0.0	
0	19	NULL	NULL	NULL	0	0	0.0	
27	0	NULL	NULL	NULL	0	0	0.0	
0	47	Facebook	NULL	NULL	0	0	0.0	
0	19	NULL	NULL	NULL	0	0	0.0	
0	0	NULL	NULL	NULL	0	0	0.0	
0	0	NULL	NULL	NULL	0	0	0.0	
1	21080	NULL	NULL	Op0S0PLJKg6nCJ9vtmVtTvn L3A02chu6/…	0	0	0.0	
0	0	NULL	NULL	NULL	0	0	0.0	
1	87989	2b186109 b8b4 473d 8ecb …	NULL	Q4V1x5JMsSwzXUYYqSysAnsHLykYQsA1YYjH…	0	0	0.0	
16	1581271157003081	NULL	Here I am!!	NULL	18	0	35.6609133	-7
16	1581271157003082	NULL	Here I am!!	NULL	212	0	35.6609423	-7
5	0	NULL	NULL	NULL	0	0	35.6609311	-7

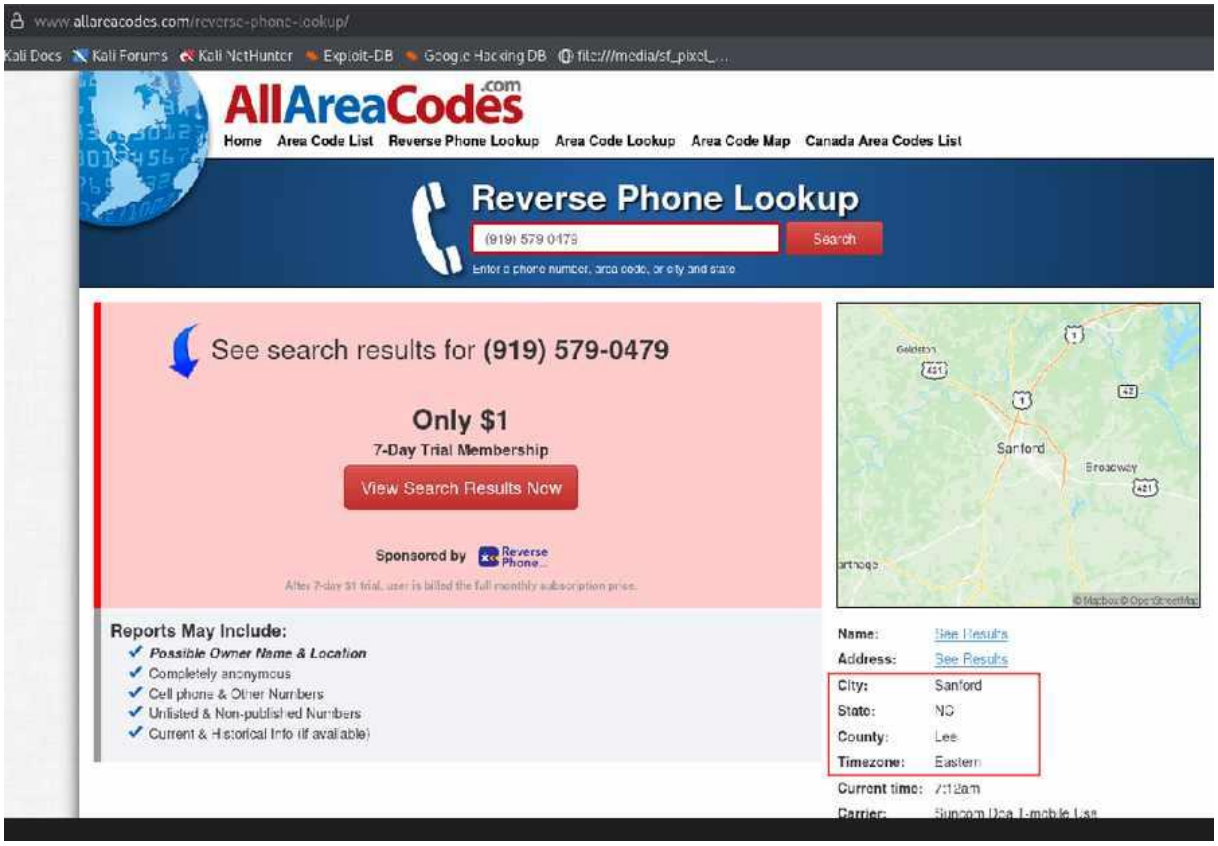
```
┌──(kali㉿kali)-[/media/…/data/data/com.whatsapp/databases]
└─$ sqlite3 'msgstore.db'
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> SELECT m.data FROM messages m JOIN jid j ON m.key_remote_jid = j.raw_string JOIN chat c ON c.jid_row_id = j._id WHERE c._id = 3;
Hi there!  I switched over.
Awesome!
The Osbourne!
```

```
┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ sqlitebrowser 'data/data/com.whatsapp/databases/wa.db'
```

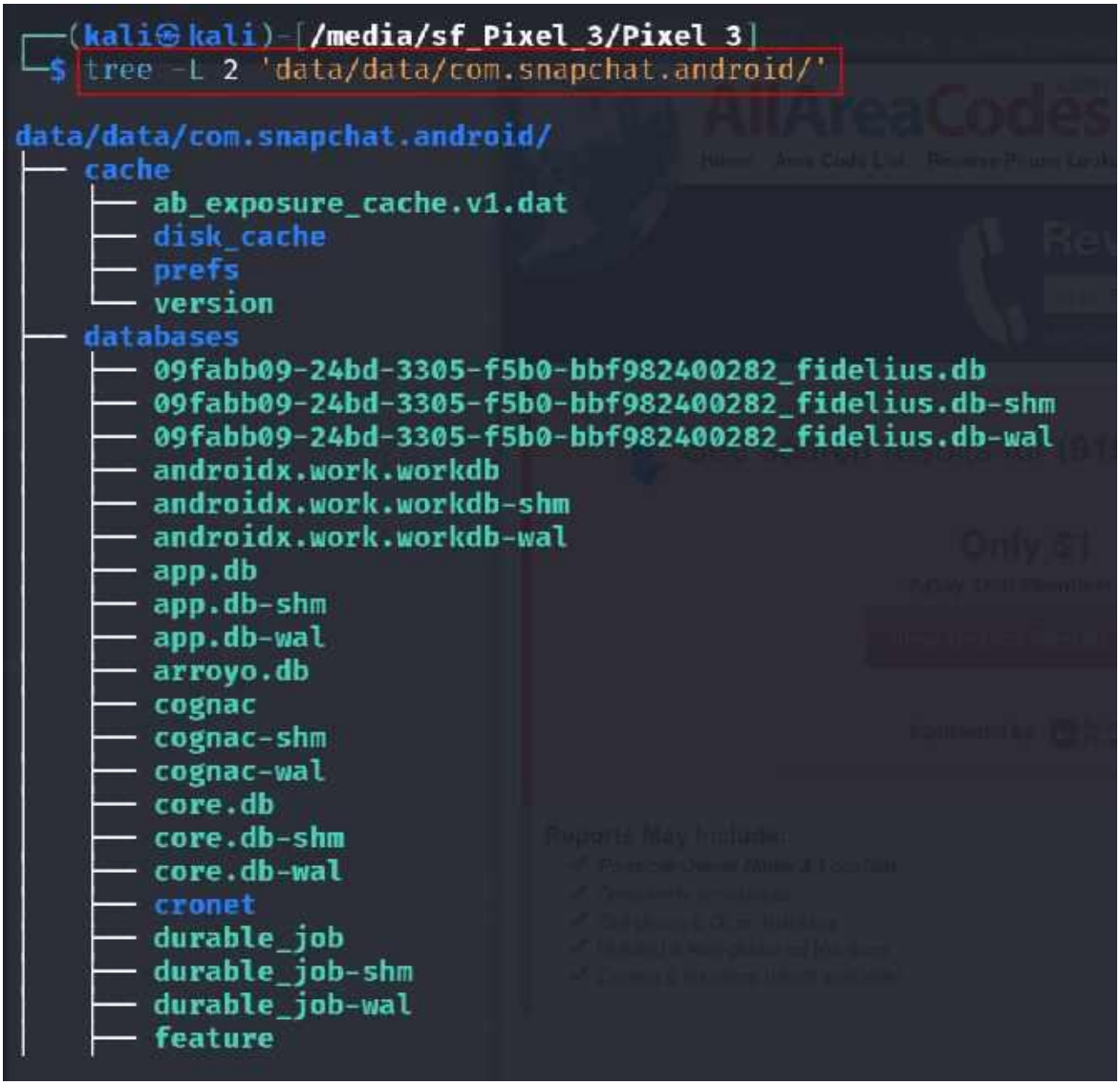


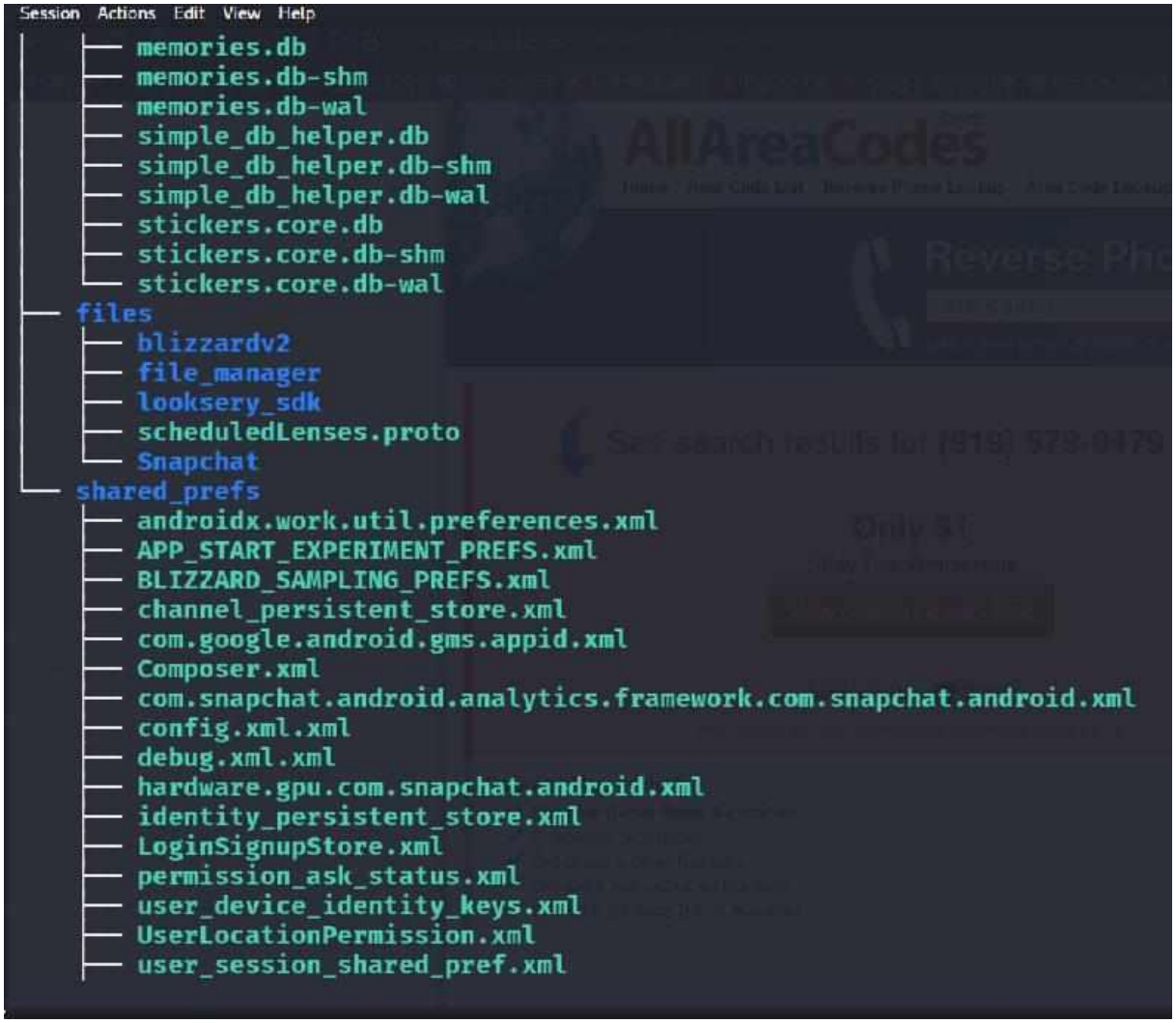

wa_contacts
_id
jid
is_whatsapp_user
status
status_timestamp
number
raw_contact_id
display_name
phone_type
phone_label
unseen_msg_count
photo_ts
thumb_ts
photo_id_timestamp
given_name
family_name
wa_name
sort_name
nickname
company
title
status_autodownload_disabled
keep_timestamp
is_spam_reported
is_sidelist_synced
is_business_synced

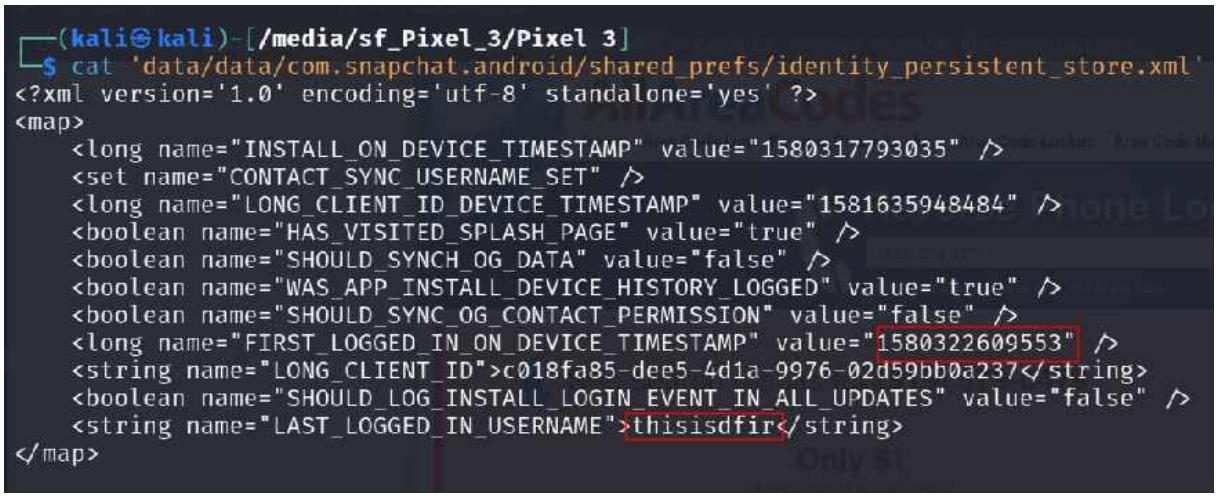

Table: wa_contacts
Filter in any column
_id
jid
is_whatsapp_user
status
status_timestamp
number
raw_contact_id
display_name
phone_type
phone_label
unseen_msg_count
1 1 status@broadcast 1 NULL 0 NULL NULL NULL NULL NULL NULL
2 2 16505434800@s.whatsapp.net 1 Hey there! I am using WhatsApp. 1550918115000 NULL NULL NULL NULL NULL NULL
3 3 19195790479@s.whatsapp.net 1 Hey there! I am using WhatsApp. 1581195270000 9195790479 1 Josh Hickman 2 NULL NULL

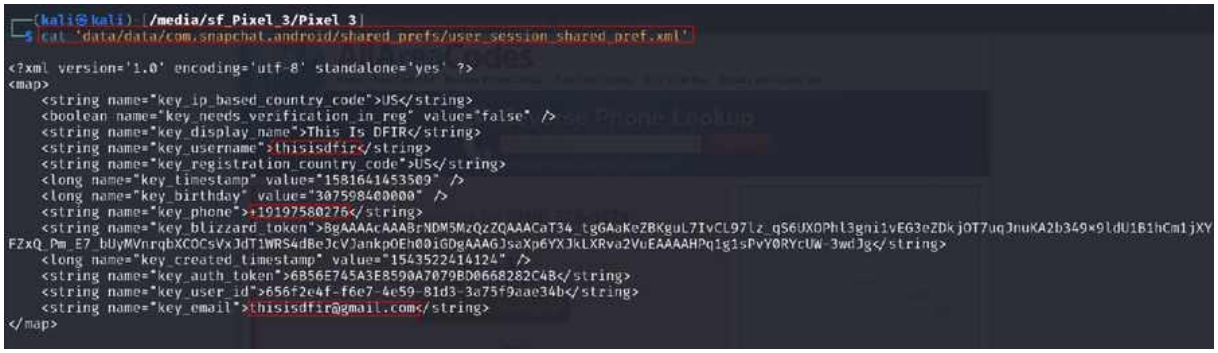

www.allareacodes.com/reverse-phone-lookup/
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
file:///media/sf_pixel_...
AllAreaCodes.com
Home
Area Code List
Reverse Phone Lookup
Area Code Lookup
Area Code Map
Canada Area Codes List
Reverse Phone Lookup
(919) 579-0479
Search
Enter a phone number, area code, or city and state
See search results for (919) 579-0479
Only $1
7-Day Trial Membership
View Search Results Now
Sponsored by
Reports May Include:
Possible Owner Name & Location
Completely anonymous
Cell phone & Other Numbers
Unlisted & Non-published Numbers
Current & Historical Info (if available)
Sanford
Broadway
Name:
See Results
Address:
See Results
City:
Sanford
State:
NC
County:
Lee
Timezone:
Eastern
Current time:
Carrier:


(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$ tree -L 1 'data/data/com.snapchat.android/'
data/data/com.snapchat.android/
cache
databases
files
shared_prefs


┌──(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
└─$ tree -L 2 'data/data/com.snapchat.android/'
data/data/com.snapchat.android/
├── cache
│   ├── ab_exposure_cache.v1.dat
│   ├── disk_cache
│   ├── prefs
│   └── version
├── databases
│   ├── 09fabb09-24bd-3305-f5b0-bbf982400282_fidelius.db
│   ├── 09fabb09-24bd-3305-f5b0-bbf982400282_fidelius.db-shm
│   ├── 09fabb09-24bd-3305-f5b0-bbf982400282_fidelius.db-wal
│   ├── androidx.work.workdb
│   ├── androidx.work.workdb-shm
│   ├── androidx.work.workdb-wal
│   ├── app.db
│   ├── app.db-shm
│   ├── app.db-wal
│   ├── arroyo.db
│   ├── cognac
│   ├── cognac-shm
│   ├── cognac-wal
│   ├── core.db
│   ├── core.db-shm
│   ├── core.db-wal
│   ├── cronet
│   ├── durable_job
│   ├── durable_job-shm
│   ├── durable_job-wal
│   ├── feature


Session Actions Edit View Help
memories.db
memories.db-shm
memories.db-wal
simple_db_helper.db
simple_db_helper.db-shm
simple_db_helper.db-wal
stickers.core.db
stickers.core.db-shm
stickers.core.db-wal
files
blizzardv2
file_manager
looksery_sdk
scheduledLenses.proto
Snapchat
shared_prefs
androidx.work.util.preferences.xml
APP_START_EXPERIMENT_PREFS.xml
BLIZZARD_SAMPLING_PREFS.xml
channel_persistent_store.xml
com.google.android.gms.appid.xml
Composer.xml
com.snapchat.android.analytics.framework.com.snapchat.android.xml
config.xml.xml
debug.xml.xml
hardware.gpu.com.snapchat.android.xml
identity_persistent_store.xml
LoginSignupStore.xml
permission_ask_status.xml
user_device_identity_keys.xml
UserLocationPermission.xml
user_session_shared_pref.xml


(kali㉿kali)-[/media/sf_Pixel_3/Pixel 3]
$ cat 'data/data/com.snapchat.android/shared_prefs/identity_persistent_store.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
<long name="INSTALL_ON_DEVICE_TIMESTAMP" value="1580317793035" />
<set name="CONTACT_SYNC_USERNAME_SET" />
<long name="LONG_CLIENT_ID_DEVICE_TIMESTAMP" value="1581635948484" />
<boolean name="HAS_VISITED_SPLASH_PAGE" value="true" />
<boolean name="SHOULD_SYNCH_OG_DATA" value="false" />
<boolean name="WAS_APP_INSTALL_DEVICE_HISTORY_LOGGED" value="true" />
<boolean name="SHOULD_SYNC_OG_CONTACT_PERMISSION" value="false" />
<long name="FIRST_LOGGED_IN_ON_DEVICE_TIMESTAMP" value="1580322609553" />
<string name="LONG_CLIENT_ID">c018fa85-dee5-4d1a-9976-02d59bb0a237</string>
<boolean name="SHOULD_LOG_INSTALL_LOGIN_EVENT_IN_ALL_UPDATES" value="false" />
<string name="LAST_LOGGED_IN_USERNAME">thisisdfir</string>
</map>



Event Chronology (13–15 February 2020, UTC)

I chose this 48-hour window because seven artefact families record activity inside it: Snapchat, Gmail, account manager, user profiles, Google Photos, Kik and the Play Store logs.

#	UTC time	Raw value	Source	Event / qualification
1	2020-02-13 23:19:08	1581635948484	Snapchat identity_persistent_store.xml	Client ID timestamp set; app state, not a confirmed user action
2	2020-02-14 01:19:15	last_password_entry	accounts_de.db › accounts	Google Duo account added
3	2020-02-14 01:34:22	1581644062878	Gmail UnifiedEmail.xml	Gmail analytics epoch; consistent with Gmail use near #2
4	2020-02-14 01:39:18	1581644358241	users/11.xml created	Secondary user 11 created; corroborates activity in #2–#3
5	2020-02-14 02:33:15	1581647595518	Photos backup_complete_times.xml	Photos backup completed for account 0
6	2020-02-14 12:50:01	1581684601922	users/11.xml lastLoggedIn	User 11 session
7	2020-02-14 15:01:23	1581692483339	Kik gms measurement prefs	Kik health monitor start
8	2020-02-14 15:51:57	1581695517124	users/0.xml lastLoggedIn	Owner user 0 last session
9	2020-02-15 12:54:02	1581771242040	Play Store play_metalog.log (xxd)	Play log written for thisisdfir@gmail.com; matched on EpochConverter

Corroboration: (a) the Duo account (#2), the Gmail epoch (#3) and the creation of user 11 (#4) come from three separate sources and fall within 20 minutes of each other. (b) The user 11 and

user 0 logins (#6, #8) come from the user XMLs. The Kik service activity (#7), from a separate source, falls between them.

Interpretation and Limits

Device and accounts: build.prop, telephony.db and the user XMLs agree that the device is a Pixel 3 on Android 10 with Google Fi SIMs, two users and the thisisdfir Google identity. Limit: the SIM and account records describe the handset. They do not identify a person.

System and app use: accounts_de.db and the app preference files show most app accounts being set up on 29 January 2020, with more on 1, 8 and 14 February. Alternative explanation: these times could come from sync or restore activity rather than manual sign-in.

Communications: Kik, TextNow and WhatsApp each contain two-way exchanges with an account labelled Josh Hickman. Limit: a contact label and a phone number do not prove who was typing. WhatsApp coordinates show what was shared in a message, not a verified position.

Chronology: the 13–15 February sequence is supported by several sources. Limit: the copy is logical and was taken after rooting. Deleted or unallocated data was not examined, and device clock accuracy was not independently verified.

Appendix A – Finding Log

ID	Finding	Source	Raw	Converted (UTC)	Qualification
F01	Model Pixel 3, Android 10	product/build.prop	ro.product.product.model	—	Manual cat; matches system props
F02	Owner last login	data/system/users/0.xml	1581695517124	2020-02-14 15:51:57	Manually converted with date -u
F03	Google account added	accounts_de.db	epoch ms	2020-01-29 15:41:07	SQLite datetime() query
F04	Kik install	Kik gms measurement prefs	1580317722000	2020-01-29 17:08:42	Corroborated by Branch install time
F05	Kik login	KikPreferences.xml kik.logintime	1580322563292	2020-01-29 18:29:23	App-level login
F06	Snapchat first login	identity_persistent_store.xml	1580322609553	2020-01-29 18:30:09	Close to F05; same username
F0	TextNow	textnow_data.db ›	1581192840000	2020-02-	Direction field

7	message	messages		08 20:14:00	recorded
F08	WhatsApp text	msgstore.db › messages	1581195314692	2020-02-08 20:55:14	sqlite3 join matched GUI view
F09	Photos backup	backup_complete_times.xml	1581647595518	2020-02-14 02:33:15	Backup state only
F10	Play metalog	play_metalog.log	1581771242040	2020-02-15 12:54:02	Manual + EpochConverter matched

Appendix B – Key Commands and Queries

```
cat 'product/build.prop'  |  grep patch 'system/build.prop'

grep -ri 'serial number' 'mnt/vendor/'

sqlite3 data/system_de/0/accounts_de.db "select
name,type,datetime(last_password_entry_time_millis_epoch/1000,'unixepoch') from
accounts;"

grep -EiorhI '([[:alnum:]_.-]+@gmail\.com)' | sort | uniq -c | sort -nr | head

SELECT m.data FROM messages m JOIN jid j ON m.key_remote_jid=j.raw_string JOIN chat c ON
c.jid_row_id=j._id WHERE c._id=3;

date -d '@1581695517' -u
```

Appendix C – Manual Validation

WhatsApp: the sqlite3 join in Appendix B returned the same three message bodies shown in the DB Browser view of msgstore.db › messages. Result: matched.

Play metalog: the epoch read from the hex dump was converted with EpochConverter and with date -u, and both gave 2020-02-15 12:54:02 UTC. Result: matched.

Archive hash: certutil SHA-256 of Pixel 3.zip returned ca6918ef…8aa00a, the same as the course brief value. Result: matched.

Academic Integrity Declaration

I confirm that this report reflects my own examination of the authorised course evidence and that I have identified all sources, tools, and assistance used.

Signed: Chukwuebuka Basil Okorie Date: 10 October 2026